

THE CHAUTAUQUAN

*A Monthly Magazine Devoted to
the Promotion of True Culture.*

*Organ of the
Chautauqua Literary and
Scientific Circle.*

Vol. V
June, 1885.
No. 9.

*Theodore L. Flood, D.D., Editor
The Chautauqua Press*

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SCIENTIFIC CIRCLE.*

VOL. V.

JUNE, 1885.

No. 9.

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REQUIRED READING FOR JUNE.

THE MECHANISM OF THE ENGLISH LANGUAGE.

BY PRESIDENT D. H. WHEELER, D.D., LL.D.

To *us* the unit of speech is the word; historically, the unit is the sentence. It matters little which of the theories respecting the first forms of speech we adopt; all such theories may be rejected, and still we shall find it most reasonable to believe that man's earliest utterances were wholes, answering in value to our sentences. A revolution has been effected and we have a *part of speech* for our unit. We construct or build our sentences out of pieces of different meaning and value. Our simplest sentence has two of these pieces—a subject or noun, and a verb; a long sentence may have a dozen or a score of pieces. The making of sentences out of parts of speech is a kind of mechanics. The sentence has its mechanism, of which we usually learn the science by analyzing sentences. This analytical process yields what we call the principles of syntax. It must be remembered, however, that we learn to talk before we learn grammar, and that multitudes of people scarcely know any unit except the sentence. Their vocabulary is a phrase-book, in which every word has a fixed and unchangeable position. These persons abound in the illiterate countries; in Italy, for example, the majority of the people speak only in sentences having invariable forms; change the order of the words and you become unintelligible to them. The same effect is produced by employing a synonym for any word in any sentence. Our people are usually more alert to variety in expression and catch meanings in forms and arrangements to which they are unaccustomed.

A long sentence falls, when we take it apart, into two large pieces; the subject and its belongings, and the predicate and its belongings. Each of these large pieces breaks up into a number of small pieces. If we look carefully at the average long sentence, we shall find that the parts are held together by a systematic and habitual principle of arrangement, and that this changes in passing from one language to another. French says “a man

good,” English “a good man.” Reverse the order of noun and adjective in either language, and the sense is obscured for the average hearer or reader. There is a number of these differences; and therefore every language has its peculiar mechanism. In mechanical type languages fall into groups. Greek and Latin, for example, use inflections to connect the words with each other; English does not employ inflections for this purpose. We have a few inflected forms, but we use them merely because they have come down to us. Greek syntax is inflectional; our syntax is said to be that of flat construction, or, as I prefer to say, it is positional. The *place* of a word determines its function and relations in the sentence. This flat construction is found in other tongues; but English abounds in it and depends upon it as a principle of arrangement. When we say “proud men,” the hearer knows that the adjective *proud* describes the noun *men*. In Latin, the adjective would have a termination to correspond in value to that of the noun, and the two might be separated by several words. Our principle requires the two to keep close together. If the adjective is to be modified, we may reverse the order and write “Men proud of their country.” If, however, the sentence is simple enough, the adjective may move to the other end of the statement and become a predicate, as when we say, “Men in that country are proud of their civilization.” These rules show the mechanics of the adjective. We expect it to precede the noun or to follow it with a dependent clause, or to follow, at an interval, the verb as a predicate. Young writers will be helped in their work by remembering that these are principles of mechanism—that they are building their sentences, and that the parts have their proper place and order, just as wood, brick and stone have in a building.

The foregoing illustrations are briefly stated to prepare the way for a few suggestions respecting some special mechanical contrivances of our language. A general principle in grammar acts as an aggressive and conquering force; it extends its domain, insensibly and gradually, but surely, as far as possible. In an inflectional age a tendency to increase and perfect inflections is discovered; in a flat-construction age the tendency to extend the domain of flat syntax is equally manifest. In our language some constructions are common now, though at one time they were scarcely allowed. This general observation is illustrated in the flat construction of a modifying clause in the nature of a relative pronoun clause. For example, “The man we saw” is a flat construction which has invaded the territory of

the relative pronoun. The sentence is cut down from “the men *whom* we saw.” Very little study has been given to these encroachments and conquests; but they will amply reward the careful student of them. The flat construction in the province of the relative is one of our devices for reducing the use of *who*, *which*, *whose*, *whom* and *that*. These words occur so frequently in the speech and on the printed page that we have quite unconsciously gone about reducing their importance, and the results are so considerable as to merit special attention. I have made some comparative studies, having for their object something like accurate measurement of the change in the use of this class of pronouns, since the year 1611, the date of the English Bible of King James. Two great changes are easily discovered. (1) The number of relative pronouns on a page has been reduced, on an average, about one half. (2) The word *that* has been almost pushed out of the relative office. The devices by which the use of relatives has been rendered unnecessary, are generally forms of the flat construction. The ousting of *that* from relative functions has been promoted by the unconscious effort to dispense with the excessive repetition of the word. When used as a conjunction, a demonstrative and a relative, its repetition becomes tiresome to both writer and reader. A careful study will show that present English employs *that* very seldom as a relative, and much less frequently than the English of the last century employed it as a connective and a demonstrative. In the case of *that* we see the operation of a principle in architectural criticism. If a particular architectural device becomes common, it becomes unfashionable. Its frequency offends the taste and the offense is punished by a change. Forty years ago the ordinary Greek column was used on small private dwellings in many sections of this country. It became so disagreeable to our taste that this column was for some time nearly out of use in public buildings. *That* is, like any piece in architecture, made so common as to become unconsciously offensive. The fact brings out a subtle principle of sentence mechanics—we require variety and dislike a dreary uniformity in this kind of architecture. Good writing in English, *readable* English, will always respond in greater or less measure to the unconscious demand of the English-reading mind. Most persons do not know what is the offending element in a dreary sentence; they only know that “the style” is tiresome, and that they can not interest themselves in the reading. The good writer overcomes the difficulty by avoiding the offending elements.

It will usually be found that the tiresome effect is produced by repetition and uniformity. The pieces used may all be good; but we do not like to see Greek pillars before every house along the road. We tire of Gibbon's periods, of Addison's perfection, of Macaulay's stateliness. We can read a little of each with delight; for daily diet we do not desire any of them. I will now give some of the results of my comparative studies of relative pronouns in the English sentence. I begin with the Bible of 1611. I notice first here that the Psalms differ from other books of the Bible, and I suppose that the difference arises from the superior directness of prayer. The same difference is discoverable between the modern prayer and the sermon. In the Psalms there is one relative in each ninety-five words, on the average; and about four fifths of these are *thats*. In the *number* of relatives, the Psalms approach closely to modern parsimony; but in the use of *that* they exaggerate the practice of the sixteenth century. Many of these *thats* are used in a formula now seldom heard, of which "*he that*" is a typical example. In St. Matthew's record of the Sermon on the Mount, there is one relative to each forty-four words; in St. John's gospel, chapters one to ten inclusive, there are two hundred and eleven relatives, and one hundred and three are *thats*. The proportion of relatives is one to each forty-six words. In the first six chapters of I. Corinthians there are sixty relatives, and of these twenty-seven are *thats*. The proportion is one relative in forty-five words. Combining the results obtained by this counting in the New Testament, the result is one relative for each forty-five words, and more than three sevenths are *thats*. It is probably safe to assume that in the New Testament sentence every forty-fifth word (on the average) is a relative pronoun, and that three times in seven this relative is the word *that*. In the Psalms the relative occurs not quite half so frequently, but four times in five this relative is *that*. We should also remember that at least one form of sentence architecture of which the relative *that* is the conspicuous piece has practically disappeared in modern English. "*He that dwelleth in the secret place of the most high,*" is a common syntax of the Psalms. There are men who say and write "*he who would be rich;*" but it is an archaic formula.

Let us turn to Shakspeare. My counting here has not been as abundant as I could wish, but I think the results are practically correct for the plays. The selections are Richard III., first and second scenes, and "*Love's Labor Lost,*" first and second acts. The proportion is one relative to each ninety-

three words; and of these relatives *that* appears a little more frequently than three times in seven. Shakspeare is, therefore, in this use of *that* almost exactly like the New Testament; while, like the Psalms, he is modern in his parsimonious use of relatives. Readers with more leisure than myself may find interesting employment in examining Addison and Samuel Johnson. In an idle hour I fell upon a copy of Burton's "Anatomy of Melancholy," and found more relatives and a larger proportion of *thats* than in the New Testament. In Samuel Johnson there are probably fewer relatives; his stately Latinity avoided these mean little hinges of clauses. Since writing the last sentences I have examined the first act of Shakspeare's "Hamlet," and I find a smaller proportion of relatives than I have found in any work except modern poetry. I find but forty-eight relatives in the whole act, and just half are *thats*. Another thing to note is that this act contains a large proportion of flat constructions. A further examination shows that the Plays differ much in the management of connective apparatus for clauses. The elevated tone and strong emotion of "Hamlet" account for infrequent use of the lifeless relative forms.

Before taking up any recent author, let me state as a general rule of proportion that present English uses relatives less frequently than the Psalms and Shakspeare, and not quite half as frequently as the New Testament. There is, however, one difference to be noted: English writers have carried this reform somewhat farther than Americans have carried it. It is still further to be noted that preachers and theological writers usually have a good deal of biblical syntax, and therefore employ relatives more freely than other writers. It is a convenient place to mention the fact that in modern English of the best type, ellipsis is more common than in older writers or inferior modern writers. The old writers and their readers had more time than we have, and the "economy of attention" was not in Shakspeare's day a recognized rule of rhetoric. The inferior modern writer is afraid to trust an ellipsis to the tender mercies of the critic, and spoils his sentences by trying to say everything. Ellipsis is one of the chief places for art and genius in writing. As a rule, American writers are in greater awe of the grammarians than Englishmen are. We shall find, then, more relatives in American than in English writers; we shall find more in sermons than in other writings. Young ministers are often advised to cultivate a biblical style. I must confess my inability to sympathize with efforts to employ

religion upon the unavailing task of continuing the use of dead words and forms. If we are to write and speak in dead tongues as a religious duty, let us go back to Greek, at least, if not to Hebrew. The truth is that we ought to put the Bible into modern English, and so end the unprofitable business of disagreeing about the claims of a biblical style upon the pulpit. At present the contention is that in order to imitate the Bible of 1611, preachers should use obsolete English.

Turning now to the usage of modern writing in the employment of relative pronouns, let us begin with a modern Englishman. I select Mr. Bagehot's books, because in him we may hope to find the high-water mark of this reform. Mr. Bagehot was an editor and a banker; he represents the directness, force, and brevity of editorial and business writing. His "Lombard Street" is a book on the financial arrangements of the business public of England. It is therefore practical; but it is also essentially scientific. In this book Mr. Bagehot employs, on the average, one relative pronoun in one hundred and twenty-seven words. This is a little more than one third as many as the New Testament employs. I call this high-water mark; two thirds of the relatives have disappeared. I am not sure of it, but I think Mr. Bagehot did not use *that* as a relative pronoun. I find that about *one fortieth* of the relatives in the American edition are *thats*; but it is probable that some of them were put in by the American printers—unconsciously, of course—and it is possible that all have a typographical parentage. In "Money and the Mechanism of Exchange," by Professor Jevons, I can not find a relative *that*; there may be a few; but in this case, too, the edition is American. Accepting, however, the count, let the readers measure the change from the Psalms of 1611, in which eighty per cent. of the relatives are *thats*, to Walter Bagehot, in whose "Lombard Street" only about two per cent. of the relatives are *thats*. The relatives occur a little more frequently in the book of Professor Jevons, just referred to. By my count there is one relative in one hundred and thirteen words. I have more carefully counted the relatives in the essays of Mr. James Anthony Froude, and find one relative in each one hundred and twenty words. I have only American editions of these essays, and in these editions *that* is employed as a relative in a very few cases. This use of *that* is so infrequent and so opposite to Mr. Froude's ordinary practice, that we may safely set it down as some one's blunder—possibly Mr. Froude's, more probably the

American printer's. If we accept these *thats* as Mr. Froude's, the per cent. of them is so small as to deprive them of importance. I thought I had caught Mr. Froude's secret when I found that in his essay on Norway he apparently wrote "trout that" and "fish that." Mr. Froude is a mighty fisherman, and it was possible that he might glorify the fish by a peculiar form of pronoun. But I turned to the essay on "A Day's Fishing at Cheney's," and found "fish which" and "trout which." This failure to find even a fanciful explanation leaves nothing to be said except that "some one has blundered" into the relative *thats* of Mr. Froude. These three English writers—Bagehot, Jevons, and Froude—probably represent very fairly the untheological writers of our generation.

For a test specimen of theological writers, I turn to a volume of sermons by the Rev. James Martineau. I have counted the relatives in three sermons, "The Bread of Life," "The Unknown Paths," and "The Finite and the Infinite in Human Nature." The relatives occur more frequently than in the non-theological writers. My count shows an average of one relative in each ninety-three words. The use of *that* is abundant. Out of one hundred and seven relative words, thirty-one are *thats*. These sermons were probably composed forty years ago, and represent an archaic type of sermonic style, a style largely affected by that of the Bible of 1611. I have noted without counting, that the sermons of Mr. Spurgeon contain a higher proportion of relatives, and that this great preacher employs *that* with as much frequency as Mr. Martineau. Turning to American preachers, I have taken up a recent sermon of Dr. John Hall, of New York, and I find one relative in each sixty-five words, and of these relatives more than one third are *thats*. Dr. Hall used in 1884 more *thats* than James Martineau used forty years earlier. But Dr. Hall is a preacher of a very biblical type, and his choice of relatives is often dictated by partial quotation of texts. Passing over to non-theological writers of our time and country, let us take the general result of countings in essays and books. The average number of relatives is about ten per cent. greater than in contemporary English writers, and *that* is relatively employed about one fifth of the time. The importance of the reduction of the use of relatives can not be properly appreciated without remembering two or three conditions of their use. One fact is that there is seldom any discernible rule which is followed in the choice of *that* in place of *which*.

The example given from Mr. Froude's practice—whether it is his or his printer's—illustrates the absence of any guidance by a rule.

That has no longer any standing place in the relative ranks; it merely relieves *which* of a part of its work; and in English writers even this supernumerary function has practically ceased to be filled by it. A second condition of the use of relatives is much more important. It has always been possible to build the best of English sentences without relatives. A peculiarly animated sentence of any age will usually contain no relatives. In Dr. John Hall's sermon, the longest sentences and the animated passages contain no relatives. When he drops into a relative, we see that the exaltation of feeling is passing off, and the sermon is sinking to a lower level of interest. It is apparently a law, then, that relatives are more rarely found in animated, elevated and perfectly clear English than in weaker and less emotional writing. A third fact is that I find the relatives of a sermon, book, or essay, occurring in groups. Often there are whole pages with none; then come three, four, five or more in about as many lines. This grouping is almost as true of the Bible as of modern English. In Dr. Hall's sermon, fifty-six occur in the first and least animated half of it, and only thirty-eight in the second half of it, and one fifth of all the relatives of the discourse occur in groups; take twenty-four printed lines out of the sermon, and there will be left only about as many as Professor Jevons employs. The effect of these groups deserves, I have thought, careful study; but the results require more space than is now at command. A single example from Mr. Froude will suffice to indicate the general conclusion. Within thirty-six lines—taken in groups of from two to fourteen lines—Mr. Froude uses twenty-five relatives—or one relative in each fourteen words, while his general average is—as above stated—one in each one hundred and twenty words. How shall we describe such a use of relatives? Plainly they are not necessary. The only explanation I can think of is that it is a careless habit. The related passages are the poorest and weakest, in all the modern English I have examined. The groups of relatives are to me very significant; they show that relative pronouns are unnecessary; the ineffectiveness of the English where they occur shows that the relative is obsolescent.

To compress this study into a small space I have omitted a number of important facts. I pass to the conclusions (*a*) I have reached. (1) The relative pronoun (*b*) being essentially an inflectional device, is opposed by the

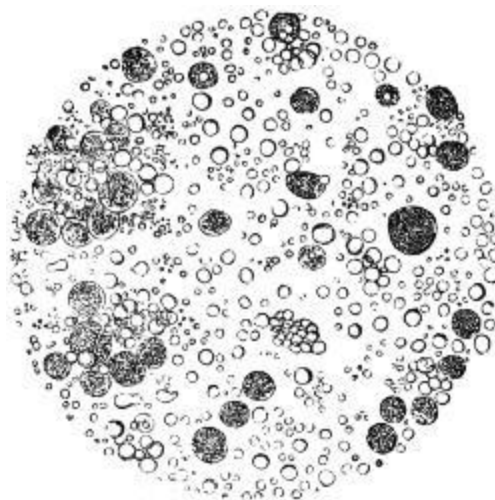
tendencies (c) prevailing in English syntax—tendencies to flat construction. If the reader will look at (a), (b) and (c) in the preceding sentences, he will see specimens of the flat construction. (2) *That* is dead as a relative pronoun. Its use is a mere carelessness. (3) In some of the so-called idioms for the relative *that*, the word is not a relative at all, and the “idiom” itself is a case of flat construction. In “All *that* we know,” the relative usually following the demonstrative *that* has been omitted. (4) The so-called compound relative *what* is not a relative at all; in *our* modern use it is another flat construction to reduce the employment of *which* and its antecedent demonstrative. (5) I infer that the flat constructions ought to be classified and studied in schools. The effect of such teaching and study will be seen in a more vigorous English, and it will not be long before we shall begin to say, “THE RELATIVE PRONOUN MUST GO.”

HOME STUDIES IN CHEMISTRY AND PHYSICS.

BY PROF. J. T. EDWARDS, D.D.
Director of the Chautauqua School of Experimental Science.

CHEMISTRY OF ORGANISMS.

An organism is a structure endowed with life, and acting by means of organs. Organic beings are of two kinds, vegetable and animal. Ordinarily there is little difficulty in discriminating between them, but there is a border line along which the two great kingdoms meet, which is as shadowy and uncertain as that uniting, in distant view, the ocean and the sky. It is usual to say that animals move their parts, and that plants do not. The former have locomotion, the latter are stationary. Animals have nerves and receive their food in cavities; plants do not. But the most important distinction of all is that the vegetable world draws its support from the mineral world, while the animal lives upon the vegetable.



SHOWING THE FAT GLOBULES IN
MILK.

Both animals and plants begin their existence with a single cell. Growth consists in the enlargement and multiplication of cells. Here the physiologist terminates his investigations, and the chemist begins.

His first step, however, results in the destruction of the organism. Of him it is emphatically true, "He murders to dissect." The moment that chemistry seeks to determine the elementary character of an organic substance, that substance ceases to have an organic form. In a sense, therefore, there is no such thing as *organic* chemistry. It is a convenient term, however, for the study of the chemistry of substances formed by life.

Until recently it has been supposed that the chemist could destroy organic substances, but that he could not create them. This idea is no longer held. A great number of the compounds formed by plants and animals have been produced in the chemist's laboratory, without the aid of vital force.

While it is undoubtedly true that many of the compounds found in plants and animals are not *necessarily* related to organisms, there are usually some plain facts which differentiate organic compounds from inorganic. Among these may be named the following: Organic substances are usually composed of but few elements; oxygen, nitrogen, hydrogen and carbon constituting almost all their material. Ten other elements are very sparingly distributed in them.

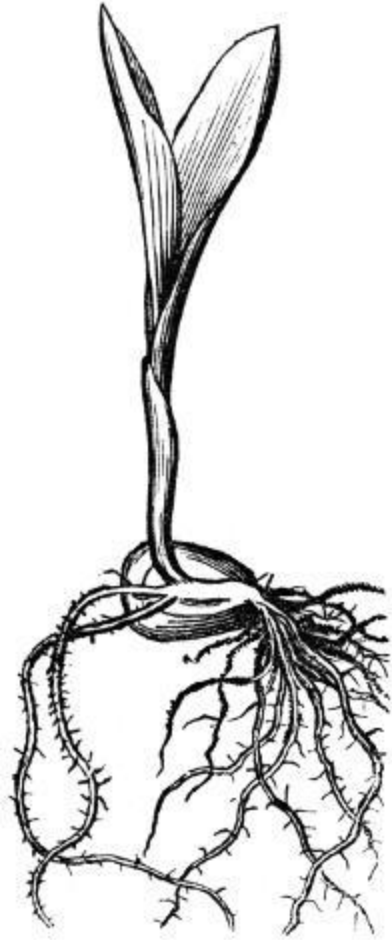
Sixty-six elements enter into the formation of inorganic matter.

The *atomic* structure of the former is very complex; that of the latter is simple. For example: A molecule of the white of egg contains 222 atoms, while a molecule of salt has but two. Again, the compounds of organic existences are innumerable. Inorganic compounds are *comparatively* few. The former are unstable, on account of the presence of nitrogen, while the latter are fixed and quite permanent. The former are also distinguished for the many examples of isomerism they furnish. Isomeric compounds are those formed of the same elements in the same proportions. Thus, camphene, the oils of bergamot, juniper, birch, black pepper, lemon, cloves, turpentine, ginger, cubebs, orange, and many others are isomeric, each one being composed of ten atoms of carbon and sixteen of hydrogen. The difference in these volatile oils is supposed to be due to a variation of the *arrangement* of the atoms composing them.

Let us now briefly consider the

FOOD OF PLANTS.

This is obtained from the air and earth. The former supplies carbonic acid, and water in the form of vapor, through the stomata of the leaves; these are little mouths or breathing pores, chiefly situated on the under side of the leaf. They vary in number from one thousand to one hundred and seventy thousand to the square inch. An apple-tree leaf of average size has one hundred thousand pores. The old elm at Cambridge, under which Washington stood while reviewing the Continental army, has been estimated to produce a crop of seven million leaves, thus exposing a surface of five acres, and therefore furnishing billions of stomata. If the amount of carbonic acid gas in the air were much increased, all higher forms of animal life would perish. If it were materially lessened, vegetation would soon wither and die, involving the death of all animals, from lack of food. Plants derive the element carbon from this gas.



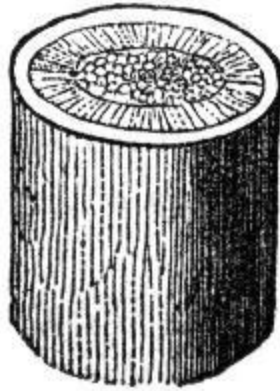
A PLANT STARTING IN LIFE.

According to Chevandier, an acre of beech forest annually absorbs three and one half tons of carbonic acid gas, and from this eliminates about one ton of carbon.

Most of the oxygen and hydrogen of plants is probably obtained from the water absorbed by leaves and roots. Recent experiments indicate that plants may sometimes absorb oxygen directly from the air. This is especially true in the case of buds, as may be shown by the following experiment:

Cut twigs of willow, oak or apple just before the buds are to unfold, and place the ends in a little holder containing a small amount of water, and set this in a saucer; partially fill the saucer with quicksilver; over the twigs invert a glass fruit jar filled with oxygen, so that its mouth will be sealed by the quicksilver. The buds will unfold, and some of the oxygen disappear,

but if the jar be filled with hydrogen or nitrogen the buds will decay. De Saussure,[\[1\]](#) by a somewhat similar experiment, proved that oxygen is absorbed by the roots of plants.



SECTION OF AN
EXOGENOUS STEM.

Both gases and moisture are taken up and distributed through the cells by osmose.[\[2\]](#) This may easily be shown; cut off the end of a carrot and scoop out the central portion of the remainder, and place in the cavity dry sugar; this will soon be converted into a syrup, and the sides of the carrot will have perceptibly shrunk, from the passage of moisture out of the cells to the sugar.

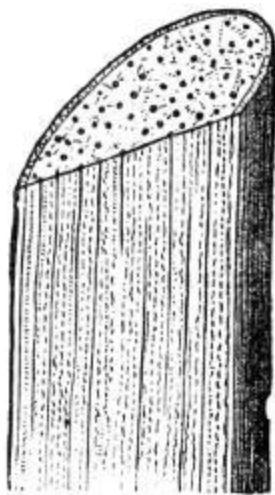
The mineral constituents of plants are all taken up by the roots in the form of solution, water being the great carrier by which plants are supplied.

The following substances are invariably present in all agricultural plants, and in many others, viz.: Potash, soda, lime, magnesia, oxide of iron, chlorine, sulphuric acid, phosphoric acid, silicic acid, and carbonic acid. The chemical composition of different specimens of the same plant is found to be quite uniform.

VEGETABLE NUTRITION.

Young plants first feed upon the store of nourishment placed in the seed, either in cotyledons,[\[3\]](#) or around them. Soon the little roots acquire the power to take their nourishment from the earth in which they are imbedded. They absorb moisture and the materials in solution, which rise through the latest formed wood as ascending sap, and in the cells of the growing parts,

especially the leaves, undergo the transformations which convert inorganic into organic substances. Hales^[4] calculated that the force which impels the sap in a grapevine in summer time is five times as great as that which drives the blood through the arteries of a horse.



SECTION OF AN
ENDOGENOUS STEM.

Much of the water is evaporated. A large sunflower was found to exhale twenty or thirty ounces during the day, but very little at night. After the sap has been elaborated in the cells, under the influence of air and light, it descends just under the bark, in the cambium layer, and furnishes the material for the growth of cells and young buds, and nourishes all growing parts of the plant. This process takes place essentially in the earlier part of the season. In late summer and autumn the circulation in the leaves is impeded by the deposition of mineral matter, so that the plant or tree becomes gorged with the fluids which are ready to flow again at the coming of spring. It is this supply which is drawn upon in the “sugar bush.” A bucketful is often obtained from a single maple tree in twenty-four hours.

The cambium layer, or mucilaginous material between the bark and wood, hardens into cellular tissue and forms an annular growth. This is the case in all exogenous plants. If a section be made of one of them its age may be easily determined by counting the rings. The other great class of plants called endogenous, has the growing masses distributed through the stem. The common cornstalk is an illustration. Few things are more

surprising than the way in which different plants manufacture from the same elements their

VARIOUS PRODUCTS.

This is noticeable in grafting. I have seen a thorn bush having one limb loaded with Bartlett pears. Now the material which ascended the stem was distributed to all the branches, but the cells in some of them manufactured it into thorn apples, while in this branch it was transformed into delicious fruit.



SHOWING A CYCAS, A YUCCA, TWO COCOANUT
PALM TREES, AN INDIAN CORN STEM, AND A
BANANA.

No doubt plants have the powers to select various materials. Upon the same acre of land a hundred different plants may feed and manufacture as many varieties of products, sweet, bitter, sour, poisonous, nutritious, fragrant, offensive, green, yellow, red, and so on through the entire list. As has already been suggested, many vegetable products which are quite diverse in character, are either identical or quite similar in chemical composition. Starch, whether obtained from the potato, the root of the carrot, the kernel of corn, the leaves of the cabbage, or the cotyledons of the bean, is composed of six atoms of carbon, ten atoms of hydrogen, and five atoms of oxygen. Sago, tapioca, bread fruit, arrowroot, and scores of other plant products have the same proportions. Woody fiber whether from the root, stem or branch, woven into cloth, built into houses, twisted into rope, made into paper, used as fuel, or manufactured into furniture, is $C_6H_{10}O_5$.

Slight variations in composition often produce marked differences. The introduction of the least ferment into sugar ($C_6H_{12}O_6$) would break it up into two deadly poisons, alcohol ($2C_2H_6O$) and carbonic acid gas ($2CO_2$ —two molecules of each). A slight addition of oxygen spoils all the sweetness of the preserves.

The rhubarb manufactures oxalic acid, the grape tartaric acid, the apple malic acid, the lemon citric acid, the oak tannic acid, from carbon, hydrogen and oxygen, by simply varying the number of their atoms.

If we add one atom of oxygen ($C_{10}H_{16}O$) to the constituents of the volatile oils previously mentioned, we form a new group comprising camphor, wintergreen, spearmint, cinnamon, bitter almonds, and many others.

Notwithstanding the great uniformity in the composition of various vegetable products, it is now well understood that one crop may restore to land what another has removed, hence the modern agricultural doctrine of

ROTATION OF CROPS.

In southeastern Virginia you find many pine forests in which may be traced the ridges of the corn rows. These fields were planted with corn continuously, until the soil became so impoverished that it would not yield a crop. They were then abandoned and allowed to grow up to pines.

A better system would have secured perpetual fertility. The soil of England produces far more than formerly, even after the cultivation of a thousand years. China furnishes a still more remarkable example of productiveness.

The amount of the earth's crust which is concerned in the support of life is exceeding small. The natural tendency is constantly to diminish this.

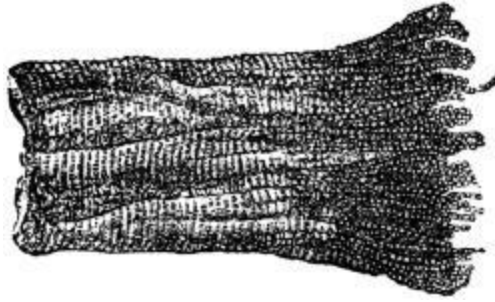
Rains and rivers bear away the best of the soil and deposit it in the lakes and seas. Some inhabitant of our earth in the far future, may secure the benefit of these stores, when the beds of the present seas and oceans shall have risen above the waters and become the continent.

Too much pressed by the demands of the present to even think of this, the wise farmer endeavors to return to his soil what it has lost.

Growing crops are plowed under, fertilizers from the thronging cities are spread upon his fields, the seaweed cast up by the waves yields its potash, phosphorus, salt and iodine. The islands of the Pacific contribute their vast stores of ammonia accumulated for ages in guano beds; marl deposited in the estuaries of ancient geologic seas feeds the cereals; and the limestone deposits are made to give verdure to the grasses of a thousand meadows. In the meantime, nature has her own processes of restoration. The crumbling of the rocks by frost, their abrasion by water, the accumulation of humus by decay, and various chemical influences conspire to convert the unproductive rocks into fertile soil. It would seem that this intelligent forethought, united with the beneficent processes of nature may secure perpetual productiveness, to the end that the earth may continue to yield its increase for the sustenance of the animal, for, as the Scriptures say,

ALL FLESH IS GRASS.

Directly or indirectly all animals live on plants. We have roast lamb for dinner to-day, but yesterday the lamb was browsing herbage. It is an interesting fact that the nutritive qualities of bread are almost the same as those of beef—each, in itself, is very nearly a perfect food.



LEAN MEAT.

Great principles of economy regulate the use of these two articles, in accordance with the scarcity and price of either. Man may live without bread if he have meat, and vice versa, but his system demands one of them, or its equivalent.

As in the cell of the plant, mineral substances become organized, so, under the influence of animal vitality does vegetable material become transformed into the constituents of a new organism. The great argument against the doctrine that alcohol is a food, lies in the fact that it does not undergo this transformation. It leaves the body as it enters it. But beef-steak ceases to be steak, and bread is no longer the same; they have become bone, tissue, nerve, and all that makes a human body

Most are familiar with the marvelous processes of mastication, digestion, absorption and aeration, by which food is converted into blood freighted with all that is essential to the nutrition of the human system.



A COLUMN, ARCHES, DOMES, SPIRES AND MINARETS.

Foods serve three great purposes—growth, restoration of waste, and supply for heat. Whether vegetable or animal, they are of two classes—nitrogenous and carbonaceous. The former consists of all seeds and vegetable tissues, and flesh in animal foods. The latter comprises the starch and sugar of vegetables, and fat in animals. Nature seems to suggest the propriety of using both as food for man. His teeth are adapted to the mastication of both, and the varied demands of different seasons and climates furnish a not less conclusive argument in its favor.

It is not our design to discuss here the dietetics or even the chemistry of food. There is, however, one branch of the subject that calls for a passing remark—the value of foods for special purposes. As the agriculturist is now carefully considering the adaptation of soils to the various kinds of vegetation, and is also inquiring into the character of those fertilizers that will continue and increase the growth-producing qualities of his land, so the physiologist is seeking to discover the special value of different aliments for all conditions of health and disease. The problem is necessarily somewhat difficult, but the end is so desirable—nothing less than human safety, comfort and development—that it is one of the most worthy of all the questions of science. Wholesome food, cheap food, and appropriate food for all classes and conditions is its aim. What does the weary brain require? What will give strength to muscle? How may the impoverished blood be

enriched? How can vigorous, symmetrical growth be secured to childhood and youth? These are vital questions. Even when applied to the wants of the lower animals they are of immense importance. What conditions are most favorable for fattening cattle? What will give greatest strength and best sustain continuous exertion.

Note a simple instance of one result of such inquiry. In ascertaining the food value of cottonseed, the revenue of our cotton crop is said to have been doubled. In medical practice physicians are more and more inclined to depend upon their knowledge of the principles of alimentation and the adjustment of proper nourishment to the sick than upon artificial stimulants or medicines.

THE CIRCLE COMPLETED.

We conclude this article on the chemistry of organisms, with the somewhat humbling reflection that to all living beings there comes a time when vitality yields to the power of those chemical forces, which resolve them again to their original inorganic forms. It can not be that this was the only and ultimate end contemplated by the Creator, in that sublime system of arrangement for life, which began with the morning of creation and ended with man. Nature is more than a cycle of change from dead matter to vegetable form, thence to animal life, and thence back again to mineral substance.

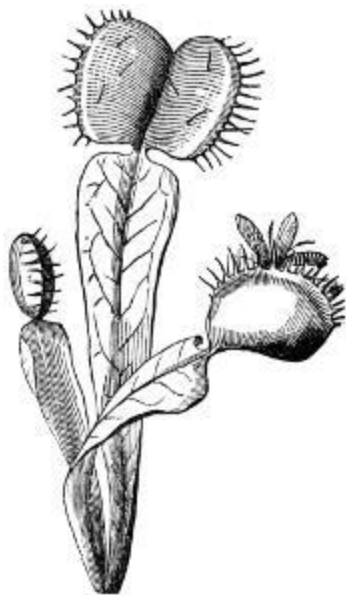
Solomon wrote: "The dust shall return to the earth as it was, and the spirit shall return unto God who gave it;" and another has said: "There remains the paramount duty of rendering worthy of survival that spiritual part of our being which no merely physical power can destroy."

PHYSICS OF ORGANISMS.

A brief discussion of some *physical* characteristics of organisms will conclude our articles on "Home Studies in Chemistry and Physics."

The abundance of metaphorical expressions even in common language, indicates the numerous resemblances between the living and inorganic

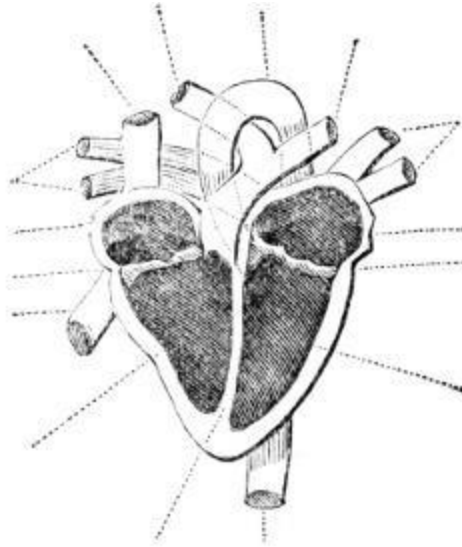
worlds. Description and poetry are full of imagery. A metaphor implies a resemblance between objects, a simile suggests it, and a comparison states it.



THE DIONÆA, OR
VENUS'S FLY-TRAP.

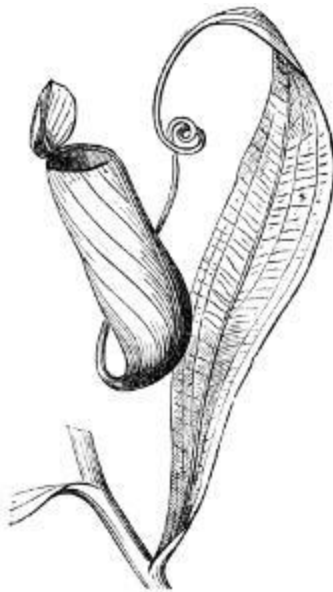
Thus to the human mind, the different departments of nature seem to reflect a light and beauty upon each other, even as the “earth-shine” lights the moon in the absence of the sun. The sky is a dome; the groves are temples; the sea moans and roars; the falling cataracts laugh and shout, and the calm lake is the smile of the Great Spirit. “Language,” says Dean Trench,^[5] “is fossil poetry.” “Architecture is frozen music.”

Many of the forms of art and devices of human invention have been suggested by Nature. The Doric column was borrowed from some stately tree shaft. The ornamented capital of the Corinthian column was decorated by carved copies of the graceful acanthus leaves. Gothic architecture found its models in the tree tops of the arching forests.



A HUMAN HEART, SHOWING
CHAMBERS AND VALVES.

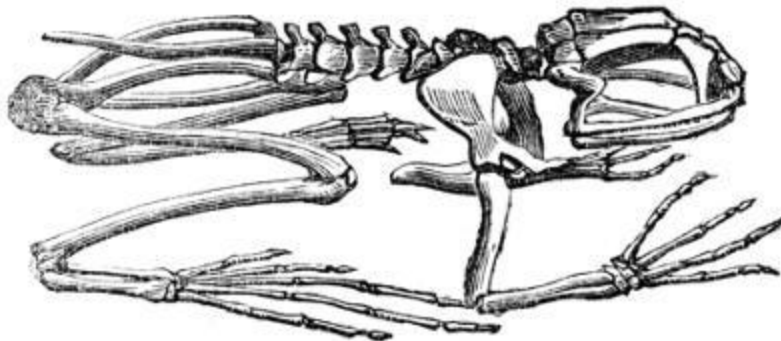
Every experimenter in science is simply one who is inquiring of Nature for her analogies, truths, forms, forces, and machines; and like the wise and good mother that she is, she has granted many a pregnant suggestion to the busy brains of discoverers and inventors.



THE NEPENTHES,^[6] A
PITCHER PLANT OF THE
INDIAN ARCHIPELAGO.

MOVEMENTS OF PLANTS.

Plants in their action illustrate many of the principles of natural philosophy, as if directed by intelligence. Turn their roots upward in the soil, and they will invariably turn down to the moisture. Bend their stems to the earth and they will seek to mount upward. The young sunflower greets the sun at his rising, and turns to behold his setting. Unwind a twining vine, and wind it in an opposite direction, and it will soon assert its right to assume its own method. Some plants shrink from touch; others, like the Venus fly-trap,[\[7\]](#) hold out their open palms to catch flies; many sleep; most seem to select special places, seasons, and conditions. They seem almost, at times, to be possessed of moral qualities. They adapt themselves to situations. If the season is dry, they are sparing of moisture; if the soil is scanty, they penetrate deeper for sustenance; if the winds are fierce, they grow strong by struggle; if gashed or broken, they have “philters for healing;” if pruned and chastened, they yield richer fruitage.



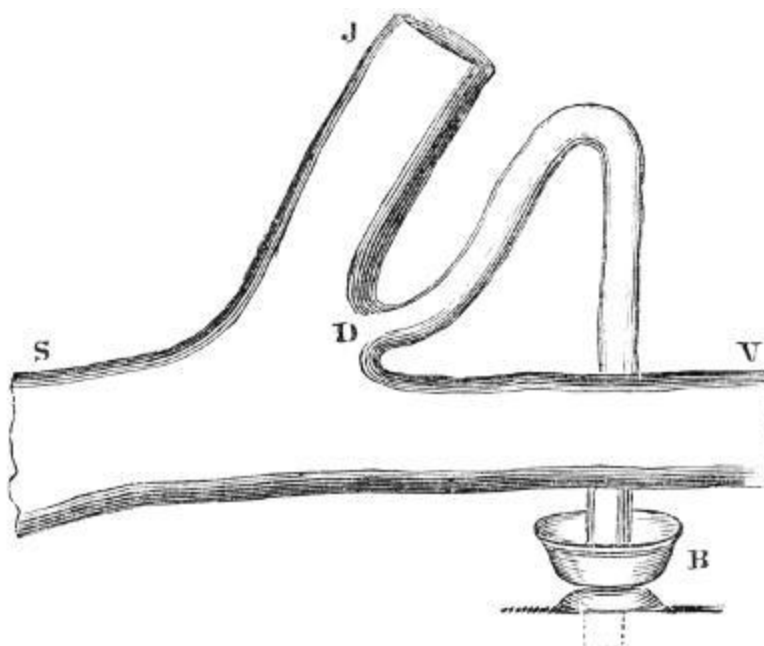
SKELETON OF A FROG.—A GOOD SET OF SPRINGS.

One can not help feeling that certain trees have a personality. They are friendly with their shade. They are proud in their loftiness, confident in their strength, satisfied in their usefulness.

Other plants are almost equally interesting. Flowers have long been chosen to express the language of sentiment. Even the lowest forms of vegetable life, like the algæ, the mosses and lichens, arrange their parts with symmetry and beauty. Even the microscopic diatoms[\[8\]](#) are exquisite in the perfection of their curves and markings.

ANIMAL MECHANISM.

Comparative anatomy long since showed us that there is great harmony in the construction of animals. A few principles seem to govern in all. For example: None violate the law of gravity with regard to the line of direction's falling within the base. They employ the lever, the inclined plane, the pulley, and the mechanical means of applying power, precisely as we do in machinery. The heart is a pump; the stomach is a churn; the backbone has springs; the elbow is a hinge; the muscles are ropes; the nerves are telegraph wires; the ear is a harp, the eye is a telescope. The most perfect mechanism characterizes the construction of all the animal kingdom, but one can do little more than suggest the interest of this most fascinating subject.



SHOWING PRINCIPLE OF VENTURI.

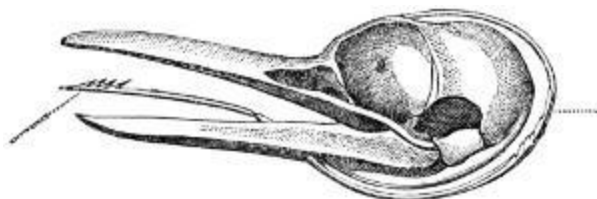
Ex.—S V is the sub-clavian vein; J is the jugular vein; D is the thoracic duct, through which the chyle is poured into the blood.

THE TRADES AMONG ANIMALS.

An ancient saying declares that “Poets are born, not made,” and classic story informs us that Minerva sprang full armed from the head of Jove. Something like this natural perfection appears in the occupations of the lower orders of creation. Man is a creature of education, absolutely unlimited in point of time in the possibilities of his development. Other

animals, within their own limited scope often attain an excellence superior to his. Note the scent of the greyhound, the hearing of the cat, the sight of the eagle. As artisans they have few apprentices, though it must be confessed that some are better workmen than others, and they are not without “bosses.” Observe a few of their trades. The brant-goose is a navigator, which may have already found the pole. The heron is a fisherman, who carries his torch upon his breast. Swallows are excellent masons; so are wasps and the caddis fly.

There is a spider that is a diver; he makes his own bell and fills it with air. The bee is a geometrician that never studied Euclid. The ant is a political economist, who, like Joseph, lays up supplies for a time of want. There is a “tailor bird.” There are hosts of hunters among the carnivora. The nautilus is a “little sailor,” and weavers are innumerable. Beavers unite the trades of lumbermen and civil engineers. There are carpenters and paper makers, indeed, time would fail in the attempt to mention all the occupations pursued in this busy world of animate creation. Yet over all these the Almighty has given man dominion. They are but organisms impelled to their appointed tasks by unreasoning instinct, but, as Sir William Hamilton has said: “*Man* is not an organism, but an *intelligence* served by organs.”



A BIRD'S HEAD.

Ex.—The mandibles form a pair of scissors. The tongue is a spear.

NOTE.—Through the courtesy of Messrs. Ivison, Blakeman, Taylor & Co., of New York, the cuts in this article are taken from two of their excellent publications, Gray's “Lessons and Manual of Botany,” and Hitchcock's “Anatomy and Physiology.”

SUNDAY READINGS.

SELECTED BY CHANCELLOR J. H. VINCENT, D.D.

[*June 7.*]

Courtesy is, strictly speaking, a Christian grace.... It is the offspring of charity; and since it derives its being from divine grace; since it is made the subject of divine command; since it is especially calculated to smooth those little asperities which sometimes hinder even “the living stones of the temple” from being so perfectly joined and so fitly framed together as they should be; since it powerfully tends, likewise, to remove the prejudices and to allay the enmity so generally entertained by the world toward the church; above all, since, in combination with other causes it may contribute to win souls to God, we surely ought not to deem it unsuitable, but to make it ... the subject of our particular and attentive consideration.... While some professed disciples of Christ seem to have substituted in the place of genuine courtesy a conformity to the manners and habits of ungodly men, which very ill consists with that simplicity of character which should distinguish the remnant of true Israelites, there are others who, through an honest disgust toward the impertinent fopperies of the world, and an ill-directed fear of becoming infected with the same spirit of guile and hypocrisies, have even run so far into the opposite extreme of churlishness as to be culpably negligent of the mere forms of civilized society.

The courtesy of the world is an imposing form.... But the courtesy of a Christian is not a mere form. It is not the phantasm of a feeling which has no real existence. It is the outward expression of an inward disposition, the conduct which a benevolent mind will on all occasions instinctively prescribe. It is the natural and unconstrained operation of unfeigned love. Let us but love our neighbor as ourselves, and it will be morally impossible to violate the laws of courtesy; for love worketh no ill to his neighbor. It will teach us cautiously to avoid whatever might unnecessarily wound his

feelings; it will dispose us assiduously to study his inclination, ease, and convenience; it will make us anxious to interpret his very looks, that we may even anticipate his requests; it will enable us cheerfully to make a sacrifice of our own gratifications with a view to his. All this is perfectly easy; it is even delightful where love exists without dissimulation; but let this heavenly principle be wanting, take away from the *form* of courtesy the *power*, and it becomes an arduous and irksome task, a yoke grievous to be borne.—*Summerfield*.[\[1\]](#)

[*June 14.*]

I would not slight this wondrous world. I love its day and night. Its flowers and its fruits are dear to me. I would not willfully lose sight of a departing cloud. Every year opens new beauty in a star; or in a purple gentian fringed with loveliness. The laws, too, of matter seem more wonderful the more I study them in the whirling eddies of the dust, in the curious shells of former life buried by thousands in a grain of chalk, or in the shining diagrams of light above my head. Even the ugly becomes beautiful when truly seen. I see the jewel in the bunchy toad. The more I live, the more I love this lovely world; feel more its Author in each little thing; in all that is great. But yet I feel my immortality the more. In childhood the consciousness of immortal life buds forth feeble, though full of promise. In the man it unfolds its fragrant petals, his most celestial flower, to mature its seed throughout eternity. The prospect of that everlasting life, the perfect justice yet to come, the infinite progress before us, cheer and comfort the heart. Sad and dissatisfied, full of self-reproach, we shall not be so forever. The light of heaven breaks upon the night of trial, sorrow, sin; the somber clouds which overhung the east, grown purple now, tell us the dawn of heaven is coming in. Our faces, gleamed on by that, smile in the new-born glow; we are beguiled of our sadness before we are aware. The certainty of this provokes us to patience, it forbids us to be thoughtfully sorrowful. It calls us to be up and doing....

There is small merit in being willing to die; it seems almost sinful in a good man to wish it when the world needs him here so much. It is weak and unmanly to be always looking and sighing voluptuously for that. But it is of

great value here and now to anticipate time and live to-day the eternal life. That we may all do. The joys of heaven will begin as soon as we attain the character of heaven and do its duties. That may begin to-day. It is everlasting life to know God, to have His Spirit dwelling in you, yourself at one with Him. Try that and prove its worth. Justice, usefulness, wisdom, religion, love, are the best things we hope for in heaven. Try them on—they will fit you here not less becomingly. They are the best things of earth. Think no outlay of goodness and piety too great. You will find your reward begin here. As much goodness and piety, so much heaven. Men will not pay you—God will; pay you now, hereafter, and forever.—*Theodore Parker*.

[June 21.]

Let us do all the business we can.... If we can't be a lighthouse, let us be a tallow candle. Some one said, "I can't be anything more than a farthing rushlight." Well, if you can't be more, be that; that is well enough. Be all you can. What makes the Dead Sea dead? Because it is all the time receiving, never giving out anything. You go every Sunday and hear good sermons, and think that is enough. You are all the time receiving these grand truths, but never give them out. When you hear it, go and scatter the sacred truth abroad. Instead of having one minister to preach to a thousand people, this thousand ought to take a sermon and spread it till it reaches those that never go to church or chapel. Instead of having a few, we ought to have thousands using the precious talents that God has given them....

If God has not given us but half a talent, let us make good use of that. When God told the people to take their seats by fifties, he told Philip to get food for them. "What," says Philip, "feed them with this little loaf? Why, there is not more than enough for the first man." "Yes, go and feed them with that." Philip thought that was a very small amount for such a multitude of hungry men. He broke off a piece for the first man, and didn't miss it; a piece for the second man, and didn't miss it; a piece for the third man, and didn't miss it. He was making good use of the loaf, and God kept increasing it. That is what the Lord wants to do with us. He will give us just as many talents as we can take care of.

There are many of us that are willing to do great things for the Lord; but few of us willing to do little things. The mighty sermon on regeneration was preached to one man. There are many who are willing to preach to thousands, but are not willing to take their seat beside one soul, and lead that soul to the blessed Jesus. We must get down to personal effort—this bringing one by one to the Son of God. We can find no better example of this than in the life of Christ himself. Look at that wonderful sermon that he preached to that lone woman at the well of Samaria. He was tired and weary, but he had time and the heart to preach to her. This is but one of many instances in the life of the Master from which we may learn a precious lesson. If the Son of God had time to preach to one soul, can not every one of us go and do the same?...

“I commend you”—and in this connection, I want to tell you how the God of all grace has kept us. For nearly twenty-one years he has watched over me. He has watched over me and stood by me in the hour of temptation and trial; he has brought light to me out of darkness; and he will do the same with you. In leaving you, young converts, I would like to leave with you two Ws—the one is Work, and the other is the Word; or, rather, the first is the Word, and the other is Work. Go out and work for him, and you will become strong Christians. There are two lives that you want to lead. The one is your inner life, that the world knows nothing of, that the wife of your bosom knows nothing of. That life is between yourself and God; and if you don’t lead this aright, the outer life will not be long right. Let me say to you, young converts, read your Bibles and you will be strong. If you don’t, you will fall; and the men who are now scoffing at this movement will say: “I told you you would fall back again; the meetings have been only an emotional excitement; only a sensation.” I pray that God Almighty may keep you. Just have those two Ws before you—the Word and Work—and make that your banner.—*D. L. Moody.*

[June 28.]

What language, “Father, forgive them!” and, in the words, what an act, greater than the most splendid miracles with which he marked his radiant path through the world....

“Forgive them!” Is it possible? With these words ... he covers the guilty heads of his murderers with the shield of his love, in order to secure them from the storm of the well deserved wrath of Almighty God. With these words, which must have produced adoring astonishment, even in the angels themselves, he takes these miscreants in the arms of his compassion, and bears them up to the steps of his Father’s throne, in order to commend them to his mercy. For know, my readers, that the words “Forgive them,” mean, in Jesus’s mouth, not merely, “Do not impute to them the murderous crime they have committed upon me.” No, when he utters “Forgive,” it comprehends something much more, and embraces the whole register of sins. In his mouth it means, “Plunge their whole sinful life into the depths of the sea, and remember no more their transgressions, but consider these sinners henceforth as dear in thy sight, and act toward them as such.”

There are individuals on earth for whom no one feels inclined to pray, because they are too depraved. There are those who even dare not pray for themselves, because their consciences testify that such worthless creatures as they are can not reckon upon being heard. What a prospect is here opened to people of this description! Ah, if no heart beats for them on earth, the heart of the King of kings may still feel for them. If among their friends not one is to be found to intercede for them, yet, possibly, the Lord of Glory is not ashamed of bearing their names before his Father’s throne. O, what hope beams on Calvary for a sinful world! And if the great Intercessor appears there for a transgressor, how does his intercession succeed? Though a whole world should protest against it, his prayer saves whom he will. His voice penetrates the heart of the eternal Father with irresistible power. His entreaties are commands. Mountains of sin vanish before his intercession. How highly characteristic and deeply significant is the fact that the Lord, with this prayer, commences the seven expressions he uttered on the cross. The words, “Forgive them!” show us not merely the heaven of loving kindness which he carries in his bosom, but it also darts like lightning through the gloom of the entire night of suffering, and deciphers the mysterious position which the Holy One of Israel here occupies as Surety, Mediator, and High Priest....

... And yet the prayer for forgiveness raises its wing from the mount of suffering and passes apparently through all those eternal and unimpeachable statutes and limitations. It puts aside even Mount Sinai and Ebal, and heeds

not the cherub of the law, who keeps the gate of paradise, and is enjoined to admit only the righteous. Careless of his flaming sword, it soars with seemingly unheard-of boldness above the brazen walls of the manifold menaces of the divine maledictions which inexorably close against sinners the entrance to the mansions above, and in a most striking contrariety with the indelible inscription over the eternal sanctuary, “Him that sinneth against me will I blot out of my book,” requests forgiveness and even admittance into the habitations of the blessed children of God, for rebels, blasphemers, and murderers.—*F. W. Krumacher*.[\[2\]](#)

THE HEART BUSY WITH THINGS ABOUT US.

BY JOSEPHINE POLLARD.

The eye may be trained so that it can detect the least flaw in a diamond; the ear be so delicately attuned that the slightest variation in the harmony will be perceptible, although there is no apparent attempt at listening, and discord will have the same effect on the sensitive nerves, as a blow on a fine strung instrument. The engineer can not see all the working parts of his engine, so he is obliged to cultivate his ear until each throb, plunge, and revolution is familiar to him, and the whole set to a rhythmic movement, any change in which betokens disaster.

An eye quick to see, and an ear quick to hear may belong to a wide-awake, successful business man; but in order to become a philanthropist, a wise, energetic, live Christian, he must have a heart to feel. A father may see that his children are poorly clad, may hear their cries of distress, but if he has no heart to feel their needs, he will go off and leave them to the care of charity, as so many fathers, and mothers too, have been known to do.

This inclination to avoid cares, to shirk responsibilities, and to live a purely selfish life, is the result of a defect in the cardiac region, which might have been corrected in early youth.

The training of the heart begins at so early an age that it can not be known with certainty just when the child is first acted upon by the influences around it, and it is more easily misdirected than guided aright.

An accomplished lady, of considerable literary fame, spent a great deal of time in preparing a lecture on "Individual Sovereignty," which was to prove that children had rights that parents and guardians ought to respect. The lecture was delivered but once, to a very small number of people who, while full of admiration and respect for the lecturer, were not in favor of putting her theories into practice, believing that a monarchy such as she

proposed would make the Land of Liberty a place that grown people would want to get away from.

The great bond of brotherhood is sympathy. "Pity and need make all flesh kin," and "sympathy is especially a Christian's duty." But there is an active sympathy and there is a passive sympathy; the one sits down and broods over the calamities of life, wrings its hands, sheds tears, and sighs over its own incapacity; while the other is up and doing all it can to relieve the necessities of those perhaps less heavily burdened than itself.

"It is not all of life to *live*," nor all of life to *love* either; for some in their excessive fondness will allow those whom they might control, to walk in evil ways and indulge in unlawful passions without putting forth a restraining hand.

"I want John to have a good time," says the indulgent mother. "I don't want Jennie to tire herself, or to soil her hands doing housework. What else am I good for?" So John grows to be a selfish, disagreeable man, and Jennie an ease-loving, self-satisfied woman, both with hearts incapable of feeling any interest in anything that does not immediately affect their physical comfort and well-being. Mothers, do not spoil your children and destroy the foundations of character. Let them wait upon you and do your errands; teach them to cultivate a self-sacrificing spirit, to feel that it is no hardship to give up their own personal comfort in order to secure the happiness of others. The sacrifices should not be all on one side; and yet we have known mothers to give up their lives rather than disappoint the children, who must have their wishes gratified at any expense.

Exacting children should be made to wait upon themselves, and to practice patience and self-denial; for the tyrannical spirit is fostered by unwise timidity and forbearance, and many a passionate man and woman lives to regret the lack of proper discipline in youth. But it lies within ourselves to correct mistakes that may have been made in our training; and it has been truthfully said, "Every person has two educations, one which he receives from others, and one, more important, which he gives to himself."

It is astonishing how much may be accomplished by one who is energetic and persevering, careful to avail himself of all opportunities, and to use all the spare time at his disposal. Ancient and modern histories and

biographies are full of illustrations showing the benefits conferred upon mankind by certain individuals whose hearts were busy with the things about them. It is interesting to read this record of General Gordon—or “Chinese Gordon,” as he is familiarly called—whose valiant deeds have won him undying fame:

From 1865 to 1875 Gordon lived at Gravesend, employed in the duty of improving the defenses of the Thames. These were his six years of quiet peace and beneficent happiness. “He lived wholly for others,” writes his friend. “His house was school and hospital and almshouse, in turn; was more like the abode of a missionary than of a commanding officer of engineers. The troubles of all interested him alike. The poor, the sick, the unfortunate, were ever welcome, and never did suppliant knock vainly at his door. He always took great delight in children, but especially in boys employed on the river or the sea. Many he rescued from the gutter, cleansed them and clothed them, and kept them for weeks in his house. For their benefit he established reading-classes, over which he himself presided, reading to and teaching the lads with as much ardor as if he were leading them to victory. He called them his ‘kings,’ and for many of them he got berths on board ships. One day a friend asked him why there were so many pins stuck into the map of the world over his mantelpiece; he was told that they marked and followed the course of the boys on their voyages; that they were moved from point to point as his youngsters advanced, and that he prayed for them as they went, night and day. The light in which he was held by those lads was shown by inscriptions in chalk on the fences. A favorite legend was ‘God bless the Kernel!’ So full did his classes at length become that the house would no longer hold them, and they had to be given up. Then it was that he attended and taught at the Ragged Schools, and it was a pleasant thing to watch the attention with which his wild scholars listened to his words.”

The workhouse and the infirmary, writes another, were his constant haunts, and of pensioners he had a countless number. Many of the dying sent for him in preference to the clergyman, and he was ever ready to visit them. His purse was always empty because of his free-handedness, and he even sent some of his medals to the melting-pot in the cause of charity.

When another appointment removed him from Gravesend, there was universal regret. The local newspaper paid him the following graceful and sincere tribute: “By general and continual beneficence to the poor, he has been so unwearied in well-doing that his departure will be felt by numbers to be a personal calamity. His charity was essentially charity, and had its root in deep philanthropic feeling and goodness of heart; shunning the light of publicity, but coming even as the rain in the night-time, that in the morning is noted not, but only the flowers bloom and give a greater fragrance. All will wish him well in his new sphere, and we have less hesitation in penning these lines from the fact that laudatory notice will confer but little pleasure upon him who gave with the heart and cared not for commendation.”

Military glory pales before this display of missionary zeal.

In order to achieve any success the heart must be in the work, and from this center of our being all true Christian culture must begin. Every one can make his own destiny, and

“Taught by time the heart has learned to glow
For others good, and melt at other’s woe.”

It is good to indulge the habit of looking out of ourselves, to study the ways and needs of others, and to keep the heart busy with things about us—the trifles, as they are considered, which are apt to be overlooked or made light of. For there is great danger that the small philanthropies and courtesies of life will be neglected because of the large schemes that are so absorbing. This kind of outlook requires, of course, a certain amount of *insight*, or intuition, without which we can not bring ourselves into sympathetic relations, or prove ourselves the friend of humanity.

Our home, the place where we spend the most of our time, should be the field in which to exercise our best gifts, wherein we both sow and reap, and which is left to us to brighten and beautify, or to darken and disgrace. The heart renewed by grace is zealous of good works, fond of home, and anxious to do its full duty therein. It has a word of cheer for those who are cast down; it comforts the sick; speaks tenderly to those in trouble; has

patience with the erring; seeks out ways of interesting the young; is mindful of the aged, and helpful to everybody.

“A heart at leisure from itself,
To soothe and sympathize,”

need never want for an opportunity; and it is often the small attentions, the unconsidered trifles, that are most highly valued.

The heart that begins its labors in too wide a sphere will find it hard to concentrate its interests, while a more gradual expansion will result in more satisfactory work, and the strengthening of the magnetic current, for on our personal magnetism depends, to a great extent, our influence on those with whom we associate. What responsibilities rest upon fathers and mothers, on sisters and brothers! How much good and how much evil they can do in their special fields of operation! That son needs a little wholesome correction from the father, a few kind words may be all that is requisite, but the father is busy about other affairs, his heart is interested in things outside of his family, and the boy slips downward for want of a restraining hand. That girl would have turned out, oh! so differently, if she had only had the right kind of a mother. Some boys and girls are not fit to be left to themselves. They seem to be born without any inward monitor, or strong moral sense. The germ may be there, but it has never been properly cultivated.

At a school examination the question was put, “What is conscience?” But one pupil could give the definition, “An inward monitor.”

“What is your idea of an inward monitor?” asked the inspector, and away down at the lower end of the room a hand was stretched forth, and a voice proclaimed, “It’s an iron-clad, sir!” So it is. Fortified to resist evil, but to assist good; and a “tender conscience,” to keep up the nautical figure, is the small convoy that supplies the soul with spiritual sustenance.

A conscience alive to duty will serve as an electrical alarm to notify us what is to be done and when and how we are to do it. If you have sick neighbors, and can not conveniently call upon them yourself, send to

inquire how they are and in what way you can be of service. An offering of fruit or flowers will often be most acceptable to invalids or their families, who are cheered and sustained by the thought that other hearts are sympathizing with them. Keep down your own sensitiveness, and learn to make generous allowance for other people. Show that your zeal is not of the offensive sort, that your politeness is deep and genuine, that “out of the abundance of the heart the mouth speaketh,” and you will always be on good terms with your neighbors, and with yourself.

There is a variety of ways of helping people, and no two people may be helped in precisely the same way. The most deserving are the most modest, and it may take considerable study and observation for you to discover that Mrs. Needlewise, whose children always look so neat and well dressed, would be thankful for the cast-off clothes which are taking up the room in your garret, or which you give away recklessly just for the sake of getting rid of them. Mrs. Needlewise would be offended at your offering her the garments, but friendliness, and the exercise of a little tact, will remove the barriers and enable you to relieve anxieties that were a continual burden.

A dear, good woman whose heart was always open to the necessities of those about her, was in her old age given to somewhat erratic impulses. One morning, much to the mortification of her family, she seized the coffee-pot and went into a near neighbor's, a poor but proud little woman, who would go without rather than beg. The old lady, in her sweet way, said to the farmer's wife that it was a pity to waste so much good coffee, and she had brought it over for her to use. The gift was accepted with a smile, and to please the old lady the coffee was used by the good man of the house, who sent word that he had never tasted any quite so delicious, and should be glad to be so favored again. It was the entering wedge of neighborly kindness, and the beginning of better days for the poor family whose fortunes at that period were at a low ebb. The right kind of a lift at the right time will put human nature on its feet, and reestablish the foundations that were in danger of giving way.

O, the magnetizing power of love! beginning first with the love of Christ, and then reaching out toward all our fellow-creatures! How wide spreading, how far-reaching in its influences! Home missions, foreign missions, small charities or large ones, incidental acts of kindness,

thoughtful consideration for the welfare of others, all that a self-sacrificing spirit can do is done cheerfully and without hope of reward.

A saintly young woman whose earthly pilgrimage ended at thirty-four, is thus eulogized by the pastor of the church for which she labored lovingly and assiduously: "She gave out so much to others that she has left herself broken in fragments here and there, and you and I hold this or that fragment so really that we are only hardly persuaded to acknowledge that there is an end of her earthly life for a time. How simply, purely and patiently that life was lived, you know.... Many and many a time have I pointed to that life as an example of what people might do and might be, if they would do as she did—be content 'to live faithfully a hidden life,'" Her mother says: "Her unselfish work and devotion were marvelous. It is often said of her that she crowded more into her short life than is done or experienced in the longest. She shone the brightest in her daily life at home, always serving some one, forgetful of self. Her pure spirit grew so fast that the frail body could not retain it; yet she faded so slowly, so cheerfully and hopefully, that we hardly believed she could not rally until the day before she left us. Our sunshine is gone, but the radiance is still left in the memory of her sweet life."

Blessed memory!

In this bright way she speaks of herself when laid aside from her sphere of usefulness, and obliged to some extent to discontinue the literary work which had proved so acceptable to the public and already given her considerable fame as an author: "I've been occupied with turning a corner, round which the landscape is different, and getting used to the change. I suppose it has been coming on for a good while; at all events, after some ten years of ruddy health and active exercise, I am not to sweep any more for a while, not to walk, not to sing, read aloud, talk much, not to hurry, not to get tired at anything.... Of course it is a little queer and painful sometimes, because, particularly, my father is always an invalid, and increasingly so. One dreads to be another anxiety. But I have no real reason to fret. Something good will come of it, I will expect. I am so thankful that I need not give up writing that I will not mind the rest of my denials;" and then when her own health was declining, and the sunset hour of her life was nearer than she thought, she closes her letter with "Good night, dear. Health

and God's blessing!" and it was a final "good night," for word soon came of her entering into communion with the saints above.

The more heart we put into our work, whether it be domestic drudgery, the care of the sick, or "whatsoever the hand findeth to do," the more perfect and satisfactory it will be; and according to the measure with which we serve others, is meted out to us the happiness we derive from that service. The sending of a letter full of kindly thought and sympathy has often brought a return far beyond the expectation of the sender. A little gift, a token of good will, insignificant in itself, has spoken volumes to the recipient, and brightened a day that was full of clouds. To do no more than our duty does not fill the measure of Christian usefulness. We would grieve if compelled to walk a narrow path, fenced in on either side, and not allowed to look to the right or left, or to pluck the fruits that lined the road; and God and his holy angels must grieve when we neglect to turn out of our path to assist others, and make excuses for the non-performance of heart service.

A young lady, very much interested in mission work, and an active worker in a large school connected with a flourishing church, felt offended at some of the officers of the school, and decided to send in her resignation. Each Sunday she had been accustomed to place before the children some text that they might carry with them through the week, absorbing its teachings and principles so that they would be "wrought out in living characters."

All through the week her mind dwelt upon the injustice that she felt had been done her, and she went to the mission school the following Sunday fully determined to give up the work which had been her delight for so many years. As she entered the room her gaze rested on the text which stood out in bold lettering, as she had printed it the previous Sunday:

"Even Jesus pleased not himself."

The arrow struck home. It would never do to have that text uppermost when she handed in her resignation, so she reversed the roll, and there in as bold type appeared:

“Jesus, the same yesterday, to-day, and forever.”

There was no use trying to avoid the situation, or to escape the responsibilities. The teacher’s work was there. She knew it; she felt it.

“To doubt would be disloyalty;
To falter would be sin;”

so she roused herself to greater endurance, put more heart into work, and had the satisfaction of hearing it said that never since her connection with the school had she given such a splendid lesson. The approval of her own conscience was not the least of her compensations, and there is no further talk of her giving up her position in the mission school.

We may be so situated that we can not do any great work in the world. By temperament, by education, or by reason of ill-health we may be restricted from carrying out our ambitious schemes, but there are none so weak, so ignorant, or so poor that they can not do some good in the world. The ladder that reaches to heaven is not composed of wooden rungs, or of cold, senseless material, but God has made every human being so dependent on his fellow-creatures that each one is lifted up by some one above him, some busy heart that feels another’s need and reaches out; and when there is no looking up nor reaching out there is no growth nor spiritual attainment.

If you want to know people you must get near them; first go down to their level, and then bring them up to yours, not waiting for any great occasion, or a more direct revelation, but taking advantage of small opportunities, and making your influence felt in quiet, unobtrusive ways.

“And when it is all over, and our feet will run no more, and our hands are helpless, and we have scarcely strength to murmur a last prayer, then we shall see that instead of needing a larger field we have left untilled many corners of our single acre, and that none of it is fit for the Master’s eye were it not for the softening shadow of the cross.”

“It was only a cup of water, with a gentle grace bestowed,
But it cheered a lonely traveler upon the dusty road;
For the way was long and dreary, and the resting places few,
And the sun had dried the streamlets and drank up the sparkling dew;
None noticed the cup of water as a beautiful act of love,
Save the angels keeping the record, away in that land above;
But the record shall never perish, and the trifling deed shall live,
For heaven demands but little from those who have least to give!

“It was only a kind word spoken to a weeping little child,
But the thread of its grief was broken, and the little one sweetly smiled;
And nobody stayed to notice so tiny an act of love,
Save the angels keeping the record in the wonderful book above.
And she who had spoken kindly went on her quiet way,
Nor dreamt such a simple action should count at the last great day.
But the pitying words of comfort were heard with a song of joy,
And the listening angels blessed her from their beautiful home on high.

“It isn’t the world-praised wonders that are best in our Father’s sight,
Nor the wreaths of fading laurels that garnish fame’s dizzy height;
But the pitying love and kindness, the work of the warm caress,
The beautiful hope and patience and self-forgetfulness;
The trifle in secret given, the prayer in the quiet night,
And the little unnoticed nothings, are good in our Father’s sight.”

There is always some one to smile at, somebody to give your chair to, somebody to whom a book, a flower, or even an old paper, will be a boon. These small attentions will open the way to confidence, will make it possible that in need these friends will give you opportunities to help them which, unless you had shown thoughtfulness and regard for them, they could never have done. A quiet, sympathetic look or smile many a time unbars a heart that needs help which you can give.

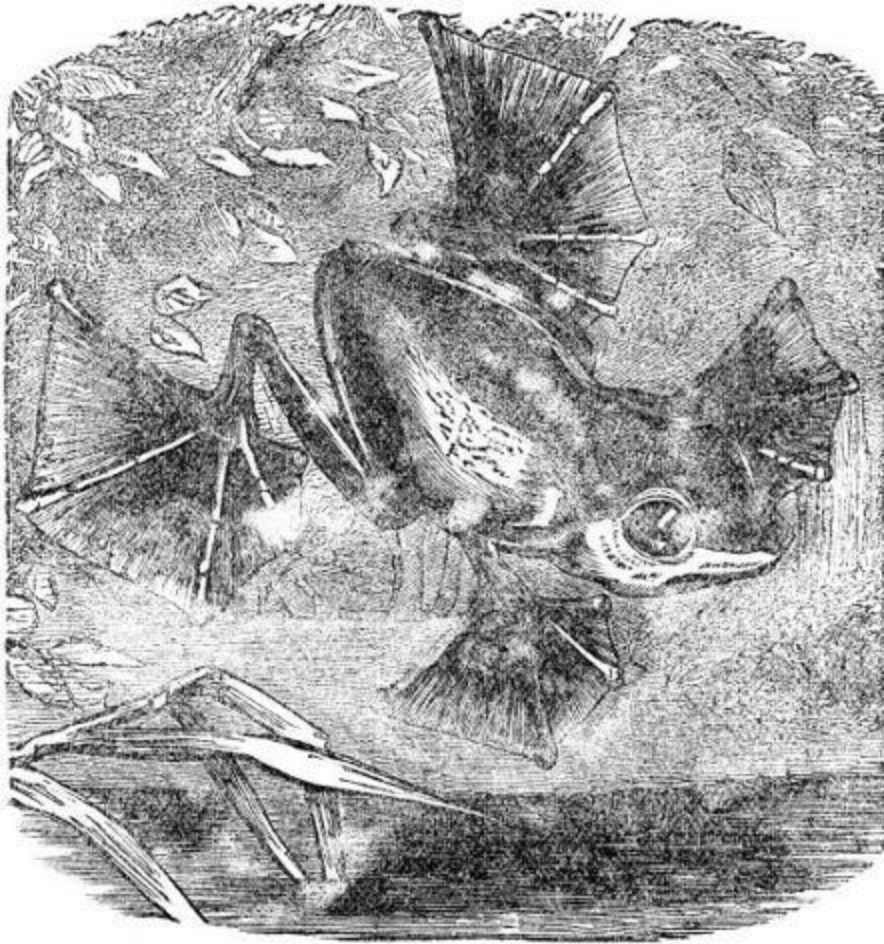
EASY LESSONS IN ANIMAL BIOLOGY.



CHAPTER III.

SUB-KINGDOM IX.—VERTEBRATA.

Though reptiles are generally regarded with aversion, there is much of interest in their natural history, and even they may serve some useful purpose. There are four genera, and about fifteen hundred species. The respiratory organs are the same in all, and are such as belong to air-breathing animals. With few exceptions they are carnivorous; generally sluggish in their habits, and their sensations dull. Nearly all the winter they are in a state of lethargy.



FLYING TOAD.

AMPHIBIA.—There are several species of cold blooded amphibious vertebrates which are properly neither fishes nor reptiles, but, having some things in common with both, should be mentioned here. Their respiration is most peculiar, as they have only gills when young, and, when adult, lungs. In their immature state they are fish-like, but develop no fins; and their limbs, when they at length have them, show the same articulations as those of higher animals. These amphibia, in the progress of their development undergo a series of remarkable metamorphoses. They begin life as water-breathing larvæ. In some of the lower orders the gills remain through life; others, when mature, have lungs only. Aquatic Newts, two-legged Mud-Eels, Toads, and Salamanders, are their representatives. The *Batrachia*[\[1\]](#) include the frog, a well known, tailless amphibian. Its metamorphose is peculiar. The ova are deposited in a jelly-like mass at the bottom of the water, and, if the temperature is suitable, develop rapidly. In a few days

very interesting microscopic observations are possible. The embryo presents four distinct appearances before leaving the egg, and five more before assuming its complete organization. For a time it seems a soft-skinned, scaleless fish, breathing only as fishes do, and having a long, flexible tail, used as an oar. Then it develops the palmated hind legs, as assistants in its locomotion, the caudal process diminishes, the forelegs protrude, true lungs are developed, the wriggling tadpole is a frog; abjures kinship with its still aquatic relatives, hops to the land, and enters on pursuits befitting its new mode of life. The common frog is well known, and we present one of a different family. It is distinguished by having the ends of its toes dilated into discs, or suckers, by which it sticks to the smooth bark, and is capable of its arboreal life. It is often called the “flying frog” and the “tree frog.” The webbed feet are large, and when spread out serve to bear up the body, as would paper kites of the same size. But the animal does not fly. It is active, and leaps from branch to branch expertly, and, when it wants to descend, by spreading those fan-like feet, it comes down with less violence. Tree toads are noisy little creatures, and seem to be ventriloquists, as their voices do not indicate their true position. This deception, and their color, make their capture more difficult.

CLASS II.—REPTILIA.

ORDER I.—*Ophidia*,^[2] or serpents, are oviparous, air-breathing vertebrates, having round, tapering bodies. They crawl or glide on their ventral surface. The serpent’s body is very flexible. Its numerous vertebræ, concave in front, and hemispherically convex behind, are so jointed as to allow a free horizontal motion. Its progress is usually by lateral undulations, made practicable by the great number of the vertebræ, attached flexible ribs, and muscles along the sides, specially arranged for prompt action on the spinal column. The bones of the head are loosely jointed, and movable, making the mouth very dilatable. The sharp teeth are hooked backward, so that whatever is seized is likely to go down the capacious throat. The entire skeleton and skin are so elastic that objects much larger than the serpent, in its normal condition, can be swallowed whole. Though without limbs, they perform with dexterity a great variety of movements; creep, climb, swim, raise the body almost erect, and spring from the ground. There are two classes, the poisonous and the harmless. Of the former the Cobra,

Copperhead, Viper, and Rattlesnake are representatives. They have extensile fangs, and, along the side of the upper jaw, a large gland, which secrets a poisonous liquid that is injected through the hollow or grooved fangs into the wounds they inflict. Harmless snakes have much the same structure, but nothing of this special arrangement.

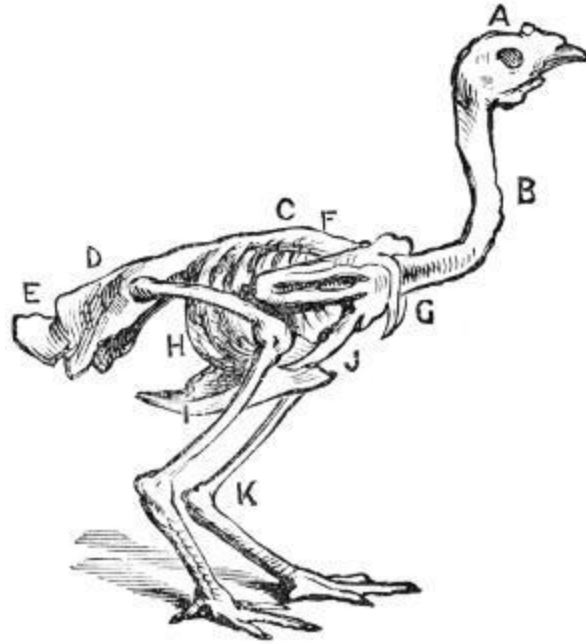
ORDER II.—*Lacertilia*^[3] or lizards, are snakes with short legs, and usually much quicker in their movements than most other reptiles. Most species are harmless, though a few are said to be venomous, and, because of them, all are dreaded. They differ much in appearance, some being really beautiful, and others about as hideous as reptiles can be. There are an immense number and variety of lizards, having some distinctive characteristics. Several extinct species, as the Iguanodon,^[4] Megalosaurus,^[5] Plesiosaurus,^[6] and Ichthyosaurus,^[7] are known only by their gigantic fossils, casts of which may be seen in many of our college museums.

ORDER III.—*Chelonia*.^[8] This name is derived from the Greek word for tortoise. The class includes all the varieties of tortoises or turtles. Tortoise means crooked, or twisted, and refers to the awkward manner in which the limbs are twisted about when in motion. There are two skeletons, the one external, including the viscera and the whole muscular system, which is internally attached to it. The upper plate of this covering, called the carapace, is more or less convex, and made up of strong elastic plates. The lower, or plastron, is flat and smooth, being apparently an extension of the sternum or breast bone. These plates are joined together at their lateral edges, leaving anterior and posterior openings through which the head and limbs are extended. When at rest, these are usually under the covering. The head and neck are covered with a rough, corrugated skin, and the horny upper jaw terminates in a strong, hooked beak. They are usually distinguished as marine, fresh water, and land turtles. The former are good swimmers, but their movements on land are slow and awkward, and if turned on their backs they are quite helpless. Some marine turtles are very large, being from four to six feet in diameter, and weighing from 600 to 2,000 pounds. Usually they weigh much less. All are oviparous, the female producing about one hundred and fifty hard-shelled eggs at a time. For this purpose she cautiously comes ashore, and, dextrously using her hind flippers, excavates a hole in the sand, from a foot to eighteen inches in depth, lays her eggs, and, having carefully covered, leaves them to be

hatched by the heat of the sun. The green, edible turtle is very valuable, and furnishes much wholesome and delicious food. The “hawks-bill” is seldom used for food, but supplies the greater part of the well known “tortoise shell” of commerce. There are many other species, both marine and fresh-water, whose flesh or shells are of considerable value. Soft Tortoises, Snappers, Mud-Turtles, and Terrapins, all have some peculiar characteristic.

ORDER IV.—*Loricata*.^[9] This corselet or armor-covered family includes Crocodiles and their allied species. Their upper parts are covered with a corselet of bony plates set in the tough, leathery skin. The jaws are long, and have many strong, conical teeth, fixed in sockets. The lower jaw extends back of the cranium, and the upper is hinged and movable. Crocodiles belong in tropical climates, are sluggish animals, and live a long time if not destroyed by violence. Their legs are short but powerful, and when on land they manage to wriggle, or drag, their immense bodies along with considerable speed. They are found in India, and in all the large rivers of Africa. The Gavial of the Ganges, the Crocodile, and the Mississippi Alligator are closely related, though there are some structural differences. Of whatever variety, whether of the Old World or of the New, they are more numerous than desirable, not being noted for either their beauty or usefulness.

CLASS III.—*Aves*.^[10] Birds furnish a delightful study, and their leading characteristics are easily stated. Widely as they differ, they all have a common type of structure, and are essentially alike in those particulars that distinguish them from other orders in the animal kingdom. They are all warm blooded, feathered, biped vertebrates, mostly with wings fitted for flight, and with either webbed feet for swimming, or claws for seizing, scratching, and perching.

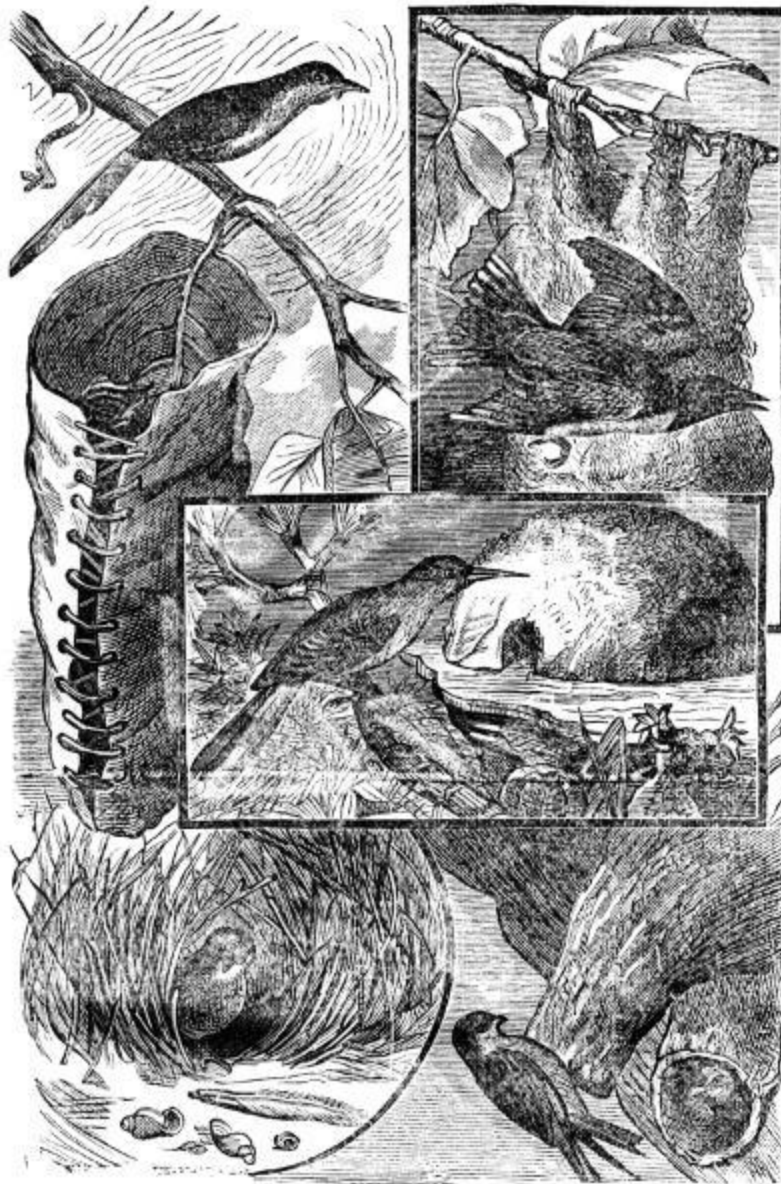


ANATOMY OF A HEN.

Ex.—A, cranium; B, cervical vertebræ; G, furcula, or merry thought; J, sternum, or breast bone; H, cloaca, surrounded by the pelvic bones; E, caudal vertebræ, terminating in plowshare bone; D, lumbar vertebræ, or rump; C, F, dorsal vertebræ, to which the ribs are attached.

There are several points of interest in the anatomy of a bird that may be used to bring out general characteristics: (1) The long neck, with vertebræ so adjusted as to allow great freedom of movement, makes the head a convenient prehensile organ. (2) The skeleton is remarkably light. It has fewer bones than other vertebrates; the thin skull bones, the dorsal vertebræ, and bones of the feet being anchylosed.^[11] They are also harder, of lighter material, and filled with air, thus having the greatest possible strength with the least weight. (3) To make respiration sufficient during rapid flight, an abundant supply of air not only inflates the lungs as in other animals, but fills little membranous sacs, or cells, distributed through the body, and extends even into the wing feathers. (4) The digestive apparatus differs materially from that of either fishes, reptiles, or mammals. The æsophagus^[12] before reaching the sternum is dilated into a large sack, or crop, and serves as a first stomach, in which the food is softened, and prepared for other digestive organs. Below this is another slight enlargement, the walls of which are thicker, and secrete gastric juice. Still

further down, the canal is enlarged into a third stomach, the muscular gizzard, in which the process is completed. (5) There are no teeth set in the bone sockets, as their weight would be inconvenient, but the mandibles are sheathed in a horny case, which has sharp edges, and terminates in the beak, or bill.



NESTS OF BIRDS.

Both genera and species are very numerous, but any two birds that can be selected differ less in their anatomy than some reptiles of the same order differ. The leading orders only of this class have been selected. *Raptors*[\[13\]](#)

(robbers). This is a mild term when used to indicate the ferocity of most birds of prey which not only plunder but destroy. They are almost constantly committing murderous assaults on their weaker neighbors. This is their nature, and accords with their physical structure. The strong hooked bill, powerful legs, feet armed with sharp claws to seize and hold their victims, while the murderous beak is tearing off bits of flesh, are some of their chief characteristics. These murderers have representatives in nearly all countries, but as civilization advances they become less numerous.

They are usually divided into three great families: Falcons, Vultures, and Owls. The first includes all kinds of Hawks and Eagles. True falcons reveal a predatory character, not only by their general structure as described, but have a special arrangement for keeping their formidable weapons in order. The continual sharpness of their claws is necessary, and to maintain it they must be kept from coming in contact with hard substances. To make this practicable, though they are not retractile, like a cat's claws, there not being sufficient integuments to cover them, the bird has power to elevate their points when stepping or perching on anything likely to dull them. Being very powerful and rapid flyers, falcons were in former times tamed, and trained for catching other birds and small game. Falconry, in the middle ages, was one of the principal diversions of kings and noblemen, lords and ladies, and in later times the same sport was practiced in England under the name of "hawking." The training of the birds was a profession, and there were teachers who became proficient in it. The Eagle, king of birds, belongs to this class; he is, perhaps, inferior in activity and enterprise, but is more powerful, and his supremacy is undisputed. The Bald Eagle adorns our American flag, and is a fit emblem of national sovereignty, though there was, it is confessed, some ground for Franklin's protest, "The bird has a bad moral character, and does not get his living honestly." The Gypætos, or Vulture Eagle, is the largest bird of Europe, and often a terror to the peasants near the Pyrenees and Swiss Alps. Its great strength, bold, predatory habits, and impetuosity in pouncing on animals exceeding itself in size, make it more formidable than most eagles are.



THE FLAMINGO.

Vultures are cowardly, disgusting creatures, and feed mostly on putrid flesh, yet they are useful scavengers in hot climates, devouring much animal matter that would otherwise cause pestilence and death.

Owls are nocturnal birds of prey, and though seldom abroad except in the darkness, or twilight, being often captured and confined, they are pretty well known.

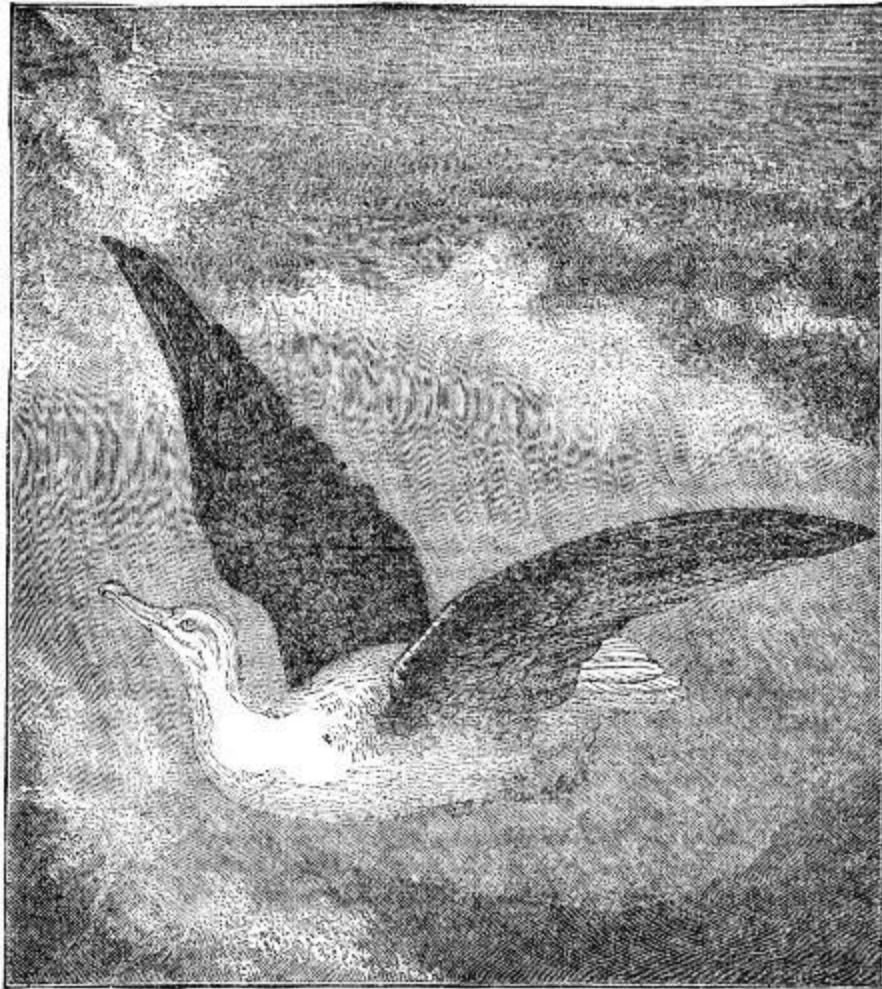
Insectores^[14] (perching birds). This term is adopted as vaguely descriptive of many species, of which no particular mention can be made. They include our common singing birds and skillful nest builders. Some of these build low, among the grass, others on branches of trees. Some are solitary, and during the nesting season isolate themselves. Others, as the “sociable weavers,” unite in building them cities, in which each pair claims a private residence.

Cursores^[15] (runners). Birds of this class run rather than fly, have no keel on the breast bone, no well developed wings, and their feathers are loosely put together, without the connecting barbs that strengthen the wing feathers of others. But they have strong legs, and feet with nails rather than

claws. The Ostrich is most distinguished for its size, strength, speed, and peculiar feathers. The Emeu and Apteryx belong to the same class, the latter being tailless.

Grallatores[\[16\]](#) (waders). These aquatic birds are distinguished by their very long, bare legs, also necks and bills of like proportions. Herons, Storks, and Flamingoes are representatives.

The latter is a peculiar bird, common in some portions of America, Southern Europe, Asia, and Africa. When walking erect it is about five feet high, the body of a light color, and the wings red. When a number go in single file, as such birds do, they appear like a company of British soldiers in uniform. As their legs are quite too long for convenience when incubating, they construct, of grass, rushes, and mud, a little cone of sufficient height, make their nest on the top, and sit astride, the long legs hanging down the sides. This also protects the eggs and young from moisture in wet weather.



THE ALBATROSS.

Natatores[\[17\]](#) (swimmers). Aquatic birds being keel breasted, with short legs, webbed feet, and thin, light, warm covering, delight in swimming. Some of these have also good flying qualities, as Wild Geese, Swans, Sea-Gulls, and Petrels, which spend much time in skimming over the water, either on or above the surface.

This family includes the wandering Albatross, which is among the largest of the sea birds, having from ten to fifteen feet expanse of the wings, and weighing from twenty to thirty pounds; yet it sustains itself in the air for many hours together without any apparent difficulty. It is often met far out at sea, and sailors have many superstitions respecting it. Like all petrels and sea-gulls, the albatross makes its nest on crags and cliffs along the coast, choosing places that seem inaccessible. To gather the eggs of sea-fowls, which are esteemed a delicacy, and to capture the birds

themselves for the feathers, is one of the most perilous employments. The adventurous fowlers are swung over precipices from five hundred to a thousand feet above the sea, sustained simply by a rope, that, by breaking, or slipping from the hands that hold it, must hurl them to certain destruction.

CLASS IV.—*Mammalia* (milk givers). Though there are some very low species in this order, it reaches upward and includes the highest forms of animal organization, all of what ever degree, that have the mammary glands, and secrete milk for the nourishment of their young. There are more than two thousand species of these in North America alone. They have several distinguishing characteristics. The heart is four chambered, having two auricles and two ventricles. They are warm blooded, the blood having two kinds of corpuscles, the red predominating, and being globular. The impure blood from the body pours into the right auricle, passes to the right ventricle, and thence to the lungs. In the lungs, which are spongy and full of air cells, it is supplied with oxygen from the air freely circulating through them. Being thus changed to purer arterial blood, it passes back through the left auricle to the left ventricle, and, by a muscular action, is forced through the great aorta and its innumerable branches to every part of the body. Mammals always breathe by lungs enclosed in a membranous sack called the pleura. They are viviparous, or produce their young alive, though some when in a very immature, helpless state. They are for convenience subdivided:

ORDER I. *Monotremata*.[\[18\]](#) These animals, the lowest of mammals, have long, flattened beaks, webbed feet, and other bird-like characteristics. But they are of little importance, few in number, and not widely distributed, being confined mostly to Australia and Tasmania.

Belonging to this class is the Water-Mole, with a broad, duck-like, horny bill. It lives on worms and vegetables, and burrows in the banks of streams, having the opening to its quarters under the surface of the water.

ORDER II. *Marsupial*, or pouched animals. The young are brought forth in a very premature state, but are immediately placed in a marsupium or pouch under the abdomen, where, attached to the little teats, they receive their nourishment. They are thus protected and carried by the mother as

long as such care is necessary. The opossum, the only native marsupial of the United States, is about twenty inches long, exclusive of the peculiar rat-like, prehensile tail, by which the animal often suspends itself from the branches of trees. When in danger, it feigns death as a way of escape—and can survive injuries that would be fatal to most animals. They were once very numerous in this country, and though destructive to fruit and poultry, are of some value. Their flesh is used by some, and the skins are in demand, the hair or coarse fur being wrought into felt.

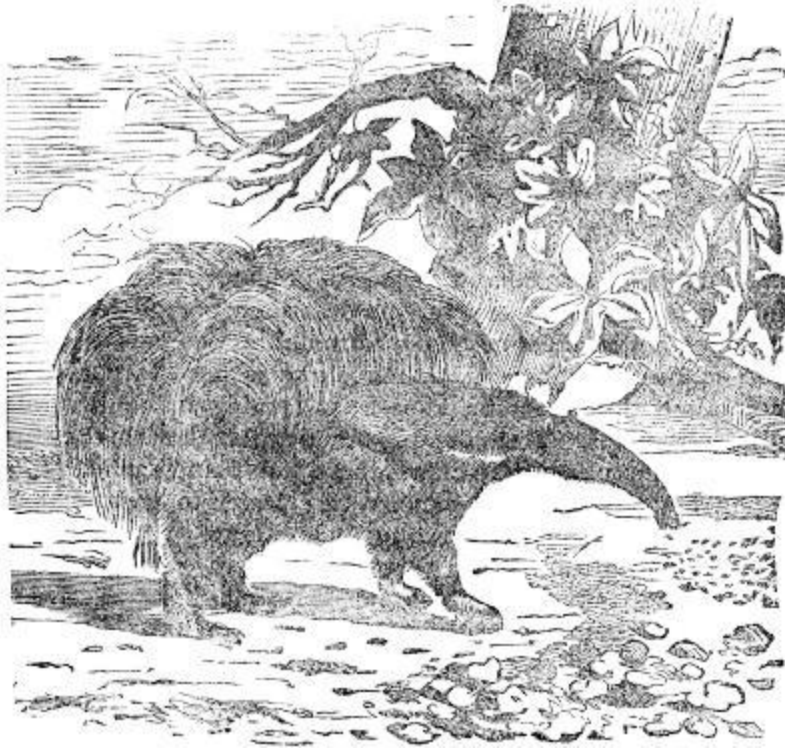
ORDER III. *Edentata*[\[19\]](#) (toothless). Animals without incisors, and having separate clawed toes, are included under this order. The chief representatives are the Armadillos, Sloths and Ant-eaters.

Armadillos[\[20\]](#) are remarkable animals, with a covering of horny plates, not unlike a tortoise shell, but arranged in sections in a way that allows more freedom of motion. Some of the extinct species were of gigantic proportions. A fossil armadillo found near the La Plate was as large as a full grown rhinoceros.

The Sloths are natives of South America. They are covered with long, coarse, gray-and-black hair, resembling the moss of the trees in which they live, for these animals are arboreal, while the other members of the order burrow.

Ant-eaters, of which there are several varieties, are covered with spines like the hedge-hog. Even the long, rough tongue and palate are provided with the sharp little spines with which they spear and hold their prey. Their claws also are fitted for digging into the ant hills, where the food is obtained. They are very inferior mammals, but have their place, and are useful in destroying noxious insects.

The giant ant-eater of South America is much the largest of the genus, and an animal of considerable strength. From its long, bill-like snout it thrusts out its longer tongue, and the frantic ants, disturbed in their quarters, on rushing out stick to it, and are rapidly swept into the mouth. When sleeping, it is coiled up under its immense bushy tail and looks more like a little heap of dried grass than an animal.

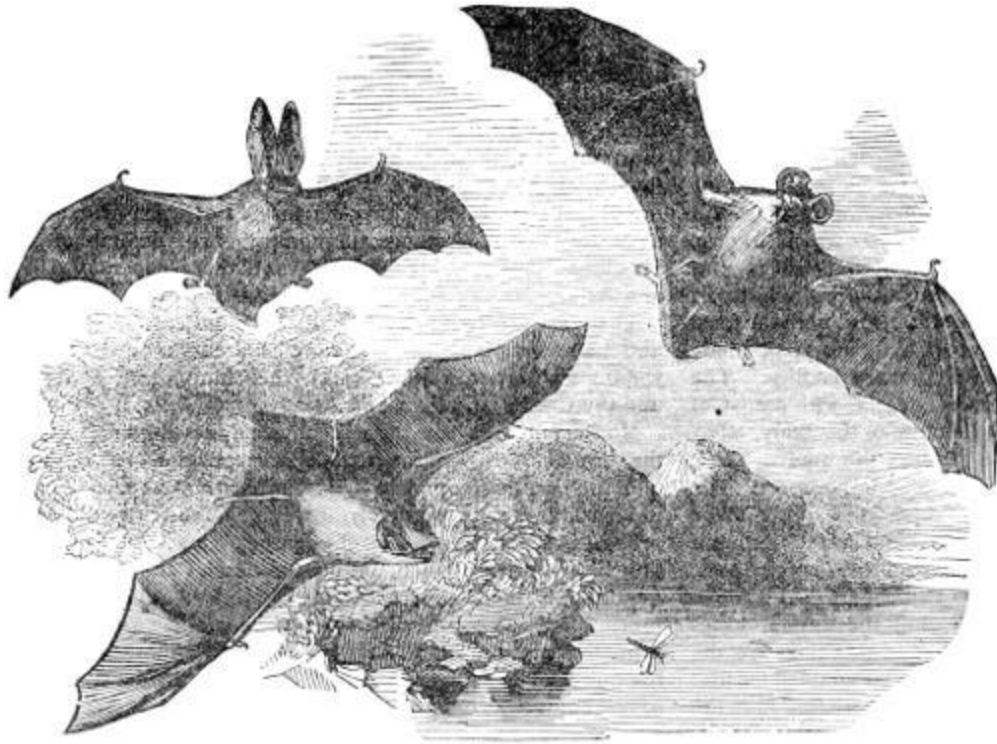


THE ANT-EATER.

ORDER IV.—*Sirenia*[\[21\]](#) (sea cows), include the Manatee, found between the Amazon and Southern Florida, and the Dugong, of India. They are amphibious milk givers. In appearance and structure they are very unlike most mammals. Their forelimbs resemble fins, having fingers or toes. The hind legs are wanting, and the tail like that of a fish, but they suckle their young, sometimes supporting them for the purpose with their flippers.

ORDER V.—*Cetacea*[\[22\]](#) (Whales). They are much the largest of mammals. They live entirely in the sea, and have in general a fish-like appearance. Their forelimbs are huge paddles, the others being only rudimentary. But they produce their young well developed, and nourish them with an abundance of rich, creamy milk. The amount of blood in a large whale must be immense, the great artery being a foot in diameter, every pulsation of the heart forces many gallons along the channels prepared for it. A single whale, captured in 1884 by the crew of a New London vessel, produced whalebone and oil worth \$15,720, beside other products of considerable value.

ORDER VI.—*Insectivora* (insect eating). An order of small animals, having well developed teeth, the molars remarkable for their sharp cusps; and five-toed feet furnished with claws. The Mole, Hedge-hog and Shrew are examples.



VARIETIES OF BATS.

ORDER VII.—*Cheiroptera*[\[23\]](#) (wing handed). Bats are not generally favorites, and are by many regarded with aversion. Their strange and rather uncomely forms are seen on wing only in the dim twilight, as they spend the day mostly in deserted buildings and gloomy caverns. In eastern countries, where such receptacles of the dead are common, bats are often found in sepulchers or catacombs, and are regarded as fit dwellers with desolation and death. There are many species, with enough variety in their appearance; and the study of their natural history softens prejudice and reveals much of interest in their structure and habits.

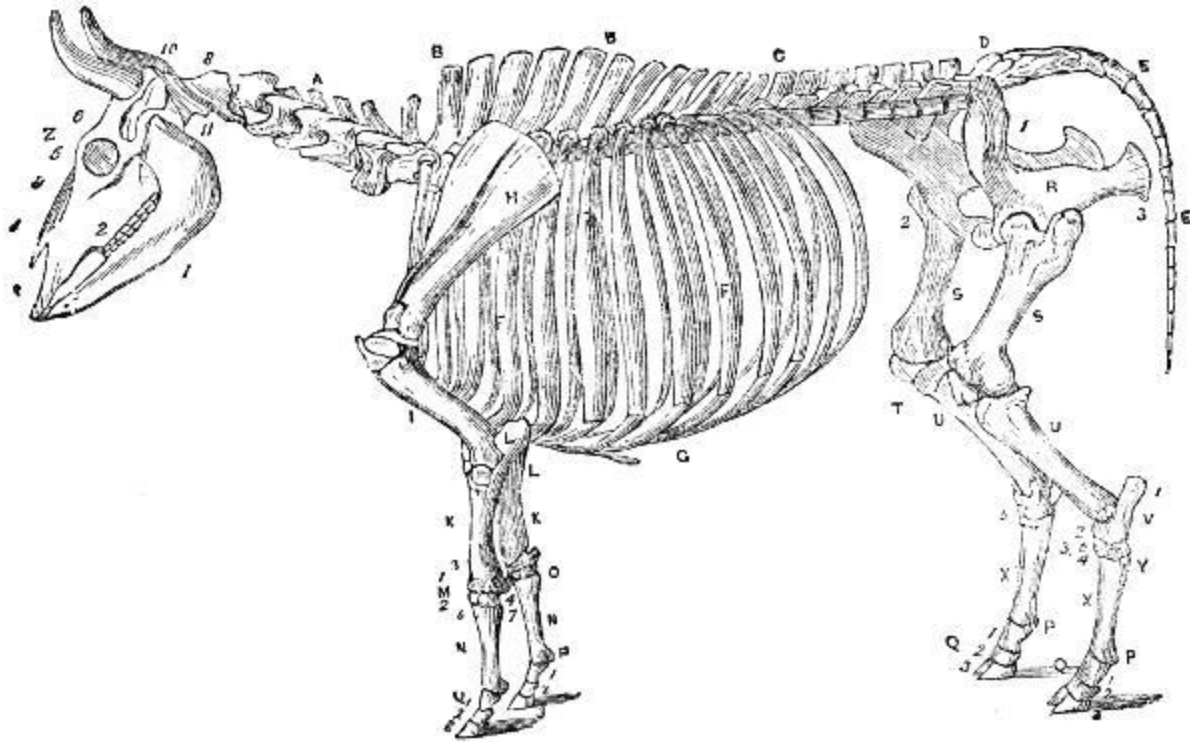
ORDER VIII.—*Rodentia*[\[24\]](#) (gnawing animals). Animals of this order have two large, chisel-like incisors in each jaw, and, separated from them by a wide space, are the molar teeth. The incisors never cease growing from the roots, but are constantly worn away by nibbling. The lower jaw moves

backward and forward. This order includes more than half of the known mammals, and its representatives range from the equator to the poles. Hares, Mice, Rats, Squirrels, and Beavers are among those best known.

ORDER IX.—*Ungulata*^[25] (hoofed animals). An order of mammals most valuable to man. They are grouped in two divisions, according to the number of toes. The uneven-toed ungulates include the Elephant, Rhinoceros and Horse. The elephant is marked by the prolongation of the nose and upper lip into a trunk or proboscis, which is said to contain over 40,000 muscles. It has no canine teeth, and incisors only in the lower jaw. Elephants are found in Asia and Africa. There are two extinct species, the mastodon and mammoth. The rhinoceros is a native of Africa and India. It is an immense animal, covered by a hairless skin, which lies in folds on the body. The nose bears one or two horns, which grow sometimes three feet in length. The horse includes animals having one toe upon each foot, upon which they walk. The family includes the Ass, Zebra and Quagga.

The even-toed ungulates include all the *Ruminates*, or cud chewing animals, with the Swine and Hippopotamus. The ruminants are remarkable for their peculiar method of digesting their food. They possess a very peculiar stomach, of four compartments. The food, mixed with saliva, is swallowed and passes into a paunch where it is mingled with water, and then forced into what is called the honey-comb stomach, a sack in which the food is formed into cuds, and by a muscular arrangement is forced back into the mouth to be masticated a second time before passing directly into the third stomach or manyplies, where it is strained, and then driven into the true stomach to be acted upon by the gastric juice and assimilated.

The deer is a fine representative of the ruminant, with solid branching horns. Like all ruminants it has two toes. There is a large class of cud-chewers having hollow horns, which usually are not shed, as are the solid horns. The Buffalo, Ox, Sheep, Goat and Antelope belong to this division.



OX SKELETON.

10, horns; 8, spine; A, cervical, B B, dorsal, C, lumbar, D, sacral, and E E caudal vertebræ; F F, ribs; G, sternum and cartilages; R, ossa innominata; H, scapula; I, humerus; K, radius; L, ulna; N, metacarpal; S, femur; T, patella; U, tibia; V, hock; X, metatarsal; Y, small metatarsal; P, sesamoids;

Y, bones of tarsus, nine in number. Figures near Z: 1, infero maxilla; 2, supero maxilla; 3, premaxilla; 4, nasal; 5, lachrymal; 6, frontal. Figures near letter M, bones of the carpus or knee, 1, trapezium; 2, cuniform; 3, lunar; 4, scaphoid; 5, unciform; 6, magnum. Figures near the letter Q, mark the three phalanges, or small bones of the foot.

The Giraffe, an inhabitant of Central Africa, and remarkable for its long neck, is another ruminant. The camel also belongs here. The true camel has two fatty humps upon its back; another species, called the dromedary, has but a single hump. The camel has a peculiar modification of one compartment of the ruminant stomach. The paunch is divided into cells which hold supplies of water.

ORDER X.—*Carnivora* (flesh-eaters). The distinguishing characteristics of this order are sharp canine teeth, and one molar on each side in the upper and lower jaw, longer and sharper than the rest; and feet which are provided with toes, generally supplied with claws. According to the modifications of the feet they are divided into *Pinnigrades*, *Plantigrades* and *Digitigrades*. The first have short, webbed feet which they use as paddles for swimming,

and are represented by Seals and Walruses. These are amphibious mammals of a high order. They spend much of their time in the water, and both the form and covering of their bodies are adapted to their aquatic mode of life. They swim and dive with the greatest facility. The soft woolly down, close to the skin, is covered with a coat of long, smooth, shining hairs which lie close to the body, offering no resistance to their passage through the water. Their skins and oil are of considerable mercantile importance. The Plantigrades include those animals which in walking place the sole of the foot flat upon the ground. Such are Bears and Raccoons. The Digitigrades are those which walk on the toes, as Weasels, Foxes, Dogs, Cats, Lions and Tigers.

ORDER XI.—*Primates* are at the head of the animal kingdom. The distinguishing characteristics of this class are the more erect carriage of the body, a hand better adapted for use, its fingers being furnished with nails, and a thigh free from the body. The order includes the Lemurs, Monkeys, Apes and Man. The lowest division of the primates, the lemurs, are small animals whose bodies are covered with hair. They have a fox-like face, a pointed nose, large ears, and a long tail, and they walk on all fours. The monkeys have long prehensile tails, and though their hands can be used for grasping, the thumbs are not opposable. A distinguishing mark is the number of teeth; they have thirty-six instead of thirty-two, as the apes and man. Among the monkeys are the Baboon, the Howling Monkeys, the Mandril, and the Sleepers of Africa and Asia.

The apes are more erect than the monkeys, have no tail, and have longer arms. The division includes the Orang-Outang of Borneo and Sumatra, the Chimpanzee, of Africa, and the Gorilla of Africa and Asia.

Man commands the highest physical development among the primates. In him the partially erect position of the monkey and ape becomes complete. His limbs are more nearly equal in length and more perfectly developed than those of other primates, the skull is larger, the forehead more rounded, and the brain nearly twice the size of that of the gorilla, the dentition is more perfect, the teeth being regular and rarely protruding. The great distinction between man and the lower primates is that of mind. He alone, of all animals, possesses the power of articulate speech, of forming abstract ideas, and of reasoning.

End of Required Reading for the Year 1884-85.

SUMMER HOMES FOR THE CITY POOR.

BY HELEN CAMPBELL.

Who began it?

The germ theory.

This answer may be regarded as not strictly to the point, yet it is certain that it holds the truth more nearly than any statement which might give New York or Philadelphia as the first cause. By this mysterious law, which from time to time we encounter in settling precisely such questions, minds entirely remote and with no kinship of faith or mutual purpose, felt the sudden moving toward action practically simultaneous. A hint of things to come had been given half a dozen years before, but full action waited for the Centennial year of 1876.

Up to this time summer rest, save for the rich, had been regarded as a needless luxury. Increased knowledge of sanitary laws had demonstrated that change of air might be as vital a necessity for poor as for rich. But that this applied also to the lowest and most helpless classes had no place save in the mind of a practical philanthropist here and there. Nor had it yet become a part of the creed of such workers, and, far less, a subject of common discussion among those who sought the most practical methods of help, that the children were the ones to whom such help would mean the most. Charity concerned itself chiefly with alleviation, and looked in more and more hopeless dismay on the ever-increasing numbers of this army of incapables. It was evident to all who walked through city streets as August sunshine blazed on squalid home and reeking gutter, and gaunt women and pale, unchildlike children came and went in noisome street and alley, that something must be done, but what? A good woman had opened a summer home for very young children on Staten Island, and in 1873 transferred its management to the Children's Aid Society on the understanding that four thousand dollars should be collected to insure its success. The founder, Mr.

A. P. Stokes, at once gave half the amount, and the rest was quickly made up, the headquarters for the new home being made at Bath, Long Island, where the work still goes on with largely increased facilities. Beginning with the rental of a private house, in which every inch of space was so utilized that the parlors became a sleeping room for twenty-four persons, and the carriage house was so made over as to contain nearly sixty beds, it has grown into one of the most efficient and beneficent of charities. A week is given to each detachment, chosen most generally from the members of the industrial schools under the direction of the Children's Aid Society. The children are abundantly fed, the dietary including milk, bread, butter, oatmeal, fruits, vegetables, fish and meats. The salt water is almost at the door, while on the other side are woods and wild flowers, and the children change even in a week of such life, sometimes almost beyond recognition. But there are objections to these homes, beautiful as is their mission in many points, and a writer in the *New York Evening Post*, after faithful examination of this and other summer homes, writes: "Neither the excursions, which are only for a day, nor the seaside homes quite reach the best results. What the pinched sufferers in alley-ways and courts, garrets and basements need is to be sent to the country. It is not enough to give them a day on the river, though that is good as far as it goes. Even in seaside homes they are housed with children of their own sort; they have the same conversation, the same plays, the same depressing companionship with disease and want. What they need is to meet the healthy life of the country; to make acquaintance with nature, to learn the difference between a calf and a pig; to have a whole new set of objects before their eyes and mind. The country is a bit of heaven to such children. They catch a new life from it; they bring back into town a better tone of mind, body and morals."

Such had long been the conviction of a woman whose name is synonymous with every advance that has been made in wise dealing with social problems, and who, as she came and went in the narrow streets and stifling heat of Philadelphia, yearned always more and more, to give the children some hint of what lay almost within their reach, yet inaccessible as the poles. At the meeting of Progressive Friends in 1876, one of the earliest addresses, and one of the most powerful in its effects, made an impassioned appeal for an end of dreaming and theorizing, and a life of action—action which should mean help to every human soul in need of help. There was a

flutter of interest and excitement as the speaker ended, but no one made suggestion as to what form such action might take. Great issues were apparently ended. To a gathering made up in great part of veteran Abolitionists, any other struggle seemed weak and puerile, and they looked doubtfully at one another as the words ceased.

The wise woman saw her time, and when the afternoon meeting brought them together, rose in her place, and in few words told what her eyes had seen and her heart desired. A buzz of interest and of opposition was heard at once. One and another stated objections; objections made by many since then, but that have proved no obstacle to the progress of the work. Such children would bring in their train, dirt, disease, vermin, foul language and general demoralization for every child in the neighborhood who came in contact with them.

“Such things are all possible, and may all be anticipated,” said the wise woman, who had already in her own experience demonstrated that each and all could be overcome, “but I think you will find that there are ways of meeting them.”

Still objections were made, and a warm argument was under way, when a Quaker wife whose eyes flashed from the shade of the close bonnet, and whose voice held a ring not unknown even to Quaker gentleness, rose before the debating broad-brims:

“I’m going to have two of those children, if I have to tie my own to a tree,” she said, and said no more.

More was not needed. The tide had turned, and one and another volunteered to open the doors and at least make an experiment. And now, curiously enough, came a difficulty not even imagined; the difficulty of securing children for the offered and waiting places. Fathers and mothers looked with dark suspicion on the people who requested the loan of the children.

“Shure it’s none o’ mine ye’ll be gettin’,” said one of them. “Makin’ out it’s for play ye’s want ’em, an’ settin’ them to work soon as our eyes is off of ’em.”

“It’s not work I’ll have child o’ mine do, long’s I’ve hands that’ll earn the bit an’ sup she’ll be takin’,” said another, and the children themselves, hollow-eyed and haggard, watched the fracas with small interest. The first who went out had to be compelled, as it were, to come in to the feast, but never again were like measures necessary. The good work went on with courage, even with glee. Baths, clean clothes, plenty to eat, sweet air and sunshine did their work, and the dirty, forlorn little wretches chosen in the beginning returned home, imploring for longer time, and with a new sense of life born in them. Season after season the same doors opened, and as the work grew, funds poured in; a method formulated itself, and the management became a model for all similar work elsewhere.

This is what had happened at one point, but the germ was not a single one, and at another in the same state, another mind had felt the same touch, and the same result had come. The project had been talked over before it took positive form, and talked over with a woman, who, from the upper chamber where long years had held her prisoner to pain, looked out upon the world through others’ eyes, but with an insight that went to the heart of all possibilities for help. The young minister who counted her word as equivalent to the united force of a dozen elders, went home to his flock among the Pennsylvania mountains—hard-working farmers from whom quick response could hardly have been expected—and this is the letter he wrote on a Sunday when he had spoken to them his first word of the wish born a year before:

“SHERMAN, PA., June 3, 1877.

“MY DEAR MRS. L.:—The ball is set in motion. I took for my text this morning, “Inasmuch as ye have done it unto one of the least of these, ye have done it unto me,” and made the practical bearing of my words the bringing out into our homes some of the waifs and outcasts from the city. One man stopped on his way home to say that he would take four. In another house there is a call for a mother and a baby, and so on through the town. The enthusiasm and response of my people have delighted me.

“Next to get the money; then to tell the children. Must not two weeks in this pure mountain air be felt by them in after life? It seems to me that they are all but here.

“Now may I have the introduction you promised me to Dr. Eggleston? I shall try for a pass over the road to go back and forth with the children myself, and perhaps I can arrange with some of these good people on the way to bring us a country lunch as the train comes along. Some good angel whisper it in the ear of a little one! Tell a tired mother there is life for her child in this fresh country air.

“WILLARD PARSONS.”

It was an unknown name then, but through Dr. Eggleston, then on the point of sailing for Europe, interest was roused. The Erie Railway officers proved that corporations have sometimes a soul, and full fares were reduced to half fares, and half fares to quarter fares, and a pass was given to Mr. Parsons, and on July 19th went out the first group of nine. They were mere wraiths of children, crippled, in consumption, enfeebled from whooping cough; each one stamped with disease and pinched and thin for want of food. It was doubtful how they could bear the journey.

The children who swarmed out next day “to catch raspberries” proved perfectly manageable, and when their two weeks ended, returned home transformed from sad-eyed, prematurely old little figures, into live children, loaded with gifts and crying to stay longer. The story has become a familiar one, but it can never become tedious. The second group gave less anxiety.

The work was better understood. Seventeen boys and girls, each wearing a blue ribbon bow as the badge of "country week" children, gathered from all quarters, and all, delicate, half-starved, suffering with hip-disease, asthma, and a dozen ailments, met at the Erie train.

The diary of the summer's work runs over with small absurdities, with pathos, with promise. Sixty in all shared the good provided for them at a total cost of only one hundred and eighty-seven dollars and sixty-two cents. But for New York as for Philadelphia, it was easier to get the money than to get the children. Often a pale and care-worn child was the breadwinner. "Sometimes the mother had a fear of separation, or the feeble, childish hands must tend baby and do the housework while mother goes out by the day. 'It is harder for Jack than for any one else when the baby comes,' one mother said. 'The care comes on him.' Baby was in her arms as she spoke, but Jack was close by, thirteen years old, under size, and washing stockings at the tub!"

In many cases the children made friends for life, in a few the attachment formed being so strong that adoption followed. For all of them was the same experience; a fortnight or more of bliss and revelation and a return, loaded down with bundles and boxes and bags of the things that each one chose to gather.

That ticks had to be washed and straw burned after the occupation of many of them, made no apparent difference, and even to-day, when it is all an old story, and board must in many cases be paid, there is unfailing consideration for the tastes and whims of the strangers. Now and then there was fright and tears and long wails for mother, soon quieted. Now and then rebellion and ingratitude; sometimes lying and petty thefts, but all yielding to kindness.

In Philadelphia, to whose markets the farmers for miles about come in, the children were in many cases able to keep up their acquaintance, and were often found behind the stalls, in long talks with the friends who now and then bundled them into the great Conestoga wagons and gave them an unexpected country week. The work has grown at this point, as in New York, beyond any expectation, and rooms and officers have both become necessary, the modest reports giving small hint of the patient labor

bestowed. The work was long confined solely to children, with now and then a worn-out mother smuggled in, but the same eyes that had seen their needs were studying now into possibilities for the class just above them—the working girls. Of all grades, from factory to store, all were living on the least sum on which body and soul could be kept together, and all needed quite as strenuously as the children, the change from narrow, stifling homes to country air and sights. To accomplish this has been far more difficult than the first undertaking, but has resulted in a success quite as complete. The objections were natural. Children could be disciplined and taught even in a week's stay, but growing or grown girls, probably pert, self-sufficient and generally unpleasant, were quite another matter. It was impossible to say what airs they might not put on, or what demands they might make, and quiet housewives turned in dismay from even the thought of such inmates. The same woman who had decided that her own children should be temporarily tied up rather than to stand in the way of more needy ones, opened her doors again, not as a charity, but on the lowest terms that could well be fixed, and half a dozen girls came to her for a fortnight, each paying two dollars per week, and finding such interest on the investment as no dollars of their earning had ever known before. The girls were gentle, quiet, over-worked and timid, and so far from putting on airs, required all the assurances that could be given to make them willing to take the good before them. Other doors opened at once, and the neighborhood soon found even more interest in this phase than had attended work for the children. Girls in many cases clubbed together, and it has been found, wherever attempted, that three dollars per week for board and washing still leaves a margin of profit for the entertainers. Home after home has sprung up by the seashore or in the country proper; but the same objection applies to massing girls together that has already been mentioned as affecting the children. Numbers seem always to include inevitable demoralization, and to develop unpleasant possibilities in even the most inoffensive, and the conclusion is the same for both, that the best results follow where private families open their doors and the workings of home life can be part of the vacation experience. Hints have come in silent ways stronger than any words, that have borne fruit in many lives. A new sense has been born—a sense of the beauty of order and many a quiet virtue unknown to the crowded and scrambling city life, and the lesson has as yet reached but the alphabet.

It is not the intention of this article to describe any one work at length, or to define more than the possibilities before us all. More and more we have come to recognize that only in dealing directly with the individual, can any efficient work be done, and this principle now underlies the best that the Associated Charities has given us. Summer rest is but a phase of the wide-reaching work, but not one holds a larger significance. Two or three years ago the writer recorded one case, which, while hardly typical, still shows what may lie in wait for the young soul of whose possibilities we can never judge in full, and the story is given again, as the best illustration of what a country week may mean.

Long ago in a dull, old street, making part of an equally dull and colorless part of old New York, a very solitary child extracted such amusement from life as forty feet of back yard could afford. He sat in his small rocking-chair and listened to the talk about him, growing a little paler, a little more uncanny all the time, till one day when a country cousin appeared, and, horrified that anything so old and weazened could call itself a boy, begged that he might go home with her.

There was infinite objection, but her point was finally carried, and the child found himself suddenly in a country village, a great garden about the house, a family dog and cat, a cow, an old horse, and all the belongings of village life. Old-fashioned flowers were all about, and the old-fashioned boy sat down in the path by a bed of spice pinks and looked at them, his hands folded and a species of adoration on his face.

“Pick some,” said the cousin; “pick as many as you want.”

“Pick them?” repeated the old-fashioned boy. “I’m afraid to. Ain’t they God’s?”

An hour later the seven years’ crust had broken once for all, and the child, who had to be put to bed exhausted from his scrambles through and over every unaccustomed thing, began to live the first day of real child life. When the time came for his return he begged with such a passion of eagerness, such storm of sobs and cries for longer stay that the unwilling aunt and grandmother left him there, and finding the transformation, when he did return, beyond either comprehension or management, sent him back to the life he craved.

To-day he takes rank among American painters, though only heaven knows how the possibility of such development found place in this strange off-shoot of a Philistine race. But he counts his own birthday from the hour when the first sense of sky and grass and flowers dawned upon him, and he looked upon the garden that he thought truly God had planted.

Such revelation is the portion of few, but for all it comes in degree. To aid such revelation is hardly a charity. Is it not rather self-protection? Men and women in the slums are beyond much power of ours for reconstruction or reformation, but the children can be influenced still. And so let every one who looks with apprehension at the daily criminal record, and wonders what should be done, remember that a very small sum will be one means of giving a chance to some child born to all evil, whose first sense of something better will come, not through school or mission, but through the silent teaching unconsciously working in them, through every breath of fresh air, every sight of blue sky and sunshine, and green grass and trees. A "country week" may come from a very small sum, but it is an investment on which interest is unending, and whoever has once made it will find that the pleasure is not for the child alone, and that life opens up more possibilities than had come even to one's deepest dreams.

For those who desire more specific knowledge than that afforded by the story of the undertaking as a whole, it is sufficient to give one or two details which will enable any one interested in further examination of the matter to obtain reports and information from headquarters. For Philadelphia these headquarters are at 1112 Girard Street, from which the companies of children, registered and numbered, are sent out. A letter addressed "Officers of the Country Week," at this number, will always receive prompt attention, the work having received formal organization some years since, and owning a regular board of untiring and devoted officers. They have come to include now not only children, but tired shop-girls, young mothers worn with care, and working women of all orders, and there is opportunity not only for those who are willing to open their houses without charge, but also for those who must cover expenses, the Board paying wherever necessary, a sum per child, sufficient to cover these. For New York there is perhaps a trifle less system, but a work equally beneficent, and for all information regarding it, it is sufficient to address either "The Tribune Fresh Air Fund," THE TRIBUNE,

Park Row, New York, or the Rev. WILLARD PARSONS, care of New York *Tribune*.

Contributions or inquiries either will be welcomed, and matters are now so perfectly systematized, that distance to be traveled proves no obstacle, and even the remotest village on the lines of the great railroads may have its share in this most beautiful and essential form of service.

LEARN TO ENJOY PEOPLE.

BY MARGARET MEREDITH.

There is one surpassing beauty of manner which, I think, might be attained by cultivation—that of taking an interest in people who are talking to you, with the subtle added charm of seeming at leisure to enjoy them as long as they choose to stay.

With some it is natural. I was struck to-day with the sweet graciousness of a young girl, ill dressed (for her) and very busy, who almost deceived me into wasting ten extra minutes of her precious time, by the satisfied air with which she sat down beside me in her parlor and welcomed my inopportune appearance. It was not politic, to be sure, for getting her cake making done, but, ah how politic for winning admiration!

Of the few whom I have remarked, in our busy American life, as possessing this faculty supremely, one family had long lived abroad, and were people of leisure. The look of content, and of being established for an indefinite time, with which any one of them sat on the sofa before a dull chance acquaintance, was an imprint, no doubt, of their idle life, but it was also the imprint of an exquisite training. It was the quintessence of the aristocratic manner.

We do not want gilded uselessness; far better remain as we are; busy, even if chiefly for ourselves; but there are some who, in the midst of hard work, have been wise enough to leave spare time in their plan of life for casual meetings with their fellows. I think of two women, both, as it happened, the main stay of churches—churches widely removed in geography and opinion; both were responsible in the care of those churches, and both entered with marvelous industry into the details and into the drudgery of their work. I heard a servant say of one that she had never known her to utter an impatient word when cards were handed in; while her cordial ways won the hearts of all. The other, older and poorer, and not

versed in the ways of the world, it was even more wonderful to see sitting hour after hour engrossed and pleased as one and another used up her morning; though the standing puzzle among all who knew her was, how she possibly found time for all she did.

This heartiness is more attractive to the small than to the great among one's friends; for those full of attractions themselves are not so apt to be received with indifference. It is those modest as to their own fascinations who are grateful for being made welcome and at home; the shy girl just emerged from the school room to do her difficult part in society, the old lady who is painfully conscious that she knows very little about the topics of the day, the estimable young man of moderate capacity. These must be tolerated—why should they not be made happy?

I know that I am playing with edged tools, advising an attempt that is all too apt to end in affectation, even in deceit, an overdone aspect of admiration which would disgust; but you can avoid that. Really, about all you need is to deal strictly with yourself as you walk down stairs to the parlor; force yourself to stop inwardly fuming that you are interrupted, to accept the fact that this is not an interruption, but probably a better occupation for you for twenty minutes than that from which you were called, at any rate the inexorable duty of the moment, and to be discharged as such. Instead of sitting half numb and self-absorbed, talking only in the *rôle* of a machine for receiving a caller, try at once to throw yourself quite away from the life upstairs out into the life of your visitor. The calls will consume a little more of your time, will average a little longer, even in spite of waiting carriages, but they will prove a recreation instead of a weariness, and you will gain twenty friends where you now gain one.

A little girl once traveled in the stage-coach with Madame La Vert, and the child's indignant contradiction afterward: "She isn't a fine lady at all! She's just like me, and I love her," made plain in a sentence the secret of her power.

Why is not this sympathetic treatment of others aimed at by all? Why should the reigning belle be so nice with her tongue and her smile, and the older woman, who probably feels the want of affection far more keenly, be quite oblivious that this same winning kindness which she has so long

ceased to exhibit, and has replaced by the dry manner of formal endurance, would, perhaps as promptly as the wand of Cinderella's god-mother, transform some of these prim callers into life-long friends.

OUR LADIES OF SORROW.

BY MRS. E. A. MATTHEWS.

Three sisters guard the lives of mortal men
With care unvarying, from youth to age.
Ever around us, though beyond our ken,
Do they in silent ministry engage.
Mysterious Trio—hand in hand, they go
Through the sad realm of human pain and woe!

Mother of Tears, of bitterness, and grief,
Thou art, to-day, in many a silent room—
Before thy steps, death stealeth, like a thief,
And turns life's sunshine into blackest gloom.
When hearts are rent with unavailing prayer
For life's lost treasures, Mother, thou art there!

Dark-browed and dreadful, Tenebrarum, thou!
Bringer of unbelief, doubt, and despair,
Mother of suicide, revenge, and gloom—
With frenzied look, and maniac's awful glare—
When man is hated, and when God denied,
Thou standest, mocking, at the wretch's side!

But thou, O Goddess, mild Mother of Sighs,
Sweet source of pity, patient sorrow's balm,
At thy mild bidding all our anguish dies,
And grief's wild billows soften into calm.
When human hearts bring sympathy and share
The woes of others, Mother, thou art there!

When the sad penitent laments in vain
O'er wasted moments—thou dost hope restore—
To the pale captive, and the child of pain,
Thou bringest liberty and health once more!
All generous deeds and tender charities
Are in thy hands—O, blessed Mother of Sighs!

THE NICARAGUA AND PANAMA ROUTES TO THE PACIFIC.

BY FELIX L. OSWALD, M. D.

It is curious how often the first impression of experts has been confirmed by the verdict of posterity. During one of his journeys of inspection Peter the Great visited the mouth of the Volga River and pointed out the advantages of fortifying a certain promontory which a military commission, after long controversies, has now selected for the site of a bombproof arsenal. The gold discoveries in Upper California were predicted by Sir Francis Drake; those of the Ural by Baron Humboldt; and after fifty years of coast surveys, mountain surveys, negotiations, reports and counter reports, it seems now probable that two American Republics will ratify the opinion of Don Rodrigo Contreras, who more than three hundred years ago called the attention of the Spanish Government to the advantages of the Nicaragua Lake system, and its navigable effluent, as the rudiment of an inter-oceanic canal, and the superiority of that route over those both of Darien and Panama—Tehuantepec having then not yet entered into competition. This Rodrigo was the son-in-law of Davila Pedrarias, the first governor of Panama, and in his coasting trips between the landings of the southern isthmus had probably noticed a circumstance which may yet turn the scales in the decision of the canal problem, though it had escaped the attention of the routine sailors of a latter age, and perhaps even of some professional engineers who confined themselves to the comparison of altimetrical surveys.

The matter is this: Along the north shore of the Caribbean Sea the coasts are deep and rocky, but further south, as the Cordilleras decrease in elevation, the shore is lined with sandbanks, and can be approached only through shallow estuaries, just as the coasts of the Mediterranean become sandy at their southeastern extremity, the only point where the circle of lofty coast ranges is broken by the delta of a swamp river—the depth of the

shore waters being apparently proportioned to the height of the shore lands. Even single depressions in the chain of a long-stretched coast range are often confronted by isolated sandbanks—as if a collapse of the mountain walls had shoaled the littoral sea. But at Panama that tendency is aggravated by another cause. For the last two hundred years the line of the overland route has followed the Rio Chagres and its western tributaries; the adjoining hill country became studded with settlements, and, as usual in Spanish colonies, “civilization” led to forest destruction and progressive aridity. The abundant rains of the summer solstice, which were formerly absorbed by the not less exuberant vegetation of a tropical coast region, now reach the valley in the form of raging torrents, saturated with the mould of the treeless hill-slopes; and the alluvium deposited at the river mouth has thus gradually formed silt banks, against which the repeated improvements of the estuary have proved only a temporary remedy. The Russians had a similar experience with the port of Azof, once the best harbor in the basin of the Euxine; but after the destruction of the inland forests the naked hills of the Don revenged themselves by a hydra growth of dunes that defied all expedients of human skill, and after diking and dredging away some ninety million roubles, the government yielded to Nemesis and removed the wharves to the harbor of Taganrog. In order to enable vessels of deeper draught to enter the mouth of the Chagres, the canal itself would have to be supplemented by a *channel mole*, a marine canal of nine miles, protected by dikes which in their turn might become a source of peril to sailing vessels approaching the estuary during the prevalence of the frequent gales which make the headlands of the southern Caribbean so many Guardafuis.

The Bay of San Juan de Nicaragua, on the contrary, is remarkably free from storms, and the San Juan is not a swamp river. It is the effluent of a chain of rockbound lakes, one of them of sufficient extent to equalize the drainage of torrents from above, so that an overflow of the San Juan could be caused only by the simultaneous rising of all its lower tributaries, aided by a northwest gale that would drive the waters of the lake toward the estuary. A conjunction of that sort occurred in the summer of 1872, when the San Juan rose some twenty feet in as many hours; but even then the river did not shoal its delta, but tore out a new channel through Costa Rican territory, which now receives a lion’s share of the outflow; but there is no doubt that an isolated mishap of that sort can be remedied by a short-line

canal from the coast to a point above the divergence of the rival streams, or by the same means that reclaimed the channel of the Mississippi at New Orleans. Further up all serious difficulties cease. The rapids of the San Juan can be passed by locks of moderate depth, the total difference between the level of Lake Nicaragua and that of the Atlantic being hardly twenty-five feet. The lake itself will need no improvements. Throughout its breadth of seventy-five miles (three fifths of the distance from ocean to ocean) it maintains an average depth of fifteen fathoms, has no dangerous sandbanks, no driftwood islands, while its undercurrents secure it against the danger of being shoaled by the floods of its affluents. The project of locating the western terminus at Port Brito, on the Pacific, would reduce the length of the canal proper to about twenty-eight miles, and shorten the trip for vessels from our ports by nearly seven hundred miles—the distance from Port Brito to Panama.

The question is, if all these advantages would justify the construction of a second canal. For we need not delude ourselves with the fear—or hope—that the Panama project would be abandoned; the *present* interests of the French stockholders, if not the name of Monsieur de Lesseps, is a guarantee that their work will be completed. They have gone too far[A] to retreat before the completion of a single rival; for on the other hand it is equally sure that the Tehuantepec scheme has lost its last hope of practical support, since even the promise of a monopoly could hardly override the veto of Nature. It is enough to say that the Mexican projectors admit that the proposed route through the lowest gap of the Chimalapa Range would require the construction of *one hundred and forty locks*—with the alternative of tunneling through thirty miles of granite rocks—for we know how nearly a difference of fifteen locks defeated the chances of the Wellington canal against its American rivals.

Nor should we, in this respect, underrate the advantages of the Panama route. The distance from ocean to ocean is only forty-eight miles, against one hundred and fifteen at Nicaragua, and two hundred and twelve at Tehuantepec, and neither hills nor rocks oppose any obstacle that could not be overcome by the conqueror of Suez. Yet there remains a consideration which in connection with the result of other comparisons must leave a preponderance of arguments in favor of Nicaragua. For three fifths of the year the valley of the Rio Chagres is a reeking hotbed of malaria,[B] and

from Aspinwall to Panama only dreary sandhills alternate with festering swamps. Nicaragua, on the other hand, is the healthiest region between Chili and California, and offers scenic attractions absolutely unequaled in any other part of the New World, and which can hardly have been surpassed when the Mediterranean peninsulas were still clad in the glory of their primeval forests. I visited the lake region of Nicaragua in 1875, after having seen California, the West Indies, and the highlands of Mexico, and only here I relinquished my doubt that the international park of Switzerland had, after all, a transatlantic rival. The harbor of San Juan is nearly a thousand miles from anything an Anglo-American would call civilization, and since their Walker and Spalding experiences the natives are, on the whole, rather inclined to dispense with the patronage of enterprising foreigners; but, for all that, the time is near when the islands of Lake Nicaragua will be studded with international hotels. The luxuriance of the tropical hill forests is equaled in the mountains of Oaxaca; the climate is not superior to that of the northern Antilles; but the scenery of the highlands, with their fourteen active volcanoes, would turn the scales. Of the forty-six craters in the maritime Cordilleras, five are eruptive and nine smouldering and smoking volcanoes, besides a score of *infernillos*—smoking caves and fissures, opening in the rocks of the foothills, as if the obstruction of the roof-chimneys had forced the subterranean fires to break a vent through the basement of the Sierra. And, moreover, these craters are, with few exceptions, of easy access from both sides of the mountains, and at all seasons of the year. The climate of Nicaragua enjoys the advantages of the real tropics. In the western hemisphere that term is as ambiguous as the “beginning of spring.” At Brownsville, Texas, nearly twelve hundred miles further south than Naples, I have seen snowstorms that would appall a Shetlander; and even in Mexico frosts are not confined to the upper tablelands. But the traveler who reaches the fourteenth parallel has left the winter zone behind. In Leon, near the northwest end of the lake system, he will find butterflies in December, and when he sits down to his Christmas dinner of sweetmeats and broiled bananas, mine host will introduce a *caza-moscas*, or fly-brush boy. *Moscas* of a larger variety also infest the lagoons of the old town, but mosquitoes are rare, and further inland unknown, for Leon is the gate city of the western highlands, one of its suburbs being only a short distance from the foothills of the Cerro de las Pilas, a naked peak whose summit affords a complete panorama of the lake region and the coast

range. In the north the Sierra Madre of Honduras looms on the horizon like a white-crested cloud, flanked by the blue ridge of the coast range, and the volcanic domes of the central plateau which attains its maximum altitude in the uplands of Matagalpa. But forty miles southeast of Leon the great tableland which stretches in an almost unbroken line from Denver across Mexico and the highlands of Central America is intersected by the basin of the Isthmus lakes; and here, as if the disruption of the Sierra had opened a vent for the furnaces of the nether world, volcanoes and hot springs are massed in a way which to my knowledge is paralleled only in the Island of Java, where a highland region of twelve hundred square miles is veiled with the almost perpetual smoke of its burning mountains. About twenty miles north of Las Pilas the continuity of the coast range is broken, and the main chain seems to be segregated into numerous isolated mountain groups, each of them crowned with two or three volcanic cones. At the end of the cape which forms the breakwater of the sound known as the Estero Real stands the volcano of Consequina, the Vesuvius of the New World. In one of its last eruptions it caused an inundation by completely obstructing the outlet of the Rio Casco, and in 1835 it scattered its ashes as far as La Guayra and Tehuantepec, *i. e.*, over a circle fourteen hundred miles in diameter. The head of the bay is begirt by the dolerite cliffs of two other volcanoes, one of which fronts the shore with a sheer precipice of 4,400 feet, rising from a pedestal of rock colonnades with cavernous interspaces where trickling springs have generated a rank vegetation of climbing evergreens. The same arrangement repeats itself at the head of Lake Managua, where the volcano of Monotombo rises in tower-like basalt cliffs to a vertical height of 7,400 feet. Professor Vanhouten, of the canal survey, makes the highest point of the peak 7,700—at all events a thousand feet more than the summit of Mount Washington. The “South Dome” of the Yosemite, if my memory serves me right, rises hardly 5,000 feet above the valley, and the granite front of the “Captain” considerably less than 4,000, while the grandeur of the basalt coliseum is yet doubled by the mirror of the Island Sea, as the first colonists translated the Indian name of Lake Managua. The lake is dotted with wooded islands which accompany the south shore into the strait of Panalaya, and where that strait opens into the broad basin of the lower lake, the islands, too, expand, two of them so much, indeed, as to have mountain ranges, broad valleys and secondary lakes of their own. Like all the higher ridges those island mountains culminate in volcanoes; the steep

peak of Ometepe rises to a height of 5,200 feet, and gives its island the form of a pyramid. West, east, and north of both lakes volcanoes rise above the shore mountains in all directions, and northeast of Lake Managua not less than seven of them stand close together, like the peaks of the Sieben-gebirge, on the upper Rhine. The volcano of Chiltepec can be recognized by its cloven peak, the upper part of the cone having collapsed during one of its eruptions; Tellica and Santa Clara by the smoke trails of their ever-seething craters. The "Hell of Mesaya," sixty miles further south, has the deepest crater of any known volcano, as Kilauea has the widest, and Stromboli the most obstreperous. The upper two thousand feet of the cone form a mere shell, and from the brink of the crater the spectator looks down into a dizzy abyss of trachite cliffs, rent by vertical fissures, and now and then veiled by the eruption of a smoke whirl. From the hills on the east shore of the lake the prospect toward the peaks of the island mountains affords views of marvelous and almost incomparable beauty. Yet all along the shores of that lake land, with the exception of the richest alluvial creek estuaries, can be bought for three dollars an acre, even on flat topped bluffs and near rock springs in sheltered dells, more inviting than the finest artificial parks. Thus far only camping British sportsmen have now and then availed themselves of such opportunities, though Nicaragua offers hunting grounds to the lovers of nearly every kind of outdoor pastime and science. Geologists can watch the active operations of the Titans that have transformed so large a portion of our planet. Antiquarians can here find treasures, from the tomb of a Toltec warrior to the monuments of a buried city; the neighborhood of San Carlos and Mayagalpa is covered with ruins, and on the island of Zapotera colossal idols rise like sphinxes from the ground and from the debris of fallen temples. Naturalists can watch the jaguar and the tapir in their native haunts, catch butterflies enough to stock a museum, study the habits of five different species of quadrupeds, including the large Coati (*Lagothrix paniscus*), with his spider arms and strange, flute-like cries, can visit the bat caves of Dipilta or the bird islands of Bluefield's Lagoon, that make Nicaragua a zoological epitome of the American tropics, and offer a winter asylum to legions of feathered refugees from the North.

In Nicaragua the rainy season ends in November, and the winters are dry and pleasant, and far less capricious than in Florida, where sultry Gulf

winds so often alternate with very perceptible polar waves. Nicaragua will become the winter rendezvous of American tourists. An international highway, built by American engineers and American capital, would soon aggregate North American settlements that would add to the charms of a tropical garden-land a feeling of security which the traveler in other parts of Spanish America must generally forego; villas, watering places, and hotels for the accommodation of winter guests will spring up by scores, and the canal will soon become something more than a thoroughfare of international travel. The stockholders of such an investment could well dispense with a formal protectorate, and the obstacle of the Clayton-Bulwer treaty[C] is precisely analogous to the scruple which no politician would blame Prince Bismarck for setting aside in the Congo affair. In both cases England had for years the refusal of an opportunity which she failed to improve, and in the unwritten by-laws of international ethics the *iners non obstat* is a rule which held good even in the Black Sea controversy. Besides, the alternative of a French monopoly and eventual protectorate in Panama may help to moderate the opponents of the Monroe Doctrine.[D]

While Nicaragua will be visited for its own sake, Panama, like Vera Cruz and Cayenne, will be shunned by northern travelers, though considerations of proximity will secure it the inter-oceanic traffic of South America. Vessels carrying tobacco from Brazil to Peru, or guano from Peru to Caracas, will prefer the risk of the Chagres fever swamps to the certain disadvantages of a voyage around the distant south cape of the continent.

Nicaragua will compete with Switzerland, Italy, and the winter resorts of southern California, as well as with our transcontinental railways. Panama will compete with Cape Horn.

[A] The *Panama Star and Herald* reports that \$68,000,000 of the total appropriation of \$120,000,000 have already been expended. The preliminary surveys alone cost 7,000,000 francs. After the plan of Monsieur de Lesseps the canal will be thirty feet deep, one hundred and twenty feet wide, and about forty-six miles long.

[B] Panama (the city), too, is one of the unhealthiest spots on earth. Two months ago (January 2, 1885) the wife of Monsieur Jules Dingler, the Director-General of the Canal Company, succumbed to the same fever that had cost the lives of all her children. Dr. Ferdinand Lahr, the Sanitary Superintendent, died on the same day.

[C] Captain Bedford Pim, of the British navy, states various reasons that would insure the tolerance, or even the coöperation of Great Britain. He estimates the aggregate cost at \$200,000,000, and believes that England, under certain conditions, would assume one half of the demanded guarantee of three per cent. interest.

[D] It is a curious fact that the originator of that doctrine is not an American, but a British statesman. George Canning, on the eve of his departure for Verona, having got an inkling of a South-European federation for the purpose of assisting Spain in the reconquest of her lost transatlantic possessions, postponed his trip and put himself in communication with the American minister, in order to call the attention of our government to the favorable opportunity for asserting the claims of a counter alliance.

GEOGRAPHY OF THE HEAVENS FOR JUNE.

BY PROF. M. B. GOFF,
Western University of Pennsylvania.

THE SUN.

“’Tis by thy secret, strong attractive force,
As with a chain indissoluble bound,
Thy system rolls entire; from the far bourne
Of utmost Saturn, wheeling wide his round
Of thirty years, to Mercury, whose disk
Can scarce be caught by philosophic eye,
Lost in the near effulgence of thy blaze.”

Had our poet written his *Summer* in 1781, the year in which Uranus was discovered, instead of 1727; or could he have waited till 1846, his “utmost Saturn wheeling wide his round of thirty years” would probably have been changed and in beautifully flowing verse would have been expressed the wonderful fact of “utmost Neptune, wheeling wide his round, whose years could only be by four and sixty and one hundred told.” So much in one respect had astronomy grown in a little more than a century. But we could have no heart to blame our poet’s neglect of our “utmost” planets, even had he known of their existence, since he was so evidently in earnest in giving its due to our glorious orb, recognized to-day as the source of all our light, and heat, and life.

“From brightening fields of ether fair disclosed,
CHILD of the SUN, refulgent Summer comes,
In pride of youth, and felt through Nature’s depth;
He comes, attended by the *Hours*
And ever-fanning *Breezes*, on his way;
While from his ardent look, the turning *Spring*
Averts her blushful face; and earth and skies,
All-smiling, to his hot dominion leaves.

...

“When now, no more the alternate Twins are fired,
And Cancer reddens with the Solar blaze,
Short is the doubtful empire of the Night;
And soon, observant of approaching Day,
The meek-eyed Morn appears, mother of Dews,
At first faint-gleaming in the dappled East;
Till far o’er Ether spreads the widening glow;
And, from before the luster of her face,
White break the clouds away.”

Had our poetic friend, Thompson, lived in Meadville or New York, instead of London, he would not even on the 21st of June, have been so much disposed to sing:

“Short is the doubtful empire of the Night;”

for the shortest night of New York is over an hour longer than the shortest night of London. But what were Meadville and New York in 1727? June 21st is our longest day, in latitude 41° 30', a little more than fifteen hours from sunrise to sunset; the night, of course, a little less than nine hours. On this day the sun reaches his northern limit, 23° 27' 3.2", and at 2:42 a. m. begins his southern journey; or, to put it astronomically, the sun enters *Gemini*, and summer begins, on June 21st, at 1:42 a. m. On the 1st, sun rises at 4:31 a. m., sets at 7:24 p. m.; on the 16th, rises at 4:29 a. m., sets at 7:32

p. m.; and on the 30th rises at 4:33 a. m., sets at 7:34 p. m. Many persons judging from the temperature will be inclined to think on the 21st that the sun must be at midday directly above their head; but it will have an elevation of not more than 73° .

THE MOON

Enters upon its last quarter on the 5th, at 6:56 p. m.; new moon on the 12th, at 5:34 p. m.; first quarter, 19th, at 8:40 a. m.; and full on the 27th, at 6:09 a. m. Farthest from the earth on the 28th, at 12:54 a. m.; nearest the earth on the 13th, at 11:12 a. m. Greatest elevation, $66^{\circ} 56' 25''$, on the 13th; least elevation, $30^{\circ} 2' 40''$, on the 27th. Rises on the 1st, at 10:15 p. m.; sets on the 16th, at 10:41 p. m.; and rises on the 30th, at 9:27 p. m.

MERCURY

Will be our morning star till the 27th, after which it will be evening star till the end of the month. On the 1st it will rise at 3:34 a. m., or about one hour earlier than the sun, and can probably be seen by good eyes. On the 16th, it will rise at 3:46 a. m.; and on the 30th, will set at 7:51 p. m., or a few minutes after sunset, but will be too near the sun to be visible. Its motion will be direct, amounting to $59^{\circ} 50'$. Its diameter will decrease from 7" to 5". On the 5th, at 2:00 p. m., will be $48'$ south of Neptune; on the 11th, at 10:57 a. m., $2^{\circ} 57'$ north of the moon; on the 23d, at 11:00 p. m., $1^{\circ} 41'$ north of Saturn; on the 24th, at 3:00 a. m., in perihelion (nearest the sun); and on the 27th, at 10:00 a. m., in superior conjunction with the sun.

VENUS,

Although she acts the part of an evening star, is rather "mild" this month. She rises after the sun, and sets on the 1st, at 7:56; on the 16th, at 8:20; and on the 30th, at 8:33 p. m. Her diameter increases four tenths of a second of arc; and she makes a direct motion of $40^{\circ} 3'$. On the 7th, at 5:00 p. m., she is $1^{\circ} 32'$ north of Saturn; on the 13th, at 10:54 a. m., $5^{\circ} 48'$ north of the moon; and on the 26th, at 3:00 p. m., in perihelion.

MARS

Has a direct motion of $22^{\circ} 6'$; his diameter remains constant, 4.4", during the month. Can be seen at the early dawn. He rises at the following times: On the 1st, at 3:22; on the 16th, at 2:59; and on the 30th, at 2:33 a. m. On the 10th, at 6:00 p. m., $1^{\circ} 29'$ north of Neptune; on the 10th, at 7:55 p. m., $3^{\circ} 51'$ north of the moon.

JUPITER

"Speaks for himself" every evening this month. On the 1st, he rises at 10:34 a. m., and on the 2d sets at 12:06 a. m.; on the 16th, rises at 9:50 a. m., and sets at 11:16 p. m.; and on the 30th, rises at 9:03 a. m., and sets at 10:23 p. m. He has a direct motion of $4^{\circ} 2' 31''$; and his diameter decreases from 34" to 31.6". On the 17th, at 9:44 a. m., he is $3^{\circ} 44'$ north of the moon. It may interest the general reader to know that while he is admiring, night after night, this beautiful body making its way through the "lesser lights" of the heavens, that the astronomer is laboring diligently to discover its properties and learn with exactness its motions. From the last report of Prof. G. W. Hough, Director of the Dearborn Observatory, Chicago, we find that "the disk of Jupiter was observed on every favorable occasion, and micrometric measures made on the principal spots and markings, including the great red spot first remarked in 1878." These observations were made principally with a view to obtaining the time of revolution of the planet on its axis; and the result of the observations from September 12, 1883, to June 11, 1884, a period during which the planet made 660 revolutions, was a mean of 9h. 55m. 38.5s., which differs from the mean of five years' observation by 1.5s.; the former mean being that much greater than the latter.

SATURN

Will be evening star till the 18th; after which date, morning star till the end of the month. On the 1st, rises at 5:40 a. m., and sets at 8:24 p. m.; on the 16th, rises at 4:47 a. m., sets at 7:31 p. m.; on the 30th, rises at 4:00 a. m., sets at 6:44 p. m.; from which it will be seen that except during the first part of the month it will be invisible to the naked eye. Its diameter is 15.6" and does not change to the amount of one tenth of a second during the month. On the 7th, at 5:00 p. m., $1^{\circ} 32'$ south of Venus; on the 13th, at 1:19 a. m., $4^{\circ} 3'$ north of the moon; on the 18th, at 6:00 p. m., is in conjunction with the

sun; on the 23d, at 11:00 p. m., $1^{\circ} 41'$ south of Mercury. This planet has also been the subject of observation from the Dearborn Observatory, for the purpose of detecting markings on the rings; but “nothing indicating a division in the outer ring has ever been noticed.”

URANUS.

On the 1st, this planet rises at 1:10 p. m.; on the 2nd, it sets at 1:18 a. m.; on the 16th, rises at 12:12 p. m., and on the 17th, sets at 12:18 a. m.; on the 30th, rises at 11:18 a. m., and sets at 11:24 p. m. Decreases in diameter two tenths of a second; and has a direct motion of $14'' 30'$. On the 5th, at 1.00 p. m., it appears stationary; on the 19th, at 10:15 a. m., is $55'$ north of the moon; on same date, at 10:00 p. m., is 90° east of the sun.

NEPTUNE

Is a morning star throughout the month, and has a direct motion of $1^{\circ} 0' 16''$. Diameter increases from $2.5''$ to $2.6''$. It rises as follows: On the 1st, at 3:47; on the 16th, at 2:49; and on the 30th, at 1:56 a. m. On the 5th, at 2:00 p. m., is $48'$ north of Mercury; on the 10th, at 6:00 p. m., $1^{\circ} 29'$ south of Mars; on the 10th, at 7:49 p. m., $2^{\circ} 21'$ north of the moon. The 26-inch equatorial of the National Observatory at Washington, D. C., was during the past year chiefly employed in observations of the satellites of Neptune, Uranus, Saturn and Mars.

HOW TO WIN.

BY FRANCES E. WILLARD,
President National W. C. T. U.

CHAPTER IV.

Thus far I have been trying to impress upon you the reasons why you should cultivate individuality and independence in word and deed. I have claimed that each one of you has a “call” to some specific work, indicated by God’s gifts to you of brain, or heart, or hand. But I would not have you only, or indeed chiefly, concerned with the evolution of your powers for your own sake. If you acquire, let it be that you may dispense; if you achieve, that others may sun themselves in the kind glow of your prosperity. The people who spend all their strength in absorbing are failures and parasites. It is alike the business of the sun and of the soul to radiate every particle of light that they can muster. There is reason to believe that this is precisely what they are for. And so, having made sure of your light, strength and discipline, strike out from the warm and radiant center of a self-poised brain and heart, into the lives about you, and you will find that “What is good for the hive is also good for the bee.” The luminous characters of history have done this, always. Losing their lives in those of other men, they have found them in the crest of the world’s gratitude and fame. What they have done on a grand scale, we, from identical motives, may do on a small one. Such natures are as different from those who cultivate their strongest gift simply for their own sake, as a *lighthouse* is different from a *dark lantern*. “Self-culture” is much in vogue nowadays, and has for its high priests some of the most incisive minds of this or any age. But self-culture stops in the middle of the sentence I would fain help you to utter. It says: “Make the most of your powers;” it does not say “*for others’ sake* as well as your own.” It claims that if we set the candle of our gifts upon the candlestick of modern society, its light will inevitably radiate according to its power of shining, and thus while brightening ourselves we shall have

done our utmost toward lighting up the general gloom. But self-culture forgets that a candle is no type of you and me. We are human spirit-lamps, whose rays should be directed and intensified by the blow-pipe of an unceasing purpose; for we are all so made that unless we *will* to light up other lives, we can never do so to the limit of our power. Self-culture is never base; it is often noble, but it can never be the noblest aim of all.

Why is the memory of Mrs. Browning loved beyond that of almost any poet who has sung? Because “the cry of the human” is so strong in that wondrous voice of hers. Why is the name carved deepest on the republic’s heart that of its martyr President? Because he gave their manhood back to four millions of slaves, and lived and toiled for his people’s sake, “with malice toward none, with charity for all.” Why was the lamentation well nigh universal when under the sea flashed the telegraphic message, “John Stuart Mill is dead?” Because this quiet thinker lived for other men; because he “struck out from the center,” from himself, that pitiful pivot on which so many human wind-mills turn, and measured, in the swift flight of its benignant thought, the long radius between him and the remotest circle of human need; because, more than any other philosopher of his day, he labored for the time when “all men’s weal shall be each man’s care.”

Nay, while I mourn, as I have seldom mourned for an historic character, the cloud that early dimmed, for Stuart Mill, the Star of Bethlehem, I will not, as a woman, withhold from his memory the tribute of my humble gratitude. But while I speak of all these lives, shining like beacon lights of our own day, I would not fail to point you in conclusion toward a wide-armed cross upon a lonely hillside, while I repeat his words who said, “And I, if I be lifted up, will draw all men unto me.” Dear girls, CHRIST is the magnet of humanity, and she has found the best vocation, and the highest, who brings most souls diseased within the healing power of his immortal gospel. This is a work for which women have gifts preëminent. The Saxon word for lady means “a giver of bread,” and is full of beautiful significance, but America’s new century shall evolve another meaning, freighted with greater blessing for humanity: lady, giver of the bread of life! In later years we have had a revelation of our duty to the ungospelled masses, the “elbow heathen,” as an evangelist has called them, to the intemperate (who, as a rule, are quite beyond the hearing of the pulpit’s voice), and to the dusky dwellers in the Zenana, whose faces are misty with the unshed tears of

generations passed in misery and shame. Two thirds of the Church of Christ are women. By the freer life and richer opportunity which you and I enjoy; by society's growing tolerance, not to say its kindly appreciation, of our activities; by the heart transformed and the peace imparted through the gospel, the voice of our Redeemer pleads for our consecrated service. I would not undervalue the culture of the intellect, but would exalt the culture of the heart.

In all that has been written until now, I have simply tried to outline the new horizon opened up to the gentler sex as the supreme outgrowth of that civilization which He introduced, of whom history records the significant fact that women were "last at his cross, first at his sepulchre." To attempt some delineation of the landscape enclosed by that far-reaching horizon is my more difficult task in the chapters now to follow. Many letters have come to me as a result of the articles thus far; they bristle with questions and are eloquent with aspiration. Later on I may ask space for a "symposium" with these "inquiring friends" in the genial pages of our tolerant CHAUTAUQUAN. One speaking with authority exhorts me to "Be practical—that's what we want!" As if I hadn't been! But every mind is the prisoner of its own material and method. I can but give of such things as I have; can only tell what life has told to me. According to my own habits of thought, the sequence seems logical, when I turn for a while from the presentation of the modern outlook for women, with its opportunity and hope, to the *rationale* of this new horizon stretching so far away. Let us note the pathway that has led up to this more hopeful point of view, asking the inevitable question, "Why does that seem natural and fitting for a young woman to do and to aspire to now, which would have been no less improper than impossible, a hundred years ago?" Sweet friends, it is because *the ideal of woman's place in the world is changing in the average mind*. For as the artist's ideal precedes his picture, so the ideal woman must be transformed before the actual one can be. In an age of brute force, the warrior galloping away to his adventures waved his mailed hand to the lady fair who was enclosed for safe keeping in a grim castle with moat and drawbridge. But to-day, when spirit force grows regnant, a woman can circumnavigate the globe alone, without danger of an uncivil word, much less of violence. We shall never span a wider chasm than this change implies. All our inventions have led up to it, and have in nothing else

wrought out beneficence so great as they have accomplished here, purely by indirection. In brief, the barriers that have hedged women into one pathway and men into another, altogether different, are growing thin, as physical strength plays a less determining part in our life drama. All through the vegetable and animal kingdoms the fact of sex does not widely differentiate the broader fact of life, its environment and its pursuits. Hence, the immense separateness which sex is called in to explain when we reach the plane of humanity, is to be accounted for largely on other purely artificial grounds. In Eden it did not exist, nor in the original plan of creation, as stated in these just and fatherly words: "And God said, 'Let us make man in our own image, after our own likeness. Let them have dominion.' ... So God created man in his own image, in the image of God created he him, male and female created he them, and God blessed them, and said unto them, '... replenish the earth and subdue it ... and have dominion over every living thing.'" After the fall came the curse, which was no part of the original design, and from which the gospel's triumph is releasing us, for there is "neither male nor female in Christ Jesus." I believe that the origin of evil came in with man's supremacy over one who was meant to be his comrade, and that Paradise regained will come only when the laureate's prophecy is realized:

"Two heads in council, two beside the hearth,
Two in the noisy business of the world,
Two in the liberal offices of life;
Two plummets dropped to sound the abyss of science
And the secrets of the mind."

The times when a new ideal is moulded, in church, state, or society, mark the epochs of history. Amid what throes did Europe pass from that of supreme authority in the church to the incomparably higher one of supreme liberty in conscience; from the divine right of kings to the divine right of the people. But there was to come a wider evolution of the same ideal, namely, the coequal power of the copartners, man and woman, in working out the problem of human destiny. This newest and noblest of ideals marks the transition from physical force ruling, to spiritual force recognized. The gradual adjustment of everyday occupation, custom and law, to this new

ideal, marks ours as a transition period. Those who have the most enlargement of opportunity to hope from the change, will, in the nature of the case, move on most rapidly into the new conditions, and this helps to explain, I think, why women seem to be climbing more rapidly than men, to-day, the heights of spiritual power, with souls more open to the “skyey influences” of the oncoming age.

More women study to-day than men; a greater proportion travel abroad for purposes of culture; a larger share are moral and religious. Half of the world’s wisdom, three fourths of its purity, and nearly all its gentleness, are to-day to be set down on woman’s credit side. Weighted with the alcohol and tobacco habits, Brother Jonathan will have to make better time than he is doing now, if he keeps step with Sister Deborah across the threshold of the twentieth century. For the law of survival of the fittest will inevitably choose that member of the firm who is cleanliest, most wholesome, most accordant with God’s laws of nature and of grace, to survive. To the blindness or fatuity which renders him oblivious of the fact that the coming woman is already here, our current writer of the W. D. Howells and Henry James school owes the dreary monotony of his “society novel.” Not more “conventional” was the style of art known as “Byzantine,” which repeated with barren iteration its placid and colorless “type,” than are the dudesque pages of this pair of literary martinets. The “American novel” will not be written until the American woman, a type now to be found in Michigan, Madison, Boston, Cornell, and other universities, shall have taken her place, twentieth century product that she is, beside the best survivals of young men in similar institutions, and wrought out the Home, the Church, the State that are to be. Measuring each other on all planes, these life partners will know each other’s value, and no appeal to the divorce court will be made to relieve them, a few years after marriage, from an incompatibility that has ripened into open war. Happy homes will dot the country from shore to shore, in which both the man and the woman will do their best to lift the world toward God.

“Self-knowledge, self-reverence, and self-control; these three alone lead life to a sovereign power,” and these are fast becoming essential to any ideal of womanly character which the modern age will recognize as the product of its institutions. Of self-knowledge, these talks have said much. Self-reverence I would fain help you to develop in your character as a

woman. If my dear mother did me one crowning kindness it was in making me believe that next to being an angel, the greatest bestowment of God is to make one a woman. With what contempt she referred to the old Jewish formula in which the less refined sex rolled out the words, "I thank thee, O God, that thou hast not made me a woman," and with what pathos she repeated the gentle prayer of the other, "I thank thee, O God, that thou hast made me as it pleased thee," with the pithy comment, "What could have pleased Him better, I should like to know, than to make one so rare, so choice, so spiritual as woman is?" Perhaps some of you may have thought you wanted to be a boy, but I seriously doubt it. You may have wanted a boy's freedom, his independence, his healthful, unimpeding style of dress, but I do not believe any true girl could ever have been coaxed to be a boy. Reverence yourself, then, if you would learn one of the first elements of "How to Win" in this great world race, with its "go-as-you-please" terms, but its "Lucifer-may-take-the-hindmost" penalty for failure.

What will the new ideal of woman *not* be? She will never be written down in the hotel register by her husband after this fashion: "John Smith and Wife." He would as soon think of her writing "Mrs. John Smith and Husband." Why does it not occur to any one to designate him thus? Simply because he is so much more than that. He is a leading force in the affairs of the church; he helps decide who shall be pastor. (So will she.) He is perhaps the village physician, or merchant (so she will be, perhaps—indeed, they are oftentimes in partnership, nowadays, and I have found their home a blessed one). He is the village editor. (Very likely she will be associate.) He is a voter. (She will be, beyond a peradventure.) For the same reason you will never read of her marriage that "the minister pronounced them *man* and *wife*" for that functionary would have been just as likely to pronounce them "husband and woman," a form of expression into which the regulation reporter will be likely to fall one of these days, it being, really, not one whit more ridiculous than the time-worn phrase, "man and wife." The ideal woman of the future will never be designated as "the Widow Jones," because she will be so much more than that—"a provider" for her children, "a power" in the church, "a felt force" in the state. I think George Eliot is the first woman to attain the post-mortem honor of having her husband called "her widower," John W. Cross having been thus indicated in English papers of the period. A turn about is fair play, and the phrase is really quite

refreshing to one's sense of justice. The ideal woman will not write upon her visiting card, nor insist on having her letters addressed, to Mrs. John Smith, or Mrs. Gen. Smith, as the case may be, but will, if her maiden name was Jones, fling her banner to the breeze as "Mrs. Mary Jones Smith," and will be sure to make it honorable. She will not be the lay figure made and provided to illustrate the fashions of Monsieur Worth and lesser lights of the same guild, but will insist that the goddess Hygiea is the only true modiste, and will dutifully obey her orders. As the Louvre gallery proves that when men were but the parasites of the court they too decked themselves with ear-rings, high heels, powdered hair and gaudy garments, so the distorted figures in the detestable fashion plates of to-day are the irrefutable proofs of woman's fractional estate; but this will not be so to-morrow, when she finds her kingdom—which is her own true self. The ideal woman will cease to heed the cruel "Thus far and no farther," which has issued from the pinched lips of old Dame Custom, checking her ardent steps throughout all the ages past, and will be studious only to hear the kindly "Thus far and no farther" of God.

The ideal woman will play Beatrice to man's Dante in the Inferno of his passions. She will give him the clue out of materialism's Labyrinth. She will be civilization's Una, taming the Lion of disease and misery. The State shall no longer go limping on one foot through the years, but shall march off with steps firm and equipoised. The keen eye and deft hand of the housekeeper shall help to make its everyday walks wholesome; the skill in detail, trustworthiness in finance, motherliness in sympathy, so long extolled in private life, shall exalt public station. Indeed, if I were asked the mission of the ideal woman, I would reply: IT IS TO MAKE THE WHOLE WORLD HOMELIKE. Some one has said that "Temperament is the climate of the individual," but home is woman's climate, her vital breath, her native air. A true woman carries home with her everywhere. Its atmosphere surrounds her; its mirror is her face; its music attunes her gentle voice; its longitude may be reckoned from wherever you happen to find her. But

"Home's not merely four square walls."

Some people once thought it was, and they thought, also, that you might as well throw down its Lares and Penates as to carry away its weaving loom and spinning wheel. But it survived this spoliation; and when women ceased to pick their own geese and do their own dyeing, it still serenely smiled. The sewing machine took away much of its occupation; the French and Chinese laundries have intruded upon its domain; indeed, men, by their “witty inventions,” are perpetually encroaching on “woman’s sphere,” so that the next generation will no doubt turn the cook stove out of doors, and the housekeeper, standing at the telephone, will order better cooked meals than almost any one has nowadays, sent from scientific caterers by pneumatic tubes, and the debris thereof returned to a general cleaning-up establishment; while houses will be heated, as they are now lighted and supplied with water, from general reservoirs.

Women are fortunate in belonging to the less tainted half of the race. Dr. Benjamin Ward Richardson says that but for this conserving fact it would deteriorate to the point of failure. A bright old lady said, after viewing a brewery, distillery and tobacco factory: “Ain’t I thankful that the women folks hain’t got all that stuff to chew and smoke and swallow down!” It behooves us to offset force of muscle by force of heart, that what our strong brothers have done to subdue the material world for us, who are not their equals in physical strength, may be offset by what we shall achieve for them in bringing in the reign of “Sweeter manners, purer laws.” For the world is slowly making the immense discovery that not what woman does, but what she is, makes home a possible creation. It is the Lord’s ark, and does not need steadying; it will survive the wreck of systems and the crash of theories, for the home is but the efflorescence of woman’s nature under the nurture of Christ’s gospel. She came into the college and elevated it, into literature and hallowed it, into the business world and ennobled it. She will come into government and purify it, into politics and cleanse that Stygian pool as the waters of Marah were cleansed; for woman will make homelike every place she enters, and she will enter every place on this round earth. Any custom, or traffic, or party, on which a woman can not look with favor is irrevocably doomed. Its welcome of her presence and her power is to be the final test of its fitness to survive. All Gospel civilization is radiant with the demonstration of this truth:

“It is not good for man to be alone.”

The most vivid object lesson on history's page is the fact that his deterioration is in exact proportion to his isolation from the home of woman's pure companionship. To my own grateful thought, the most sacred significance of woman's work to-day lies in the fact that she occupies the outer circle in this tremendous evolution of the Christian idea of home. Ours is a high and sacred calling. Out of pure hearts, fervently let us love God and humanity; so shall we be Christ's disciples, and so shall we safely follow on to know the work whereunto we have been called.

“'Tis home where'er the heart is,”

and no true mother, sister, daughter or wife, can fail to go in spirit after her beloved and tempted ones, as their adventurous steps enter the labyrinth of the world's temptations. We can not call them back.

“All before them lies the way.”

There is but one remedy; we must bring the home to them, for they will not return to it. Still must their mothers walk beside them, sweet and serious, and clad in the garments of power. The occupations, pleasures and ambitions of men and women must not diverge so widely from each other. Potent beyond all other facts of everyday experience is the rapidly increasing similarity between the pursuits of these two factions that make up the human integer. When brute force reigned, this rapport was at zero. “Impediments to the rear,” was the command of Cæsar and the rule of every warrior—women and children being the hindrances referred to. But to-day there is not a motto more popular than that of the inspired old German, “Come, let us live for our children;” and as for women, “the world is all before them where to choose.”

No greater good can come to the manhood of the world than is prophesied in the increasing community of thought and works between it

and the world's womanhood. The growing individuality, independence and prestige of the gentler sex steadily require from the stronger a higher standard of character and purer habits of life. This blessed consummation, so devoutly to be wished, is hastened, dear girlish hearts, by every prayer you offer, by every hymn you sing, by every loving errand of your willing feet and gentle hands. You are the true friends of tempted manhood, bewildered youth and every little child. The steadfast faith and loyal, patient work you are to do, will be the mightiest factor in woman's contribution to the solution of this Republic's greatest problem, and will have their final significance in the thought and purpose, not that the world shall come into the home, but that the home, embodied and impersonated in its womanhood, shall go forth into the world.

I have no fears for the women of America. They will never content themselves remaining stationary in methods or in policy, much less sound a retreat in their splendid warfare against the saloon in law and in politics. The tides of the mother's heart do not change; we can count upon them always. The voice of Miriam still cheers the brave advance, and all along the line we hear the battle cry: "Speak unto the children of Israel, that they go forward."

THE CATLIN PAINTINGS.

BY O. T. MASON.

If you will enter the National Museum at Washington, and give your cane or umbrella to the venerable gentleman at the stand, you will see some wonderful old pictures. Turn abruptly to your right, cast a patriotic glance at the clothing and camp furniture of the Father of his Country, and a few steps will bring you into the museum lecture room, whose east, south, and west walls are covered with quaint sketches of American aboriginal life, mounted in the dingiest possible black frames. This is the Catlin Collection, by far the most celebrated Indian paintings in the world, since the dreadful fire in the Smithsonian Institution burned up the Stanley portraits in 1864. A few words about this wonderful painter would certainly interest you before you begin to look at his pictures.

George Catlin was born in Wilkesbarre, Luzerne county, Pennsylvania, in 1796. His father was a lawyer, and, naturally, took every pains to educate his son for that profession, sending him at last to Yale College to finish his course. But that which has often happened to boys was true in George's case—he loved fishing and painting more than he loved the bar. His law practice lasted two years, after which he established himself in New York and Philadelphia as a portrait painter. In the year 1832 he saw a delegation of Indians from St. Louis, a town of 25,000 inhabitants, then headquarters for the Central Superintendency (see painting 311), and was so overwhelmed by their appearance that nothing could overcome his desire to visit them in their homes. Convinced that the noble savage would rapidly decline before the advance of civilization, and realizing as if by inspiration the value which a pictorial history of the dying race would possess to future students of primitive history, he set out alone for St. Louis, with pen and brush, to accomplish this noble design. Here he became acquainted with Mr. Chouteau, a gentleman thoroughly conversant with the Indian country

and character, who secured for the painter a free ride to the mouth of the Yellowstone in a little, rude steamer called the "Yellowstone."

He devoted eight years, from 1832 to 1840, to his enterprise, visiting forty-eight tribes of Indians residing within British America, the United States and Mexico. Speaking of these years in after life he says: "I have seen them in their own villages, have carried my canvas and colors the whole way, and painted my portraits from life as they now stand in the gallery. Some of them have been taken while I have been paddling my canoe, or leading my pack-horse through trackless wilds, even at the hazard of my life."

On his return to the East from this remarkable Odyssey, Mr. Catlin exhibited his sketches, together with such a collection of weapons, dress, ornaments and implements as it will never be possible to procure again, in Philadelphia, New York and Boston, to great crowds of visitors. The papers were filled with praises respecting it. The *United States Gazette* says: "There can be no mistake or exaggeration in pronouncing the exhibition of these views of the scenery and natural history of the western country the most important and interesting object for public attention which has ever been offered to the eastern division of the United States."

Many of Mr. Catlin's friends were anxious for the government or some well founded institution to buy the collection, but nothing was accomplished.

In 1840 Mr. Catlin took his pictures abroad, prospects having been held out to him of getting a handsome price for them. The Chancellor of the Exchequer admitted the packages free of duty, and the whole was set up in a room 106 feet long, in Egyptian Hall, Piccadilly, London.

Numerous assemblages, comprising many of the distinguished members of the fashionable and the literary world visited the entertainments and listened to the lectures of Mr. Catlin.

It is just a little to be feared that the painter added to the genius of the artist a modicum of the showman's vanity, in support of which theory the following is quoted from the *London Morning Post*: "This valuable collection of portraits, landscapes, scenes from savage life, weapons,

costumes, and an endless variety of illustrations of Indian life, real as well as pictorial, continues to attract crowds of spectators. We are happy to find our prediction fully borne out by fact, that the exhibition only required to be made known to the public to be fully appreciated. The most pleasing attention is paid by Mr. Catlin and his assistants to gratify the curiosity of visitors, to point out the peculiarities of the various subjects through which they wander, and to explain everything which strikes the eye and attracts the observer to inquire into its use or meaning. During our visit on Saturday the company were startled by a yell, and shortly afterward by the appearance of a stately chief of the Crow Indians stalking silently through the hall, armed to the teeth and painted to the temples, wrapped in a buffalo robe, on which all his battles were depicted, and wearing a tasteful coronet of war eagle's quills. This personation was volunteered by a nephew of Mr. Catlin, who has seen the red man in his native wilds, and presents the most proud and picturesque similitude of the savage warrior that can be conceived. His war-whoop, his warlike appearance and dignified movements seem to impress the assemblage more strikingly with a feeling of the character of the North American Indian than all the other evidences which crowded the walls. Subsequently he appeared in another splendid costume worn by the braves of the Mandan tribe, also remarkable for its costly and magnificent head-dress, in which we see the 'horns of power' assume a conspicuous place. The crowds that gathered around him on each occasion were so dense that Mr. Catlin could scarcely find space to explain in full detail all the costumes; but we are glad to find he is preparing a central stage where all may enjoy a full and fair sight of 'the Red Man,' as he issues from his wigwam, clad in the peculiar robes and ornaments of his tribe, to fight, hunt, smoke, or join in the dances, festivals, and amusements of each nation."

This, of course, smacks a little of Buffalo Bill, but it pleased the Europeans amazingly. Mr. Catlin gave exhibitions in Waterloo rooms, Edinburgh; in the Louvre, at Paris, where Louis Philippe gave him ample space; and in many other European cities, occupying in all about eight years. During this time he published "Manners and Customs," etc.; "The North American Portfolio;" "Eight Years' Travels." In 1861 he published a little work on "The Breath of Life," certainly the funniest serious book we have ever read. In 1862 appeared "Last Rambles Among the Indians," etc.

As many other people have done concerning their own handiwork, Mr. Catlin overestimated the intrinsic value of his paintings and specimens, and made the mistake of thinking that everybody would surely be as enthusiastic as himself. The British government did not buy the gallery. Even the platform and Mr. Catlin's nephew could not save the ship. In Belgium financial embarrassment overtook the painter and his works. The whole material was likely to go under the hammer, when Mr. Thomas Harrison, a wealthy Philadelphian, advanced the money and took the collection as security, with the understanding that it could be redeemed. This proving beyond Mr. Catlin's means, all of the paintings and specimens were transferred to Philadelphia and stored until Mr. Harrison's death, when his widow presented the entire gallery to the National Museum, together with such dresses, etc., as time and moths had spared.

Perhaps you have heard that the Washington pictures are not the originals. The facts are these: After transferring his material in Belgium to Mr. Harrison, Mr. Catlin traveled throughout North America, and even in South America, making aboriginal sketches. This second collection was exhibited for a time in the west corridor of the Smithsonian. On his death, in 1872, the pictures were packed up and stored in the Smithsonian building until 1876, when they were transferred to the Philadelphia Exposition. They are now to be seen in the permanent exhibition there.

Now for the pictures. A great American ethnologist says "Catlin is the great American Indian liar;" another, quite as eminent, says that when he showed one of these pictures to a Sioux Indian, the latter was affected to tears at the recognition of a dear friend long deceased. Both were right. Recently M. Achille Collin, a French sculptor, was employed to produce several busts of celebrated Indians, from photographs and portraits, for the New Orleans Exposition. Among them was Osceola, whose portrait is in the Catlin gallery. Fortunately, the Museum has also Osceola's death mask. M. Collin found that Catlin had placed the eyes too far apart, and had perpetrated several other little artistic outrages, yet the sculptor was able to rectify these and to produce a wonderful bust of the wily Seminole. In one sense everything is wrong in these paintings, in another sense they are teeming with life and spirit. A French critic said in the *Paris Constitutionnel*: "A professional painter is perfectly lost in the presence of a nature new to him, in such singular lands, such original colors of sky,

foliage and men.” You must know a language in order to appreciate its beauties, you must know Indian life to appreciate Catlin. His images do not pose, they fly across the canvas. M. Schindler, whose lifelike portraiture of fishes has given him a world-wide reputation, and who has lived among the Sioux and painted them, has the same admiration of these savage portraitures, whose shadowy looks are ominous of their fast fading originals. A workman in the Museum whose business it is to arrange Indian costumes says that in those things which anybody can do, Catlin was careless; but in the arrangement of dress, ornaments and weapons, which nobody now knows how to fix, he is an invaluable guide. It may not be known to all that the numberless tribes formerly living within our domain belonged to a few well defined stocks, recognizable by language, institutions, and customs. East of the Rocky Mountains, where most of these pictures were painted, were the Athapaskan of the north, the Algonquin and Iroquois of the east, the Cherokee and Muskokee of the south, and the Dakotan of the west. In the great interior basin were the Shoshones, and west of the Sierras the Flatheads, Chinuks, and many other little known stocks.

The Algonkin stock is represented in Mr. Catlin’s gallery by portraits and groups of Sacs and Foxes, Sheyennes, Blackfeet, Chippewas and Ottawas, Crees, Menomonees, Pottawatomies, Kickapoos, Kaskaskias, Weeahs, Peorias, Piankeshaws, Mohegans, Delawares and Shawnees.

The celebrated league of the Iroquois by representatives from the following tribes looks down upon us as Oneidas, Tuscaroras and Senecas.

John Ross, a civilized and well educated chief, with four other celebrated faces, represents the Cherokee stock. The great Muskogee or Creek confederacy includes paintings of distinguished Creeks, Choktas, Seminoles, and Yuchees. Most of the last two stocks were well instructed by Protestant missionaries, previously to their transfer into the Indian Territory. It is this fact alone which explains their steady increase while so many other tribes have melted away. It was among the Dakotas, however, that Catlin’s enthusiasm first took fire and increased most fervently. In addition to the many landscape and hunting pictures, whose scenes are laid in this romantic country, you will see staring at you from these dingy frames, men and women of many Dakotan tribes, Kansas, Osages, Ponkas, Omahas, Otoes,

Missourias, Iowas, Mandans, Blackfeet Sioux, Crows, Assiniboin, Winnebagos, and Sioux proper. A few Comanches, Pawnee Picts, Weecos, Pawnees and Arikarees, Flatheads and Chinuks complete the list.

Of all the tribes visited by Mr. Catlin, the Mandans awaken the most lively interest, not only by their impressive ceremonies, but because the whole tribe were extinguished by smallpox and suicide in 1837, excepting about forty who afterward fell victims to their enemies.

Look along on the wall until you find the pictures numbered 504, 505, 506 and 507. In these, by a series of tableaux, the painter presents to us the annual ceremony of initiation, called the Sun dance by the modern Dakotas, and witnessed two years ago by Miss Alice Fletcher. This ceremony continues four days and nights in succession, in commemoration of the subsiding of the flood, and also for the purpose of conducting all the young men, as they arrive at manhood, through an ordeal of voluntary torture, which when endured entitles them to the respect of the chiefs, to the privileges of going on war-parties, and of taking a wife. The floor and sides of the medicine lodge are ornamented with green willow boughs. The young men who are to do penance by torture lie along the sides of the lodge, their bodies covered with clay of different colors, their respective shields and weapons hanging over their heads. In the middle of the lodge the medicine man prays to the Great Spirit and watches the young men through their four days' fast, preparatory to the torture. Near the medicine man lies a scalping knife, and a bunch of splints which are to be passed through the flesh of the novitiates like belying pins. The Buffalo dance takes place several times each day outside the lodge in which the young men lie. The principal actors in this dance are eight men with the skins of buffaloes around them and a bunch of green willows on their backs. The evil spirit, Okeehedee, enters the village from the prairie, alarming the women, who cry for assistance and are relieved by the old medicine man. Okeehedee is at length disarmed of his lance, which is broken by the women, and he is driven by them in disgrace out of the village. On the fourth day of the festival the young men are subjected to the torture, which in many forms amounts essentially to this: Two gashes, parallel and near together, are cut quite through the skin, either on the breast, back or arms, looking for all the world like those on the sides of a sheep dressed for market. A wooden pin is thrust from gash to gash, under the intervening

strip of skin. One end of a long and strong rawhide line is wrapped or belayed around this peg securely. Now comes the tug of war, the problem always being either to tear this peg out by breaking the strip of flesh or to see how much pain the sufferer can stand. He is hauled up to the roof of the lodge, suspended from an elastic sapling, or dragged around the camp; finally having fastened the end of the line farthest from him to a post, he tugs away with might and main until his flesh is torn loose.

Turn now away from this dreadful scene and take a look at the funny side of Indian life. Here is a fellow whose eager haste after a buffalo ends in being thrown on the monster's back and taking a bison ride over the prairie. There the clans contend nip and tuck for mastery in the ball play. On this canvas a celebrated archer is showing how many arrows he can shoot before the first one falls. Perhaps your delicate sensibility will not enjoy the dog dance, where the Sioux braves are dancing up grotesquely and biting off pieces of a heart taken raw and bleeding from a dog. Well, here is a sham battle of Mandan boys, their school of practice every morning at sunrise, and just there a prairie-dog village. This picture, No. 337, is the celebrated Pipestone quarry on the Couteau des Prairies, 300 miles northwest of the Falls of St. Anthony, on the divide between the St. Peters and the Missouri. Here is where from time immemorial the Indians have obtained for making pipes that beautiful red steatite, which the mineralogists now call *catlinite*, after our hero. There are many more just as strange and interesting stories hanging about this dear old Smithsonian, and some day THE CHAUTAUQUAN may let you into the secret. You notice here and there that the fatal pointer has gone quite through the noble brave, and that accounts for solicitude about your cane and umbrella at the beginning. You will find them at the door, and don't fail to reclaim them with your check.

GEORGE BANCROFT.

BY PROF. W. W. GIST.

George Bancroft is the Nestor of American men of letters. Born October 3, 1800, he received his early training at Exeter Academy, and graduated at Harvard in 1817. He is now eighty-four years old, and his life has touched every administration in the history of our nation except Washington's. What mighty changes have been wrought in the land since George Bancroft, a manly youth, stepped forth from his *alma mater* a full-fledged graduate! Two generations have passed away and a third is now on the stage of action. Webster, Clay, Calhoun and Benton had not yet reached the zenith of their power. These men have passed away, and another group, equally great, of whom Abraham Lincoln was the central figure, became conspicuous leaders in the most thrilling period of our history, and have passed away likewise. Indeed, there are thousands of voters to-day who were born during the exciting events of Lincoln's administration. At the time George Bancroft graduated which, in the general acceptance of the term, marked the commencement of his life's work, Grant, Hayes, Garfield, Arthur, Blaine, Cleveland, and Generals Sherman and Sheridan had not been born. Some of these have won never-fading honors in events that have attracted the attention of the whole world, and are numbered among our heroes. At that time Harvard was a very different institution from what it is now; American literature was in its infancy; Washington Irving had scarcely gained a recognition on the other side of the waters.

Forty-five years ago Bancroft held a government office and secured for Nathaniel Hawthorne an appointment in the Boston Custom House. Hawthorne was then a literary man with some reputation, but his pen did not afford him a livelihood. His great masterpieces were written during the next quarter of a century, and twenty years have passed since the announcement of his death cast a gloom over the literary world, while his

friend and benefactor still survives in the full vigor of his intellectual powers.

Macaulay and Bancroft were born in the same year; the former has been dead nearly twenty-five years; the latter is giving finishing touches to his great history, which merits a place with Macaulay's and Gibbon's.

George Bancroft's father was the Rev. Aaron Bancroft, D.D., who as a young man participated in the battles of Lexington and Bunker Hill, and who in later years won an honorable name as a theologian and man of letters, his "Life of Washington" attracting considerable attention in Europe. The son inherited many of the admirable characteristics of the father.

After his graduation at Harvard, George Bancroft spent five years in Europe, receiving a degree from the University of Göttingen, mastering the principal modern languages, giving special attention to the study of history, visiting the most important nations of the continent, and above all communing with some of the greatest minds of the age. It was his rare privilege to meet, and to enjoy the friendship of, such men as Wolf, the distinguished classic scholar, Wilhelm von Humboldt, Goethe, Cousin, Alexander von Humboldt, Chevalier Bunsen, Niebuhr, and others scarcely less distinguished.

Returning to his native land in 1822, he spent one year as tutor of Greek in Harvard, and afterward assisted in establishing a preparatory school at Northampton. The subject of United States history already absorbed his mind, and the next few years were spent in special study for his great work.

Bancroft has held a number of offices. In 1838 President Van Buren appointed him collector at the port of Boston, and he discharged the duties of the office with marked ability. In 1845 he entered President Polk's cabinet as Secretary of the Navy. A number of important events of this administration are linked with his name. Through his influence the naval academy at Annapolis was established, and he introduced many needed reforms into the naval service. He ordered the United States fleet to assist Captain Fremont in taking possession of California, and as Acting Secretary of War he issued orders for the United States army to march into Texas at the commencement of the Mexican war. In 1846 he was appointed minister to England, and held the position for three years. While in England unusual

courtesies were extended to him, and every facility was granted for carrying on his historical researches, official state papers and many valuable private libraries being accessible. He also visited Paris for the purpose of study, and received valuable assistance from Guizot and Lamartine. In 1867 he was appointed minister to Berlin and remained abroad a number of years, calling forth a special commendation from President Grant for his wise diplomatic services.

Mr. Bancroft has done considerable literary work in addition to writing his "History of the United States." When a young man he published a volume of poems; he has contributed a great many articles to magazines, and has delivered a number of memorial addresses on prominent Americans. In 1859 he prepared a paper on "Prescott" for the New York Historical Society; also one on "Washington Irving." In 1860 he delivered an address in Cleveland at the unveiling of the statue of Commodore Oliver H. Perry, and February 12, 1866, he delivered before the two houses of Congress a memorial address on President Lincoln.

Bancroft is known most widely, however, as an historian, and his noble history is a monument more durable than granite. He brought to his task a mind philosophic in character, broad in grasp, impartial in judgment, believing firmly in God's superintending care, rich in scholarship, and with enough of the imaginative and poetical to quicken and vivify all his intellectual powers. He has bestowed nearly sixty years of conscientious labor on this great historical work, the first volume of which appeared in 1834, fifty years ago.

The historian requires peculiar talent for his work. He must have such patience and energy as will enable him to carry on any research that will throw light on the subject he is investigating; he must weigh all evidence as coolly as the most unprejudiced judge; he must not assume the part of an advocate until he has examined the subject from every standpoint and reached an unbiased conclusion; he must grasp the real ideas and principles that underlie the events and are hastening the progress of civilization; he must have sufficient imagination to see the events as real, and to make his readers see them as such; in addition, he must have a copiousness of illustration and a fluency of language that will enable him to present his subject in an attractive form. In short, he must be a scholar, an explorer, a

philosopher, and a rhetorician. Few, if any, have possessed all these qualifications in a preëminent degree; Bancroft certainly possesses them all in no small degree.

Gibbon will doubtless ever hold an honorable place as an historical writer; and yet he attempts to account for the rapid spread of Christianity entirely on human grounds, and refuses to recognize the greatest force then at work in effecting changes among the nations of the world. Macaulay well says of Gibbon: "He writes like a man who had received some personal injury from Christianity and wished to be revenged on it and all its possessors." No such charge can be made against George Bancroft. He is a firm believer in God, recognizes Christianity as the most powerful factor in the progress of civilization, and continually evinces his unfaltering belief in God's superintending care over human affairs. The opening paragraph of his address on President Lincoln may be taken as his creed on God in history. Notice how clear his statement and triumphant his faith:

"That God rules in the affairs of men is as certain as any truth of physical science. On the great moving power which is from the beginning hangs the world of the senses and the world of thought and action. Eternal wisdom marshals the great procession of the nations, working in patient continuity through the ages, never halting and never abrupt, encompassing all events in its oversight, and ever effecting its will, though mortals may slumber in apathy or oppose with madness. Kings are lifted up or thrown down, nations come and go, republics flourish and wither, dynasties pass away as a tale that is told; but nothing is by chance, though men, in their ignorance of causes, may think so. The deeds of time are governed, as well as judged, by the decrees of eternity."

A quotation from his history will show his estimate of Christianity:

"To have asserted clearly the unity of mankind was the distinctive character of the Christian religion. No more were the nations to be severed by the worship of exclusive deities. The world was instructed that all men are of one blood; that for all there is but one divine nature and but one moral law; and the renovating faith taught the singleness of the race, of which it embodied the aspirations and guided the advancement."[\[E\]](#)

Notice also this noble tribute to Christianity in his history:

“The colonists, including their philosophy in their religion, as the people up to that time had always done, were neither skeptics nor sensualists, but Christians. The school that bows to the senses as the sole interpreter of truth had little share in colonizing our America. The colonists from Maine to Carolina, the adventurous companions of Smith, the proscribed Puritans that freighted the fleet of Winthrop, the Quaker outlaws that fled from jails with a Newgate prisoner as the sovereign—all had faith in God and in the soul. The system which had been revealed in Judea—the system which combines and perfects the symbolic wisdom of the Orient and the reflective genius of Greece—the system, conforming to reason, yet kindling enthusiasm; always hastening reform, yet always conservative; proclaiming absolute equality among men, yet not suddenly abolishing the unequal institutions of society; guaranteeing absolute freedom, yet invoking the inexorable restrictions of duty; in the highest degree theoretical, yet in the highest degree practical; awakening the inner man to a consciousness of his destiny, and yet adapted with exact harmony to the outward world; at once divine and human—this system was professed in every part of our widely extended country, and cradled our freedom. Our fathers were not only Christians; they were, even in Maryland by a vast majority, elsewhere almost unanimously, protestants. Now the Protestant Reformation, considered in its largest influence on politics, was the awakening of the common people to freedom of mind.”[F]

In a recent private letter to Dr. Buckley, of the *Christian Advocate*, Bancroft uses these words quoted in that paper:

“Certainly our great united commonwealth is the child of Christianity; it may with equal truth be asserted that modern civilization sprung into life with our religion; and faith in its principles is the life-boat on which humanity has at divers times escaped the most threatening perils.”

And again:

“The principles that govern human affairs, extending like a path of light from century to century, become the highest demonstration of the superintending providence of God.”[G]

But it is not necessary to multiply quotations illustrative of his faith in the Deity. Throughout the whole of his writings he manifests a devout,

reverential state of mind, and keeps constantly before the reader the idea that God is the great power back of those mighty movements that stir the nations of the world.

The philosophic cast of his mind is clearly revealed in all his discussions of causes and results. He firmly believes that “the problems of politics can not be solved without passing behind transient forms to efficient causes,” and he ever seeks to find the real origin of an event. He dates the American Revolution back to the Reformation under Luther and Calvin, and in relating the events that led to a separation from the mother country he discusses with great clearness and elaborateness three points essential to the proper understanding of the subject: In the first place he speaks of the emancipation of the mind at the Reformation, and the consequent birth of the idea of freedom. In the second place he discusses the growth of this idea of freedom in the nations of Europe and on this continent. In the third place he describes with wonderful fairness the violent discussions that arose in England and in this country when the colonists raised a protest against the tyrannies of the mother country. Referring to the origin of our present liberty, he says explicitly:

“The Reformation was an expression of the right of the human intellect to freedom.”[\[H\]](#)

He thus speaks of the influence of Luther and the Reformation: “At his bidding truth leaped over the cloister walls and challenged every man to make her his guest; aroused every intelligence to acts of private judgment, changed a dependent, recipient people into a reflecting, inquiring people; lifted each human being out of the castes of the middle age, to endow him with individuality, and summoned man to stand forth as man. The world heaved with the fervent conflict of opinion. The people and their guides recognized the dignity of labor; the oppressed peasantry took up arms for liberty; men revered and exercised the freedom of the soul. The breath of the new spirit moved over the earth; it revived Poland, animated Germany, swayed the north; and the inquisition of Spain could not silence its whispers among the mountains of the Peninsula. It invaded France; and, though bonfires of heretics, by way of warning, were lighted at the gates of Paris, it infused itself into the French mind, and led to unwonted free discussions. Exile could not quench it. On the banks of the Lake of Geneva,

Calvin stood forth the boldest reformer of his day; not personally engaged in political intrigues, yet, by promulgating great ideas, forming the seed-plot of revolution.... Calvinism was revolutionary; wherever it came it created division.... By the side of the eternal mountains and perennial snows and arrowy rivers of Switzerland, it established a religion without a prelate, a government without a king.... It entered Holland, inspiring an industrious nation with heroic enthusiasm; enfranchising and uniting provinces; and making burghers, and weavers, and artisans, victors over the highest orders of Spanish chivalry, the power of the inquisition, and the pretended majesty of kings. It penetrated Scotland, and while its whirlwind bore along persuasion among glens and mountains, it shrunk from no danger, and hesitated at no ambition; it nerved its rugged but hearty envoy to resist the flatteries of the beautiful Queen Mary; it assumed the education of her only son; it divided the nobility; it penetrated the masses, overturned the ancient ecclesiastic establishment, planted free parochial schools, and gave a living energy to the principle of liberty in a people. It infused itself into England, and placed its plebeian sympathies in daring resistance to the courtly hierarchy; dissenting from dissent, longing to introduce the reign of righteousness, it invited every man to read the Bible, and made itself dear to the common mind, by teaching, as a divine revelation, the unity of the race and the natural equality of man.”[\[1\]](#)

It is evident that Bancroft has studied the Reformation, not simply in its outward political aspect, but so as to understand the different shades of theological belief that influenced the minds of the great reformers. His parallel between Luther and Calvin is a fine specimen of composition, noted for its vigorous English, clear, discriminating judgments, and polished style: “Both Luther and Calvin brought the individual immediate relation with God; but Calvin, under a more stern and militant form of doctrine, lifted the individual above pope and prelate, and priest and presbyter, above Catholic church and national church and general synod, above indulgencies, remissions and absolutions from fellow-mortals, and brought him into immediate dependence on God, whose eternal, irreversible choice is made by himself alone, not arbitrarily, but according to his own highest wisdom and justice. Luther spared the altar, and hesitated to deny totally the real presence; Calvin, with superior dialectics, accepted as a commemoration and a seal the rite which the Catholics revered as a

sacrifice. Luther favored magnificence in public worship, as an aid to devotion; Calvin, the guide of republics, avoided in their churches all appeals to the senses as a peril to pure religion. Luther condemned the Roman Church for its immorality; Calvin for its idolatry. Luther exposed the folly of superstition, ridiculed the hair shirt and the scourge, the purchased indulgence, and dearly bought, worthless masses for the dead; Calvin shrunk from their criminality with impatient horror. Luther permitted the cross and the taper, pictures and images, as things of indifference; Calvin demanded a spiritual worship in its utmost purity. Luther left the organization of the church to princes and governments; Calvin reformed doctrine, ritual, and practice; and, by establishing ruling elders in each church, and an elective synod, he secured to his policy a representative character, which combined authority with popular rights. Both Luther and Calvin insisted that, for each one, there is and can be no other priest than himself; and, as a consequence, both agreed in the purity of the clergy.”^[J]

While the rhetoric of Bancroft is not faultless, it certainly deserves a place in our classic English. In the discussion of grave historical and philosophical questions, his stateliness of expression and his dignity of style challenge our admiration. His descriptions are very fine, and suggest a mind keenly alive to the beautiful and the poetical; but they do not reveal that spontaneity so characteristic of Irving, nor that indefinable symmetry so noticeable in Hawthorne. If his style is sometimes declamatory, I think it is generally in a connection such that the cultivated taste will pronounce it admissible.

Thoroughly versed in the historic lore of this and other countries, broad in his general scholarship, remarkably free from prejudice, an uncompromising American, and yet not an American in a narrow and bigoted sense, careful and systematic in his methods of labor and recreation, unswerving in his belief in the superintending providence of God, George Bancroft justly merits the high place of honor and esteem so willingly accorded to him, and his noble example should be a never-failing source of inspiration.

^[E] Vol. III., p. 6.

^[F] Vol. II., p. 177.

[G] Hist., Vol. II., p. 545.

[H] Hist., Vol. III., p. 183.

[I] Hist., Vol. III., p. 99.

[J] Hist., Vol. I., p. 212.

GOING TO THE BOTTOM THE ONLY WAY TO REACH THE TOP.—First go to the bottom of everything which you have to do. Know all its principles. If it be a trade, know not only its rules, but the reasons for them. If it be merchandise in raw materials, or in one or more manufactured articles, be sure to learn the whole process, from the planting of the seed, or the digging of the ore, to the completed fabric. Do this by observation, conversation with the heads of departments, and with workmen in different specialties. This was the plan of the late William E. Dodge.—*From Dr. J. M. Buckley's "Oats or Wild Oats."*

HOW PERSEUS BEGAN TO BE GREAT.

BY ELIZABETH P. ALLAN.

Fair Perseus slept upon enchanted ground,
And to him came, but with no stir or sound,
The goddess Pallas, whose clear, shining eyes
Read all men's hearts that are beneath the skies.

"Many there be," she cried, "who dwell at ease,
Who neither do nor dare—art thou of these?
Or dost thou to some glorious deed aspire?
Hast thou a heart of clay, or soul of fire?"

Then answered Perseus, vehement, "But I
Would do great deeds, though for them I should die.
Point me, O goddess, to the monster's lair,
And give me leave to show what I can dare!"

"Nay," said the voice divine, "thy stripling arm
Must first set free from danger's fierce alarm
Thy mother's life!" With this young Perseus woke,
Nor knew the peril whereof Pallas spoke.

But straight to Seriphus he hied with speed,
To find his Mother Danaë in sore need,
And rescuing her from danger and from dread,
Then went he forth to win the Gorgon's head.

O, sons and daughters of our happy land,
Whose waking dreams are filled with actions grand,
Ere yet in search of these afar you roam,
Free every burdened heart that sighs at home!

CANADA OF TO-DAY.

BY M. VICTOR DU BLED.

An Abridged Translation for THE CHAUTAUQUAN, from the *Révue des deux Mondes*.

Canada, for so long a time apparently forgotten by her mother country, came out from her isolation and again called back to herself the attention of France by sending to the Exposition of 1855 specimens of her products. In 1856 M. de Belveze, commander of the French frigate “Capricieuse,” was sent into Canadian waters. His mission resulted in the establishment of a French consulate, and the reduction of the tariff which permitted the two countries to enter upon commercial relations.

From 1854 to 1862 material and intellectual progress here marched by the side of great political progress. Public works, canals, and interior colonization, all, during this time received a vigorous impulse.

There is no such thing as spontaneous generation in politics any more than in natural history. Questions give rise to other questions, and the philosophy of history shows them springing up, one after another, from some mysterious source, obeying a sort of atavism, and producing often a most unexpected result. Excitement over representation, fixed according to the population of the country, gave birth to the confederacy. On October 1, 1864, a conference assembled at Quebec, composed of delegates from the maritime provinces, and from the Canadian government. After a long and stormy session, during which threats of resorting to arms were now and then heard, the cause of the confederation triumphed by a large majority. A basis of federal union was submitted to the several legislatures for ratification, and on July 1, 1867, the confederacy was established in the midst of public rejoicings. They gave to the united provinces the name of the Dominion of Canada. Lower Canada was called Quebec, and Upper Canada Ontario.

The Legislature is composed of a Governor-general, a sort of a constitutional viceroy, named by the crown; of a Senate, and a House of Commons. The Senate consists of seventy-six members, appointed for life by the crown, of whom twenty-four each are from Quebec and Ontario. The House of Commons is representative, its members being elected for five years. The Dominion now includes Ontario, Quebec, Nova Scotia, New Brunswick, Prince Edward's Island, British Columbia, Manitoba, and the territory of the North-West, or Hudson's Bay Territory. Thus it is fulfilling the prediction of the great American statesman, William H. Seward: "Canada is destined to become the seat of a great empire, the Russia of North America, but a Russia with civilization more advanced than the Russia of Europe." An illustrated paper of the Dominion has published a patriotic caricature representing the Canadian Gulliver with a debonair and placid figure, without any implements save his own gigantic arms and hands, seizing and swallowing the greater part of the American continent, while a crowd of Lilliputians, armed to the teeth, Turks, Yankees, Germans and Italians, survey him with an envious and astonished air.

In order to develop her resources, and to open the way for immigration, that her immense tracts of unused land may more rapidly become the granary of the world, Canada is furrowing her domains with canals and interlacing them with lines of railroads. The Grand Trunk railway, traversing the country from Portland, Me., to Detroit, has been built, with its Victoria bridge (one of the most noted structures in the world) crossing the St. Lawrence opposite Montreal. Immense sums of money have been spent in order to convert the St. Lawrence into a canal. And now she is constructing a transcontinental road, which, binding the two oceans from Port Moody to Halifax, will cross the entire confederation. They expect to finish this route in 1886, and it is estimated that the journey from Liverpool or Havre to Japan will be a thousand miles shorter by this road than by the transcontinental routes of the United States.

Almost in the middle of the Dominion, at an equal distance from the pole to the equator, lies the territory of Manitoba. There lived in 1869, a population half nomadic, called the half-breeds, sprung from marriages between the French Canadians and the Indians. They spoke the French language, and professed the Catholic religion. After the delivery of this country by the Hudson's Bay Company to the Dominion, the government

determined to direct toward it a stream of English emigration. They sent a governor and some surveyors to reside at Winnipeg, the capital. But the natives warned them they might look for trouble if they attempted to place, without consulting them, new inhabitants upon the land which they and their ancestors, from time immemorial, had held and enjoyed. The government was not to be frightened, and so the conflict came. The half-breeds obliged the governor to leave, and constituted a provisional government, with a president at its head. They then drew up a declaration, of which the following is the preamble:

“We, the representatives of the people, assembled in council at Fort Garry, November 24, 1869, after having invoked the God of nations, ... solemnly declare, in the name of our constitution and in our own name, before God and men, that we refuse to recognize the authority of Canada, which pretends to have the right to command us and impose upon us a despotic form of government.”

Later, however, they changed their opinion, and entered into negotiations with the federal government. But at the moment when all things had been arranged without bloodshed, the English colonists, who were very numerous around Lake Winnipeg, rose in insurrection against the half-breeds. The president of the latter, Louis Riel, who took upon himself the *rôle* of dictator, had the leading mutineers seized, and their chief, named Scott, was tried, condemned and shot. Far from establishing his authority, this execution discouraged the natives themselves, and when two battalions of militia under Col. Wolseley arrived on the ground, they were welcomed as liberators by the half-breeds, and Riel, with his leading accomplices, fled to the United States. A compromise was then effected, and Manitoba was annexed to the Dominion as an autonomous province. It sends to Parliament two senators and five representatives. Winnipeg contains 30,000 inhabitants, and property has increased its value to an extraordinary degree, as the following anecdote will show: A parishioner of Archbishop Taché, obliged to leave the country, sought the archbishop, and excusing himself for not being able to pay the rent of his church pew, offered as part payment a small piece of land; “scarcely what would pay for a low mass,” timidly said the poor man. Ten years later that land brought \$14,000.

In spite of the expenses occasioned by her canals and railroads, the people of the Dominion are, perhaps, among all the tax-payers of the civilized world, those upon whom the smallest rates are levied. There is no standing army, only a simple militia of about 50,000 men. The total expenses of government in 1884 amounted to \$28,730,157. The receipts for the same year were \$36,800,000. The minister of finances, in making out his estimates for the year 1885 placed them as follows: Expenses, \$29,811,639; receipts, \$31,000,000, which were to be raised as follows: Duties, \$20,000,000; excises, \$5,500,000; postoffice returns, \$1,900,000; public works, \$3,000,000; interest on investments, \$750,000; other sources, \$800,900. Duties on goods supply the source of two thirds of all the receipts. In 1880 they adopted a very strict system, which, without any distinction, exacts duties from English goods as from any other nation. On the other hand, England has granted to them the right of concluding treaties of commerce with foreign nations.

The only difference of opinion in regard to the question of tariff existing between the two parties is that the liberals wish the laws of entrance to be more moderate than the conservatives have made them, but neither of them will adopt the cosmopolitan theories of European free trade. The English compare protection to a bullet, Canadian tariff to a museum of instruments of torture, and declare that in following the example of the United States, the Dominion has forgotten the fable of the frog which wished to become as large as an ox. The Canadians hold that they shall do what they think best for their country, and that duties are the taxes least inconvenient to raise; that they save national work; and that they not only have made up the deficits of the past, but have put into the treasury an excess, so they have been able to reduce them to the amount of two and one-fourth millions dollars. The tariff for protection has become a tariff for revenue.

Since 1853 especially, public instruction has made great progress. These people who, under the patronage of the crown of England, have realized the ideal conception of a conservative and Christian republic, hold that public schools are among the luxuries of a young nation, and do not hesitate to impose upon themselves heavy sacrifices, as they believe they will result in good to their children. In the province of Quebec alone, government expended during the year 1882-83, \$350,000 for school buildings, while the contributions paid directly by the people amounted to more than

\$2,000,000. In a population of 1,359,027 inhabitants, statistics show that there are 5,039 schools of different grades; 7,211 professors and teachers, and 245,225 scholars, making an average of one scholar for every six inhabitants. As to universities and colleges, they do not come under the school regulations, but are independent institutions, which, however, may receive appropriations from the government on condition of making a report each year to the superintendent. When in a school district there live a number of families who profess a different religion from that of the majority, they have the right to have for their children separate schools, under the care of three officials, chosen by them. Thus Catholics and Protestants have equal privileges, and everything is done to secure respect for religion, independence to the citizen, and his active and constant interest in educational matters. The circulars of the present superintendent, M. Ouimet, define in clear terms the spirit of the school laws in Quebec: "In our system of primary instruction we first teach the children the catechism of true religion, in order that they may know how to serve God; then the manuals of agriculture and of design, in order to put them in condition to serve their country. *For God and country!* Behold the words which the Canadian legislature has inscribed on the walls of her educational institutions. The state unites itself to the two systems of religion in the matter of education, and does not authorize any school to be atheistical, but demands of it to be Christian before it accords help. It does not provide that one church shall be helped rather than another. Full and entire liberty it demands, and from this comes perfect harmony among the people."

Religious liberty marches by the side of educational liberty. Each church supports itself; the state no longer takes cognizance of clergy or congregations, to protect them, to annoy them, or to persecute. They can, as the citizens, found a university, a college, or a school.

Men such as Labelle and Racine have accomplished wonderful results in planting in the most barren regions, at the peril of their lives, strong and flourishing colonies. "Go west," incessantly repeated Horace Greeley to young Americans. "Go north, French Canadians and Catholics," said Father Labelle, with a prophetic foresight.

Canadian literature only dates back to 1840. Before that time it was made up of songs. Such a literature was absolutely essential to the gay and

sociable race who consoled itself, in all its troubles, with stanzas. There was a time when France held the government under control by her songs. Did any Canadian patriot attract attention by some great deed? At once a song was written. Was the question that of elections? They addressed themselves to some crude poet, and sharp, malign couplets soon overran the country. The festival of St. Jean Baptiste has furnished many a contribution to this list, and Sir George Cartier owes in great part his popularity to the fact that he composed one for the first banquet, in 1834. Often among the remote rural districts are found people possessing magnificent tenor voices, which would make the fortune of an impresario who would come from the other side of the water to look them up.

“We are yet amateurs,” said one of their writers to me. Without endorsing this very modest judgment, one can but admit that up to the present time our American cousins have been more occupied with making history than with writing it. Action has absorbed thought. They have run, closely pressed on all sides, to the conquest of political liberties. The books which they published during their unsettled national history partook of the character of the times, as the great work of Garneau will show, which was a revelation to his countrymen, and was of more value to them than an army, since it assured them of a nation’s faith and the certainty of success. The greater part of the writers have been obliged to tax their ingenuity for a livelihood, and too often politics, that deceptive siren, keeps them from those severe studies which alone will bring talent to maturity.

In poetry, MM. Crémazie and Louis Fréchette have left behind them all rivals. High inspiration, poetic fervor, appreciation of nature, and love of country have made them true poets. One can not read without emotion some of the productions of M. Crémazie. His patriotic songs, which seem to have been breathed from the very heart of the country itself, in a language harmonious and vibrating, do not for an instant decline in interest or power. The verses of M. Fréchette are written in a graceful style, and possess a youthful freshness. In history, MM. l’Abbé Casgrain, Benjamin Sulte and Joseph Tassé have become distinguished. M. Tassé, in his book “Canadians of the West,” tells of the pioneers of the American continent, those who penetrated into the icy regions of the pole, who crossed the Rocky Mountains, and spread over the fertile plains of Mexico; and has shown that of them all the French in the Canadian settlements were the only ones who

treated the Indians honorably and kindly, and who succeeded in winning their respect and affection.

The group of prose writers and romancers is increasing every day. One of the best, without doubt, is M. J. C. Taché, the author of three legends, each of which characterizes an epoch in the history of the Indians. M. de Gaspé, with his "Ancient Canadians," and M. Joseph Marmette, with his historic romances have acquired a well merited reputation.

What, then, shall be the aim in the future of Canadian literature? To acquire new strength and vigor without ceasing the study of the past; to revive the glorious annals; to gather with a pious care its legends; to identify itself also with the present; to paint the manners and the contemporaneous social life; to note and to report the majestic symphony of their land; never to lose sight of the thought of Carlyle, that the universe is a temple as well as a workshop. Such will be the duty of Canadian writers.

The Canadians through all the years since their country passed out of the hands and the control of the French, have clung to them with great affection, drawn by some profound and mystic instinct, by the lines of heredity, the power of traditions, the religion of memory. They are not ignorant of the fact that if they had remained united to France, they would not now have, in all probability, their free social and religious institutions; they would likely have formed an administrative colony such as Algeria. They know that it was England who sent them, under hard circumstances, perhaps, to the school of liberty, and to her they are indebted for their prosperity, but they look to France still as their mother country. Why should not that country give them some more solid proof of its affection? While with South America the annual exchanges of France are counted by the hundred million, and great numbers of French people emigrate there, her total commerce with Canada does not exceed \$15,000,000, and it is with great difficulty that she has commenced to send thither a few of her citizens. Why should not French emigration direct itself toward a country where wages are good, the soil fertile, where property offers itself to all, and where a welcome is awaiting them? Why should not the French go to visit the Canadians and learn of them how a people became and remain free?

SOME AMERICAN MUSEUMS.

BY CLARENCE COOK.

Under this heading it is intended to give in successive numbers of THE CHAUTAUQUAN descriptions of the principal Art Museums of our country: The Museum of Fine Arts in Boston, the Metropolitan Museum of Art in New York, the Pennsylvania Academy of the Fine Arts in Philadelphia, and the Corcoran Art Gallery in Washington. We begin in the present number with the Boston Museum.

In the year 1870 the trustees of the Boston Museum of Fine Arts were organized under a charter from the Massachusetts legislature. It was not, however, until 1876 that a building was erected in which the pictures, casts, antiquities, engravings and objects of curiosity which formed the nucleus of its present extensive collections could be exhibited to the public.

Up to the time when the first portion of the present building was erected, the amateur or the student of the fine arts in Boston or its neighborhood had been obliged to take a good deal of trouble, and to spend much time, if he would see the few objects that existed there—in public institutions or in private houses—in the domain of painting, sculpture, antiquities, or in that of the minor arts—now classed together in popular speech under the incorrect title of bric-a-brac.

Beside the permanent exhibition of the pictures which belonged to it, the Athenæum Library had generously devoted some of its rooms every season for several successive years to the exhibition of pictures painted by American artists, an exhibition answering to those held yearly by the National Academy of Design and the Pennsylvania Academy of the Fine Arts. Beside these regular exhibitions, there had been many occasional ones of importance, such as that of the Spanish pictures belonging to the Duc de Montpensier, which gave to those of us who had not visited Europe, the opportunity of seeing respectable specimens of the works of Zurbaran,

Herrera, Morales, Murillo, Ribalta and Velasquez. There was also in the Athenæum a small but well selected collection of plaster-casts of antique sculpture, so that for a long time this institution was an art center of no little value and importance. The Athenæum was not, however, an art institution, but a library, and the time came when the increase of the library made it necessary to give up its art collections and devote all its space to books.

The collections of the Athenæum were the most important, both in number and in value, to be found in Boston, but there were many interesting objects scattered about which it was felt would be of much greater service to the community if they could be brought together under one roof, and made to work in common for the education of the whole community. The late Francis C. Gray had bequeathed to Harvard College his large and valuable collection of engravings together with a fund for its maintenance, and it was found that its usefulness, whether for purposes of enjoyment or as a means of education was very much restricted by its being so far away from the capital. Yet it had been impossible to find a proper place for it in Boston, and it therefore remained shut up in Cambridge. The Institute of Technology had formed, under the direction of Prof. William R. Ware, a collection of architectural ornament, but as this was lodged in the Institute building it could only be seen and studied at such times as suited the convenience of the professors and their pupils. The Social Science Association had called the attention of the public to the need that existed of a large and complete collection of casts of antique sculpture. But—what to do with such a collection, could it be brought together?

In a city like Boston, a want so deeply felt could not long remain unsatisfied, and the matter having been widely discussed, and a general interest created in the public mind, the first steps were taken with generous unanimity, and as has been stated, a charter was procured from the legislature, the Museum was organized by the naming of trustees, and the city having given a site, no difficulty was met with in raising a subscription of \$261,000 toward a building. The plans of Messrs. Sturgis and Brigham, submitted in competition, were selected, and on the 3d of July, 1876, one wing of the front of the building was opened to the public. This was filled by the collections of the Athenæum, and by the Gray collection of engravings, both permanently loaned to the Museum, by the casts of antique sculptures purchased with funds bequeathed by the late Charles Sumner, the

Egyptian collections presented by Mr. C. Granville Way, and valuable gifts from Mr. Lawrence and other persons. The space at the disposal of the trustees was soon overcrowded, and in 1878 a fresh subscription of \$126,000 having been raised, the front was completed and opened to the public in 1879. At the present time of writing the need is seriously felt for more room, and it is hoped that the means may soon be provided for taking a third step toward the completion of the original plan of the building.

The building containing the collections of the Boston Museum of Fine Art is constructed of red brick and terra cotta on a basement of granite. It stands in the new quarter of the city, and is built like all other structures in that part of the city, on piles. It is rather ornate in its character, and compared with its massive neighbor, Trinity Church, has a somewhat effeminate look, but it is solidly built, and planned with great good sense, and with a steady view to convenience. On entering, the visitor finds himself in a large hall, in the center of which rises an ample staircase conducting to the second floor. At the right and left are doorways leading to the rooms containing the casts from antique sculpture. By taking either of these doorways we can make the circuit of the whole series of apartments, but as the present arrangement is only temporary, awaiting the completion of the building for a logical disposition of the material, it may be better to pass at once to the rear of the hall, and taking the door at the left hand, enter

THE EGYPTIAN ROOM.

The contents of this room were chiefly collected between the years 1828 and 1833, by a Scotch gentleman, Mr. Robert Hay. After his death they were purchased by Mr. C. Granville Way, of Boston, and presented to the Museum in 1872. Several fine pieces of sculpture, collected in Egypt in 1835 by the late Mr. John Lowell, the founder of the Lowell Institute, have been added to this room by the gift of Mr. Lowell's heirs. The valuable and interesting casts from bas-reliefs and statues are the gift of General Charles G. Loring, the director of the Museum, to whose zeal and efficiency the institution owes so much of its usefulness.

The room is finely lighted by large windows, and General Loring, who is much interested in botany, generally keeps here a few fine specimens of tropical plants, especially such as belong to Egypt. Thus, on the occasion of

my last visit, I had the pleasure of seeing there a fine specimen of the papyrus plant waving its graceful fans in salute to Pasht and Amenophis, hard by. The giant figure of Amenophis III., the Memnon of the Greeks (1500 B. C.) is a cast from the granite original in the British Museum. Near it is the statue of Pasht, the Cat-headed, in black granite, with the cartouch of Amenophis III., and there are also several blocks of red granite, probably portions of a throne, with a few fragments of sculpture—the colossal head of a king, pieces of the lid of a sarcophagus in green basalt, and two capitals cut out of sandstone, showing the lotos and papyrus forms. In the center of the room are several mummy cases, and in glass cases are disposed mummied heads, skulls, hands and feet, with mummies of animals, the cat, the dog, the dog-faced ape, the hawk and the ibis. In one of these cases is a hand still bearing a ring on the fourth finger. The remaining cases contain very interesting specimens of mummy-cloth of various dates and quality, one of the most important being a robe of justification supposed to be worn in the trial of the deceased before Osiris. It is sixteen feet in length by six feet nine inches in width, and has a fringe. The remaining contents of this room consist of various objects gathered from the tombs and from the mummy cases in such number and variety as to make it impossible to describe them in the narrow space at my command. But while there can be no doubt as to the value of the collection as a means of study in a field of wide interest and importance, it may be said, so far as art is concerned, the Way collection is of less value than the Abbott collection in the Historical Society of New York City. Each collection, however, supplements the other in a most interesting way, and taken together, they enable a student to make a fair beginning in the study of Egyptian antiquity.

THE FIRST GREEK ROOM

Opens directly from the Egyptian Room, and contains casts of archaic and early Greek sculpture. Here will be found the Lions from the gate of Mycenæ, the funeral slabs of Orchomenos, of Aristion, and of the soldier of Marathon, the Dresden Pallas, the relief of Demeter, Persephone, and Triptolemus, from Athens, and the so-called Leucothea, and the infant Bacchus from the Villa Albani, with several interesting archaic reliefs from the same collection. There have lately been added to the Museum a number of the funeral slabs or *stele* discovered at Athens and preserved in the

museum there, objects of great beauty and interest, properly belonging, either in this room or in immediate connection with it, but placed for temporary convenience, in the Roman and Renaissance Room. The most important objects in this First Greek Room are the casts from the sculptures of the eastern and western Pediments of the Temple of Minerva at Egina, consisting of five figures from the eastern pediment and ten from the western, arranged as they are believed to have been originally. Passing from this room to

THE SECOND GREEK ROOM

We find ourselves in the midst of a group of statues, most of them of the Praxitelean type and making too sharp a contrast by their grace and sensuous refinement to the hardness and severity of the contents of the room just left. It must be remembered, however, that owing to the small space at the command of the Museum authorities it has not been possible to follow a strict chronological order, and we must therefore be content for the present to follow the arrangement of the separate rooms. We have, therefore, here, the casts from the Parthenon frieze, the Theseus and the Fates from the eastern and the Ilissus from the western Pediment of the same building, with the Torso of the Victory, also from the eastern Pediment, together with several figures from the temple of the Wingless Victory (Nikè Apteros) on the acropolis. But space fails us to enumerate all the casts contained in the rooms devoted to antique sculpture; and why attempt a mere catalogue? The Venus of Milo is here, and the lately discovered Hermes with the infant Dionysus, the Niobe and her daughter, the Ludovisi Mars, the Diana of the Louvre, the Apollo Belvidere, the Eirene and Plutus, the Faun of Praxiteles, and the glorious mask of the Ludovisi Juno. Indeed, we miss few works of prime importance, and there are many casts here that can not be found elsewhere in America, and which are yet essential to even a superficial study of the rise and progress of Greek sculpture. Passing on, we come to the other rooms where are the Laocoön, the Dying Gladiator, the younger Agrippina, the Sophocles, the Demosthenes, the Menander, the Æsculapius, the Discobulus, the Silenus and the infant Bacchus, and the Boy taking a Thorn from his Foot (the Spinario), with many another famous and less famous work, enabling us to carry on the study until the stream dies away to rise again in new beauty in

the art of the early Italian Renaissance. While no capital piece can be said to be wanting to this collection there remain many pieces to be added which are needed for fullness of knowledge, but every year sees important acquisitions, and there can be no doubt, judging from the past history of the Museum, that if the wished for addition to the building could be made, the missing gaps in the sculpture would speedily be filled up by gift. But before leaving the antique rooms we must mention the two sarcophagi from Vulci, now deposited in the Museum, and which, it is earnestly to be hoped, will become its property, since they are not only deeply interesting in themselves, but have an added value from their great rarity. They represent the bodies of two married pair reposing upon the lids of the two sarcophagi, as on the marriage-bed. The finer of the two groups is carved in alabaster, the other is in travertine; the one in alabaster has a monumental beauty and sweet dignity that is surpassed by nothing of the kind that exists, and considering its great beauty and rarity it is said there is only one other example of this treatment of the subject, and that is in the Vatican.

Crossing the Hall of Entrance, to which we have returned, we find ourselves in the last of the antique sculpture rooms, where are placed some of the most interesting of the Roman works just enumerated. Nothing would be gained by an attempt to catalogue the rooms at present, as their contents are likely to be changed at any time when the projected enlargement of the Museum is carried out. The space in this portion of the building, the addition built in 1879, answering to that occupied in the older portion by the first and second Greek Rooms, is here thrown into one large apartment filled with the

CASTS OF ARCHITECTURAL ORNAMENTS.

The principal object here is the cast of the Caryatid Portico of the Pandroseion, one of the portions of the complex structure generally called the Erechtheium, from the name of one of its parts dedicated to the worship of Erechtheus. The Portico with its Caryatids is given here of the full size of the original, and is so placed (until the great court can be built in which these large objects are to be shown) that a good view of it can be obtained from a considerable distance, while it is well lighted by a large window at one side. The remaining objects in this room are casts in great numbers

from Greek and Roman architectural ornament, from the ornament of the Italian Renaissance, from the Alhambra, from the Gothic buildings of France, Germany and England, the specimens from England including twelve out of the thirty angels composing the so-called angel choir of Lincoln Cathedral. These figures of angels playing on musical instruments are of the thirteenth century, and are among the most beautiful works of their time. In this room again we find it impossible to do justice to our subject; the variety is too great and the range of artistic development covered by the example too extensive to be dismissed in less than an entire article, and even that would be insufficient. Turning to the right at the end of this room we come to

THE ITALIAN RENAISSANCE,

Where the works of Michelangelo, Donatello, Ghiberti, Luca della Robbia, and a few other such names meet us in some of their best works. Here is the Lorenzo of Michelangelo with the statues of Day and Night, the David of Donatello, the Cupid of Michelangelo with his unfinished bas-relief of the Virgin and Child, the Mercury of John of Bologna, and various bas-reliefs of the time with the singing boys of Donatello and those of Luca della Robbia. Here, too, is the cast of the trial plate, "The Sacrifice of Isaac," made by Ghiberti in competition with Brunelleschi and Donatello for the Florence Baptistery Gates, interesting in itself, and in connection with that most important event in the history of modern art.

The last room on this floor is filled with specimens of Greek, Roman, and Asia Minor pottery, with a sufficient number of examples of the sculpture, pottery, and glass of Cyprus, a small but well-chosen group of figurines from Tanagra, and the results of the late researches at Assos by the members of the American Society of Archæology. This room is full of interesting objects, but it is uncomfortably crowded and necessarily ill-arranged. In the next article we shall describe the contents of the second floor of the Museum.

NATURAL HISTORY AND PEOPLE OF BORNEO.

Report of a lecture delivered March 7th in the National Museum of Washington, D. C., by Mr. Wm. T. Hornaday, Chief Taxidermist of the Museum.

The island of Borneo is the home of the Head-hunter, the land of the orang-utan, the Garden of the Sun, and perhaps even the sepulchre of the missing link. There is a possibility of its being the cradle of a great empire which shall be at the zenith of its glory when the greatness of the United States shall have passed away, like that of Greece and Rome, and Washington have become the Athens of America. The center of human progress will probably eventually move into regions now peopled by savages only, and the Kaffir or Dyak of the thirtieth century will perhaps study the archæology of the Yankee with the same interest that we now bestow upon the ruins of Carthage and Mycenæ.

Borneo is situated nearly in the middle of the Malay Archipelago. Its greatest length is 850 miles, greatest width 630 miles, and its area is 192,000 square miles. The whole of New England, the Middle States and Virginia could be set down in the evergreen forests, which everywhere cover its surface, and still be surrounded by a wide belt of jungle. The whole interior is very mountainous. The rivers and creeks are the highways of Borneo, and other roads are practically unknown. Nothing could be more arduous, and full of risk to life and limb, than overland travel through such dense forests and over such rugged mountains as confront the explorer at every step. The interior is practically an uninhabitable wilderness. Even in this age of daring and persevering travelers, no white man has ever crossed the island from one side to the other. The interior is still a land of mystery, whence come marvelous accounts of a race of men with tails, with detailed descriptions of their appearance and habits, stories implicitly believed by many natives. The climate of Borneo is what one would least expect, considering its equatorial position. The temperature is very agreeable all the year round. The mercury usually stands at 80° Fahrenheit in the morning,

88° at midday, seldom reaching 90°, and never exceeding 93°. The annual variation of temperature is only 24°—from 69° to 93°. Usually there are about 200 rainy days in the year, and from 158 to 178 inches of rain.

The vegetation of Borneo is probably unsurpassed by that of any other country in the world, either in luxuriance, economic value, or, the possession of wonderful forms. On the spurs of Mount Kina Balu are found four species of pitcher plants (*Nepenthes*), of marvelous size and form and gorgeous colors. The largest pitchers of *Nepenthes rajah* measure thirteen inches in length, twenty in circumference, and hold five pints of water. Among the curiosities of vegetation is the tapang tree, which, in lieu of spur roots, throws out enormous slab-like buttresses. The *cocoa* palm bears a bountiful crop of nuts, which in turn yield oil and a coarse kind of sugar. The *sago* palm yields the valuable pearl sago of commerce. The *areca* palm produces the betel-nut, which, together with a fresh pepper-leaf and a bit of moist lime, is in the mouth of nearly every East Indian native in lieu of tobacco. The *nipa* palm yields salt, toddy, excellent syrup and sugar, and the leaves are made into kadjangs for boat awnings and roofing material for houses. The *gomouti* palm produces the best toddy, and the cabbage is esteemed by the natives as food. The *nibong* palm is valuable for its timber. The primeval forests are rich in timber trees, one of which, the *bilian*, furnishes wood which seemingly never decays. Bamboo grows abundantly in the interior, and is of great use to the natives.

Of the many fruits of the forest we can only refer to the durian. In size and shape it resembles a roundish pineapple, and is set all over with sharp conical spines, three fourths of an inch long, and stout enough to pierce the hide of a rhinoceros. When the fruit is ripe, the pod opens of its own accord. Although the smell of the pod is most offensive, we find inside four or five large cells, in each of which are from three to five horse-chestnuts, coated thickly with the most delicious paste that ever tickled the palate of man.

The agricultural products consist of sago, gambier, rice, sugar-cane and cotton, which is grown to a limited extent by the Dyaks. The cultivation of coffee is now engaging the attention of enterprising English planters, and may eventually become the most important industry of the island.

The whole island teems with animal life in great variety of forms. It would appear, judging from the success of Mr. A. R. Wallace, to be a paradise for the entomologist. This gentleman once collected seventy-six species of beetles in one day, many of which were new and of remarkable form, and during his stay of fifteen months in Sarawak he took over 500 species. There are a number of handsome species of butterflies, including the magnificent *Ornithoptera Brookana*. This butterfly is eight inches in width, and of a rich, velvety black color, on which is a broad band of metallic green scales, resembling a humming-bird's feathers. Of all insects Borneo is richest in moths. At one place, on a mountain top, Mr. Wallace took 200 specimens in a single night, representing 130 species. In the same place he took in twenty-six nights 1,300 specimens of moths.

The fishes include quite a variety of fresh-water species, among which may be mentioned the curious tree-climbing perch, the thread-fish, the celebrated *gourami*, the jumping-fish, or *Periophthalmus*, which hops about on land in search of small crustaceans stranded by the receding tide; and the very rare and curious little fish known to ichthyologists as *Luciocephalus pulcher*. The Malays capture a great many fish in small streams by poisoning the water with an extract made from the pounded roots of the tuba plant, and either spearing or netting the fish when they rise to the surface to breathe.

Among the reptiles, the most important is the crocodile, which attains a maximum length of seventeen feet, and is very destructive to human life. It seldom happens that a person escapes or is rescued, after being seized in this burly reptile's powerful jaws. Some years ago the Sarawak government began a war of extermination against the crocodiles, by offering a reward of 35 cents a foot for all killed in the Territory. In 1878, 266 crocodiles were killed, and \$738 paid out in rewards. I discovered a crocodile's nest containing fifty-five eggs. The native crocodile hunters use hook and line. The hook, or *alir*, as it is called by the Malays, is a simple contrivance made of wood, tied at the end of a tough bark rope. Another saurian, the gavial, is found in Borneo. It is not unlike that of the Ganges, called by Dr. Gray, *Tomistoma schlegelli*. This species inhabits the headwaters of some of the rivers, and is rarely seen. In the swampy forest near the coast, small reptiles are very abundant. There are pythons in Borneo twenty-four feet long. Two twelve-foot specimens were brought to me, and a monster python

twenty feet six inches long, I purchased alive in a cage, and put to death for its skin and skeleton.

Notwithstanding the contrary opinion of many observers, I think it can not be said with truth that Borneo is rich in bird-life. There are 392 species on the island. The finest bird is the argus pheasant. In life its plumage has a soft, velvety richness which is never seen in a dry specimen. These birds are extremely shy, and are taken by the natives in snares. Hornbills of several varieties are numerous. A bird of great commercial value is the swallow which builds the edible nest, so dear to the palate of the Chinese mandarin. These nests are built in caves, and are of a gelatinous substance resembling white glue. Their shape is like a small soup ladle with a broad, flat handle about an inch long. There are two kinds of nests, the white and the black, the former being most prized. A picul (133 pounds) of these is worth from two to three thousand dollars.

Borneo is favored with a great variety of very interesting mammals. So far as is at present known, there are ninety-six species, thirty-three of which are not found elsewhere. In apes and monkeys the island is especially rich. At the head of the list is the huge, red-haired orang-utan, of which we will speak presently. Then comes the long-nosed monkey, with its immense flabby proboscis. The *Nasalis* is a large species of monkey, found in the same localities as the orang, always over the water, and usually in large troops. It is something marvelous to watch a troop of monkeys, when terrified by an attack with firearms. They head directly away from the danger, and gallop madly through the tree-tops along the larger branches. Another interesting mammal is the long-armed ape, *Hylobates concolor*. This animal is extremely wary, and so rapid in its flight as to render pursuit exceedingly difficult. The flying lemur is also found here. Another curious monkey is the tarsier, a small, nocturnal animal. The krah, *Macacus cynomolgus*, actually swarms in the low trees along the river banks. The clouded leopard, the otter-cat, and civet cats of two species occur, and also several other small members of the cat tribe. Two species of bear are found, the smallest known. The Indian elephant occurs in the extreme northeastern part of the island; also the rhinoceros and tapir. These three are very rare. The thin-haired deer is very common in Sarawak Territory, and is frequently noosed by the natives. The muntjac, or rib-faced deer, is occasionally met. Wild hogs are very abundant and destructive. They sometimes measure

forty inches at the shoulder, and are good swimmers. Many beautiful squirrels are found here, and also, remarkable bats, the bear cat, otter, porcupines, and other small mammals which fall an easy prey to the hunter-naturalist.

The orang-utan is found only in Borneo and Sumatra, but is more abundant in the former island. It is most numerous in the Sarawak Territory. This animal occupies the fourth highest place in the animal kingdom—first, man; second, gorilla; third, chimpanzee; and fourth, orang-utan. This name signifies “Jungle-man,” and is derived from two Malay words, “orang,” man, and “utan,” jungle. The latter word is usually corrupted into “otang” or “outang.” The animal itself is rare and difficult to find. In August, 1878, I went on a hunting expedition for orangs to the Sadong River, at the mouth of which I settled and commenced prospecting. One day two men arrived from the headwaters of the Simujan River. They said they had seen two *mias* (orangs), and suggested that I should go up to their village for a week or so. This I did, and was very successful, taking thirty-one orangs during my first month. In my visit of three months I secured forty-three orangs. Of these twenty-seven fell to my rifle, the remainder being shot for me by natives. Our plan of hunting was to paddle leisurely up and down the streams in a Malay sampan, or dug-out canoe, and watch the tree-tops on both sides as far back as we could see. I was armed with a Maynard rifle and field glass, while three stout Malays or Dyaks furnished the motive power at the paddles. Once in sight of an orang it was a comparatively easy matter to send a ball into its breast. On one occasion, while paddling up the Simujan River on a bright forenoon in September, the Malay suddenly exclaimed “Mias! Mias! Tuan!” The other paddlers backed water at once, but we saw nothing until the boat had been backed several yards. Then we espied simply the knee of a large orang which was lying asleep on a branch about twenty feet above the water, and twenty yards from us. Its body was completely hidden by the foliage, so I stood up in the boat and fired at its leg to arouse it. It started up instantly, growling hoarsely with pain and rage, and started to swing away with a reach that was surprising in its length. Fortunately, the water was deep; there were no screw-pines to hinder our progress, and in a moment our sampan was directly under the old fellow, who then climbed high into the tree-top to escape us. It was a huge old *mias* chappin, the species with the expanded cheeks, long-haired, big and burly.

It growled savagely at us, and one of my Malays kept saying, “Chappin! Mias Chappin! Fire, sir, fire! That’s Mias Chappin. Big—big.” My companions were all intensely excited, but I knew the old fellow was ours, and waited for a good shot. In a moment the opportunity came, and I fired twice in quick succession at the orang’s breast. It stopped suddenly, hung for a moment by its hands, then its hold gave way, and it came plunging downward, snapped off a large dead limb on the way, and fell broadside into the water, with a tremendous splash which sent the spray flying all over us. As we seized the arms and pulled the massive head up to the surface of the water, the old fellow gave a great gasp, and looked reproachfully at us out of his half-closed eyes. I will never forget the strange, and even awful sensation with which I regarded the face of the dying monster. There was nothing in it in the least suggestive of anything human, but I felt as if I had shot some grim and terrible gnome or river-god—a satyr, indeed. It was a perfect giant in size, larger than even the natives had ever seen before. Its head, body and limbs were of grand proportions, and its weight could not have been much, if any, less than 190 pounds. This individual is now in the National Museum, to the extreme left of the group of oranges in the Mammal Hall. The tallest specimen I secured measured four feet six inches, but my largest one, that just described, measured half an inch less in height.

There are two species of orang found in Borneo: *Simia Wurmbii*, characterized in the males by very broad, flat cheek calossities, and *Simia satyrus*. English naturalists recognize a third species, *Simia morio*, but without any tenable grounds for doing so.

Orangs in a state of nature are seldom if ever seen on the ground. At night this animal builds a nest in the forks of a tree or on the top of a small sapling, by breaking off a quantity of green boughs, and piling them in the crotch. On these he lies upon his back, grasping with hands and feet the largest branches within reach. Orangs are perfectly harmless to human beings unless brought to bay on the ground. They are then as fierce as tigers. Their food consists of wild fruits, particularly the durian when in season, the tender shoots of the *Pandanus*, and the leaves of certain trees.

Although the aboriginal inhabitants of Borneo are divided into several tribes and scores of sub-tribes or clans, they may with reasonable exceptions be described as one body, or sub-race, viz.: Dyaks. In general

terms, a Dyak may be described as a Bornean semi-savage, of Malay extraction, with straight black hair, a yellowish brown complexion, and smooth face of the Malay type. He is rather below medium stature, but athletic, and of active and warlike disposition. He is usually clad only in a bark loin-cloth, but sometimes wears a sleeveless jacket, and particularly in war, on which occasions it is made of skins or padded cloth. He is armed with sword and spear, and possibly the *sumpitan* also for blowing poisoned arrows. He invariably lives in the jungle, in a long house-village set up high on posts. Although he has no religion whatever, and worships nothing, he has profound regard for the rights of property, respects his wife, and treats her and his children with the highest consideration. His sustenance is rice, fowls, pigs and fruit grown by himself, wild animals slain in the forest, and wild fruit, supplemented by a few things which he receives in exchange for wax, gum, rattans and gutta, although these are generally given for brass-wire, beads, cloth and other ornaments. He has no written languages, builds no monuments, makes no pottery, and only one kind of coarse cloth, carves rather neatly in wood, and works but little in iron. His bearing is independent, dignified, respectful. He is a trustworthy friend, but a dangerous foe.

In my judgment the aboriginal inhabitants of Borneo may be divided into four great tribes: The Kyans, Mongol Dyaks, Land Dyaks and Sea Dyaks. This classification differs very widely from any hitherto proposed.

[\[K\]](#)

The Kyan tribe is numerically the greatest, probably exceeding a quarter of a million. They are less civilized than the other tribes, are exceedingly warlike and aggressive. They decapitate their slain enemies, and keep the cleaned skulls as trophies.

The Mongol Dyaks inhabit northeastern Borneo. They have been greatly influenced by contact with the Chinese, with whom they have intermarried. In appearance they resemble the other Dyaks.

The Land Dyaks inhabit the country lying between the Sadong River and the headwaters of the Sambas, extending southward to the Kapurce, and an unknown distance beyond. They live inland, and differ in certain customs from their neighbors, the Sea Dyaks. The Land Dyaks are the only people in

Borneo who burn their dead. The warriors, though brave, are not fond of war for its own sake, nor are they possessed with an insatiable desire for plunder, as are the Kyans, and formerly the Sea Dyaks also. Their social customs closely resemble those of the Sea Dyaks.

The Sea Dyaks consist of seven clans, and occupy all the territory between the Rejang and Sadong rivers, from the sea-coast southward to the Kapurce. The Sarawak government estimates their number at 90,000, and the Land Dyaks at 35,000. The color of a typical Sea Dyak is dark brown, with a strong tinge of yellow. His hair is long and of a glossy black, and falls on his shoulders in graceful locks.

The Dyaks are happy and contented. Their wants are few, their diseases fewer, and their crimes fewer still. In hospitality, human sympathy and charity, they are not outranked by any people living, as far as I know, and their morals are as much superior to ours as our intelligence is beyond theirs. If happiness is the goal of human existence, the Dyak is much nearer to it than we. In this instance, at least, the highest civilization has not evolved the most perfect state of society. Is it possible that man reaches his highest moral development in a state of savagery? Is it then really true that as we increase in civilized intelligence, our capacities and propensities for wickedness increase likewise, and if so, will this always be the case with mankind?

[K] For elaborate discussions of these tribes the reader must consult Mr. Hornaday's book, "Ten Years in the Jungle, with Rifle and Knife," which has been announced by its publishers, Messrs. Charles Scribner's Sons, as nearly ready for circulation.

I am here because God has sent me to do a work that no other being could do but myself. Had there not been room for me, God had not made me. Had I not been needed in America, God had not placed me in America. Had I not work in the nineteenth century, I had not been born.... I have a place—am sent of God on a mission, and if I perform it God shall acknowledge that I have done His will.—*From Sermons by Bishop Simpson.*

THE WHAT-TO-DO CLUB.[\[L\]](#)

The ordinary village fails to get the best out of life. A candid examination of average boys or girls of the town or country, brought up without the influence of outside advantages, too often reveals the fact that they are not, in refinement, in resources or in thought, the equal of city young people. There is a painful feeling that they are narrow. Indeed, they feel this themselves, and complain that they have “no opportunities.” At the same time the narrow life does not shield them from temptation, and there are almost as many young men in America going to ruin under the narrowing influences of country and town life as in the whirl of cities.

Among women the influence is evident. They are, it is true, largely free from the temptations of frivolity, extravagance and dissipation, but they are subject to temptations of no light weight. Their few interests lead them to gossiping, prying and criticising. Lines of class distinction are drawn so painfully tight that their lives become narrow in sympathies and associations. Very largely they lack independence of spirit to help them dare untried lines of conduct. Many of our American villages and “corners” are the most trying places in the land in which to live. Few dare to try improvements, enthusiasm meets little or no response, ideas travel slowly. Village life *looks* ideal to one wearied by the rush and wickedness of a city, but there is in it a peculiarly benumbing influence which is all the more difficult to contend against because so silent in its action. Yet there are two of the best conditions for high living in the surroundings of town and country. There are leisure and quiet. Anything which will impregnate this rare life with enthusiasm and energy will furnish the happiest conditions for noble action and steady growth.

It is not an easy problem for a reformer in such a locality, but we believe Mrs. Campbell in her “What-to-do Club” offers a solution which will rarely fail among girls and women. “The What-to-do Club” is an unpretending story, but it has a practical grip on this question. It introduces us to a dozen or more village girls of varying ranks. One has had superior opportunities;

another exceptional training; two or three have been “away to school;” some are farmers’ daughters; there is a teacher, two or three poor self-supporters—in fact, about such an assemblage as any town between New York and Chicago might give us. But while there is a large enough company to furnish a delightful coterie, there is absolutely no social life among them. The differences in their opportunities they have exaggerated until they feel that their interests are as unlike as those of Fijis and Bostonians. They look at each other with curiosity merely, and all of them are bored by the dullness of their lives. Mrs. Campbell puts a wise woman into their midst. This woman’s experience has taught her that the barrier between women of different sets is largely their ignorance of each other, their belief that they have nothing in common. She finds something in common for these girls. By a little tact, exerted at a village gathering, she interests them in herself. A second stroke of policy finds them gathered in her parlor and she clinches her work by giving them an insight into practical employments—not pleasures, mind you—but work, for women at home. The interest excited quickens them all. They become alert, capable, quick-witted, and suddenly see in each other much of which they had never before dreamed. The false barriers between women invariably fall before a common interest. Show them how strangely their minds and lives are alike and the sympathy of similarity makes friends of them. So the girls of the “What-to-do Club” found, at any rate. Their meetings became voyages of discovery. Their discoveries were El Dorados to many a one of their number perplexed by the want of pin-money, or worse still, of bread-money. Simple, practical, at-home occupations for leisure hours was the first study, and it is marvelous what a number they found. One young lady undertakes strawberry culture, and in a single season clears, off a quarter of an acre, \$154.65. Better still, her vigorous out-of-door life transforms a pair of pale, hollow cheeks until they are rosy and plump, and awakens healthful interest which soon makes a happy heart out of a very discontented one. A half acre put into small fruits, currants, raspberries and blackberries, opens the way for an active young philanthropist to start a fund for a future kindergarten for her father’s employés. It does more. It opens the young lady’s eyes to the dignity of work, puts a bond of sympathy between her and the people who work for her, and strengthens the common sense of her whole family. Our strawberry girl tries poultry and finds it the most delightful of employments. It pays her, too, one season’s work yielding a clear profit of

\$86.56 on an expenditure of \$73.40. Bees, with their fascinating history, their exciting family affairs, their industrious honey making, and their clear, unfailing profit came in for one young Busybody's attention, and in a single season this young merchant clears \$113.94. One girl tries silk worms and sends to the club this report of her summer's work:

One ounce of eggs	\$5 00
Fixtures for cocoonery	5 00
	\$10 00
36 pounds stifled cocoons at \$1 per pound	\$36 00
Profit	
	\$26 00

One of the best discoveries which the club makes is of the possibilities in fruit canning, jelly making, and, best of all, fruit evaporating. Like "Dorothy" of the "club," when we read of the wonders of the latter we burned to "live in an orchard and evaporate everything that grows." How wonderful it seemed to these girls to whom fruit preserving had been bounded by the limits of the fruit closet and the demands of the table, to put up jelly for market, to "take in" canning for people too busy to do their own, to dry fruit in that wonderful evaporator, which would sell in any market in the country.

It is not strange that these new ideas put into their lives new possibilities. It showed them that there was something to do at home, something which was more than a paying employment. For these out-of-door interests are more. They are health-giving, awakening pursuits. The girl that engages in such enterprises wins more than a few dollars; she cultivates the business faculty and arouses a dormant independence which makes a new creature out of her. This new interest in the lives of Mrs. Campbell's girls gave them an interest, at first in purely money making enterprises, but it soon knit them into friends. Their friendship spread until they found themselves reading, studying, planning, as one body. The influence in the story energizes the community. It is, perhaps, quite possible that in a real club we might meet with more discouragements, but it is impossible that we fail entirely.

Town and country need more improving, enthusiastic work to redeem them from barrenness and indolence. Our girls need a chance to do

independent work, to study practical business, to fill their minds with other thoughts than the petty doings of neighbors. A What-to-do Club is one step toward higher village life. It is one step toward disinfecting a neighborhood of the poisonous gossip which floats like a pestilence around localities which ought to furnish the most desirable homes in our country.

[\[L\]](#) The What-to-do Club. A Story for Girls. By Helen Campbell. Boston: Roberts Brothers. 1885.

CRITICISMS.

BY CHANCELLOR J. H. VINCENT, D.D.

The "Chautauqua Literary and Scientific Circle," with the general Chautauqua movement, has had its share of criticism. Its advantages have been pointed out, and sometimes magnified. Its managers have had their attention called to the dangers and defects of the system. Personally, I enjoy adverse criticism and the practical counsel which it has brought quite as much as words of praise, for praise may paralyze effort, while the goad of the critic is likely to stimulate both ingenuity and resolution.

The members of the C. L. S. C. have from the beginning been encouraged to express freely to the Superintendent of Instruction their dissatisfaction with either text-books or methods. As a result of this freedom, vigilance has been promoted, and many improvements have been from time to time introduced.

The aims of the C. L. S. C. are unique. The provision of text-books precisely adapted to these unique aims has been one of the ever-present problems. If our readers were children in the school room, and daily recitations were practicable, it would be easy to find suitable text-books on every subject in the curriculum. If these readers were chiefly high school or college graduates desiring advanced courses of reading, it would be comparatively easy to provide standard works written by specialists for specialists, and assuming on every page a large measure of knowledge already possessed by the reader. If it were the aim of the C. L. S. C. to study one subject at a time, and that for a long time, exhaustively, from its alphabet to its "last word," it would not be difficult to find numerous text-books on that subject adapted to every variety of capacity and attainment.

The C. L. S. C. is not, however, designed for school children, nor for advanced readers, nor for specialists. It has enrolled but few names of members under eighteen years of age. Its members are "out of school." It

rejoices in thousands of college graduates, but these take up its readings not for advanced study as post-graduates, but to review under favorable conditions the scholastic studies of former years, and in some cases, perchance, to make amends for carelessness and superficiality during those years of unappreciated opportunity.

The C. L. S. C. is therefore a “school of reading at home” for college graduates who desire, whatever the motive, to review the college course, and for people who, having been deprived of early educational opportunity, desire by a general course of reading to place themselves in sympathy with the school and college world; to know something of the educational courses now being pursued by their children; to test their own powers by a survey of the varied field of letters, and thus by our four years’ superficial course of reading prepare for special studies further on. The C. L. S. C. aims to provide, therefore, first for the four years’ general course, and afterward for the special studies.

The scope of the four years’ course is the usual college curriculum. With this aim we began. To this aim we adhere. The success of the scheme in promoting intellectual quickening and activity has been attested by thousands who have tried it for several years.

Here lies the chief cause of our embarrassment. It is difficult to provide books precisely adapted to the needs of our peculiar constituency. The Superintendent of Instruction and the Counselors have felt this from the beginning. Heavy and elaborate books discourage a class which we are anxious to lure into the love of literature. Books too much abridged fail to satisfy more mature minds. Old books may be behind the times, or, although acknowledged to be standards, may not for the reasons above given be fully adapted to our readers. As for new books—every one knows how hard it is to secure them, and how easily a flippant criticism may destroy the confidence of the uninitiated in them. Notwithstanding these embarrassments we have tried to do our best, providing old books where the council could agree upon them, and new books where they seemed to be absolutely necessary.

It is not to be expected that any book, especially any new book, will meet with universal approval. As for criticism—well, who knoweth the

ways of critics with the new books! Did not Samuel Taylor Coleridge say of Burke's essay on "The Sublime and the Beautiful," "It seems to me a poor thing?" Did not Horace Walpole call Goldsmith "an inspired idiot?" Did not Dr. Johnson pronounce Fielding a "blockhead?" Does not Hume affirm that "no page of Shakspeare is without glaring faults?" Was not the manuscript of Thackeray's "Vanity Fair" rejected because the critic to whom it was submitted pronounced it "without interest?"

Some books of the C. L. S. C. have excited unfavorable criticism—religious books by those who do not care to read religious books at all, and think it an impertinence to obtrude them upon the general reader; certain scientific books because "not up to the times," or the critic being himself a scientific specialist is certain that the views of our specialist are "unsound."

Concerning one of the books on the list, a correspondent says:

"It is useless—worse than useless; it is harmful. Its style is involved, obscure, bombastic, florid, 'highfalutin',' diffuse, disfigured by straining after effect, by the effort for fine writing, and by many evil features carefully to be avoided."

I do not quote the above as a specimen of classic English, but as indicating the temper of the writer, whose letter is accompanied by nineteen manuscript pages of closely written criticisms upon the condemned volume.

Concerning this same book, a high authority in English to whom it was submitted, has said: "It is a clear, compact, and readable statement of the laws and principles of speech." A Boston writer of ability had said: "This little volume is the very best text-book for any one desiring to perfect himself in the laws and principles of speech. It is grammar, rhetoric and composition combined, and is doubly worth its price." A Philadelphia critic had said: "A better treatise could not be placed in the hands of a student who has not been initiated into the intricacies which make prose composition an intolerable bore to the young."

Other and equally strong commendations of this book might be quoted, commendations which were received from trustworthy authorities before it was placed on our list. I submitted the volume to one of the best literary critics in the country, who called attention to some errors which needed

correction. Owing to the illness of the author, or for some other reason, his attention was not called to the corrections required, and therefore numerous minor defects, which would have been carefully remedied by its competent and scholarly author, appeared in the new edition. Dr. Johnson made six thousand alterations in the second edition of "The Rambler." But for the oversight, for which I fear I must acknowledge myself responsible, the volume under consideration would have been thoroughly revised.

Many local circles have as leaders men of literary ability and scholarship who, prepared for such wise service by the humility which comes from years of educational experience, have pointed out these defects, at the same time fairly representing the true value of the book, and putting emphasis upon its admirable instructions which, by hyper-criticism, may have been lost sight of.

Concerning another book on the course, a critic says:

"As a close student of the classics for years past, I must say that I think there is very little scholarship displayed or employed in Prof. Wilkinson's work on Greek literature.... Further, the arrangement is senseless, even harmful. Literature is a growth, and largely the reflex of the people's life and thought. It must then be treated historically, and not in the topsy-turvy fashion of Prof. Wilkinson."

The same writer proposed another series of works on ancient Greek literature, as a substitute for the two volumes of Prof. Wilkinson. The series he proposed, however, contained an amount of matter which would prove utterly discouraging to our readers and which would cost ten times as much as the more condensed work of Dr. Wilkinson. Besides all this, the work of Dr. Wilkinson meets the object of the C. L. S. C.

When one gets into the world of criticism, he finds himself among "doctors" who "disagree." I have quoted one view of Dr. Wilkinson's textbooks, presented by our unknown critic. Now for another. Dr. Howard Crosby, of New York, a most finished scholar and close critic, and a judge of both English and Greek literature, says: "Dr. Wilkinson's Greek course is clear, attractive, judicious in its treatment of the subject, and fills a valuable place in literature." Of the second volume, the same scholar says: "The new

volume is thoroughly attractive. It is fully up to the high standard of the other." Such commendation as this sustained our earliest judgment of the works in question. Prof. Frieze, of the Latin Department of the University of Michigan, says: "I have not yet seen the 'Preparatory Latin Course in English,' though I was favored with a copy of the Greek. I have only to say that if the Latin equals the Greek it can not fail to be a contribution to classical culture both for classical and English scholars, of very great value. I have been delighted with a perusal of Prof. Wilkinson's critical notices, his own translations, and his selections of the translations of others, and I sincerely congratulate him on the admirable style in which he has presented the matter, as well as the character of the matter itself, and the plan of the whole work."

Prof. A. C. Kendrick, D.D., head of the Department of Greek in the University of Rochester, says:

"The plan of the work is quite unique, yet certainly adapted to the wants of a large and increasing class of young persons in our country. Its execution seems to me very felicitous; it is marked by the taste and scholarship which were to be expected from its accomplished author."

Dr. Alvah Hovey, President of the Theological Institution, writes to Dr. Wilkinson:

"In these latter days I do not often read a volume through from beginning to end without omitting a chapter, paragraph, or sentence. But I have read in this way your 'Preparatory Greek Course,' simply because it is so instructive and captivating a volume that I could not persuade myself to pass over any word of it unread."

The Rev. A. P. Peabody, D.D., LL.D., late of Harvard University, writes:

"I have looked through Mr. Wilkinson's 'Preparatory Greek Course in English,' and am prepared to give it my warmest commendation. It supplies a need which is more and more felt from year to year, for two reasons, one for which I rejoice, the higher standard of culture that prevails in society at large; the

other, inevitable, yet to me a subject of regret, the diminishing disposition on the part of well-educated people to study the classical languages.”

The Boston *Watchman* adds to such commendations as the above the following testimony:

“The plan is an ingenious one, and is carried out with spirit. It need not be said that the fine literary taste and critical acumen of Prof. Wilkinson are shown to much advantage. Only by a fair trial can the practical worth of such a series be proved. We have found pleasure in reading the volume.”

In suggesting to Dr. Wilkinson the idea of this “After-school Series,” I requested him:

1. To give in two volumes the substance of what the college boy in his preparatory and college course would learn of Greek literature—not the language, but the literature;

2. To put into his books as far as possible the method and spirit of the college recitation room, discussing collateral topics, introducing biographical and classical incidents, and employing illustrations from modern life and literature; to put his individuality into the work, so that teacher and pupils might be brought into friendly relations, and thus something of the animation and enthusiasm of the recitation room be enjoyed by solitary students.

It is Dr. Wilkinson’s attempt to realize this idea that produces the impression upon one critic of the “uninstructive chattiness” of the author. To a man who has just spent eight years in the study of the classics, and who makes them a specialty, there may be some things in Dr. Wilkinson’s book which are not instructive; to people for whom the book was written, there is not an uninstructive page in the book. Perfection in the recitation room may not be possible. Qualities in the *viva voce* teacher which attract and delight and benefit one student may not so favorably impress, and may sometimes almost annoy, another. A member of the Circle writes (in reference, no doubt, to Dr. Wilkinson’s book): “One author frequently converses, as it were, with the reader, telling him in a friendly way of the

many things he will relate after a while.... The book has caused the Circle to be ridiculed, and I could not think it was not without cause.”

I can readily see how a college graduate, just released from the recitation room, with lofty ideals of scholarship, and with really a vast amount of knowledge, might depreciate with a tone of contempt such a work as that of Dr. Wilkinson. I can see, too, how that smile of contempt from a scholar with local reputation might annoy and afflict less cultivated people belonging to a local circle who, devoted to an institution, are anxious that it and its text-books should receive the commendation of cultivated men. Just such commendation Dr. Wilkinson’s books have received. There may be now and then a slight tone of “chattiness.” There may be too frequent “forecasting of plan and purpose,” but on the whole the Professor’s work is admirably done, and has received the unqualified approval of our best students, men and women of the highest culture, eminent professors of Greek and Latin who fully understand and appreciate the aim of the author. I can assure my correspondent that there is nothing in Dr. Wilkinson’s books to cause the Circle or the books to be “ridiculed” by any true scholar.

A recent university graduate, and I have no doubt a brilliant scholar, writes:

“I wonder if it is safe to hint that, while the Chautauqua Idea is a noble and praiseworthy one, it is possible that the working out of the Idea may be defective?... When the members of the Circle read so faithfully the works prescribed, giving in many cases time that can ill be spared, it is but just that the very best should be given them to study, that alone will be of profit to the members, and help them to grow.”

The same writer pleads for “vigorous supervision by scholars and authorities on the respective subjects as the only thing that will enable the Chautauqua Idea to be carried out in a way that will help, and make them better and stronger in thought and life.”

This sentiment meets my heartiest approval. Indefatigably and conscientiously have the Superintendent of Instruction and the Counselors sought to do this very work for the readers who seek their direction.

It would surprise our friends to examine our budget of criticisms *pro* and *con*, from all classes of people; from public school teachers, college professors, ministers, post-graduates, classicists, scientists, so called “self-made men,” and people who, professing to know almost nothing, seek advice and offer counsel. We have diligently sought to profit by the things which have been said.

Our readers must see the difficulties which encompass us; the wide diversity of opinion concerning certain books, and the impossibility of securing works which will receive universal approval.

There are persons who do not believe in popular education at all. A recent correspondent, a man of immense wealth, wrote: “Mechanics and sewing women should confine themselves to industrial education, and not aspire to the knowledge of literature and art.” Would it be possible to produce works on literature and art for the people which a man of that type could approve?

An author of some pretensions, without much reputation in literary lines, tried to place a work of his own on the list of the C. L. S. C. *in lieu* of one on the same subject already adopted. Failing to win a place for his own, he proceeded in another book savagely to criticise the preferred volume. Would it be possible for this disappointed author to approve any book on his specialty that might be placed upon our course?

A certain youthful professor in an American college sneered at the idea of anybody enjoying the poetry of Homer or of Virgil unless he could read it in the original. Would it be possible for this literary fop to appreciate the books which seek to present the best thoughts of the old authors in classic English?

Dear fellow-student: Feel free to offer criticisms which may be helpful to the Board of Counsel. We do not modify our policy for every criticism received. But we weigh conscientiously and carefully all that is said in favor of or against the prescribed books. From year to year our course has been modified. I stand ready at all times to accept the best books; to abandon the best we have for anything better that may be placed within our reach. And as our experience broadens, helpful criticisms multiply, and

authors understand our peculiar needs, we shall approach more and more nearly to the ideals which now shine above us.

Do not, I beseech you, fail to protest against false, querulous and impertinent criticisms, and against that hyper-criticism which delights in nothing so much as in pointing out faults and defects, losing sight of the great things in excessive eagerness to detect slight inaccuracies.

Remember that no book is placed upon the course that does not have the personal approval of the best critics, and remember, moreover, that it will never be possible to provide a book which is above criticism. As one of our Counselors writes:

“Good books have always been criticised upon some points adversely. Plato freely criticises Homer. Quintilian criticises Cicero. Cicero criticises Demosthenes. Addison criticises Milton. And in each instance no doubt real faults were pointed out. The most enlightened French critics used to pooh-pooh Shakspeare. They did likewise with Dante.”

College students, with all their admiration for the professors under whom they moved through four years of study, have some foibles and defects to report and laugh at; but on the whole they honor the men who made them and led them. The authors of our text-books are our professors. On the whole they have done their work well. It is proper to note their faults and avoid them, but in defending them, and in being proud of them, and in rejoicing in the course of reading which they have provided, we have the endorsement of wise, scholarly and experienced educators.

Finally, let us learn the characteristics of the true critic, and according to the measure of our ability let us seek to possess them:

“A critic must have breadth, accuracy, sympathy, reverence, and love. He must have no partialities, and no aversions. He must not be captious, but just.”

OUTLINE AND PROGRAMS.

OUTLINE OF REQUIRED READINGS FOR JUNE.

First Week (ending June 8).—1. “The Mechanism of English,” in THE CHAUTAUQUAN.

2. Sunday Readings for June 7, in THE CHAUTAUQUAN.

Second Week (ending June 15).—1. “Easy Lessons in Animal Biology,” in THE CHAUTAUQUAN.

2. Sunday Readings for June 14, in THE CHAUTAUQUAN.

Third Week (ending June 22).—1. “Home Studies in Chemistry,” in THE CHAUTAUQUAN.

2. Sunday Readings for June 21, in THE CHAUTAUQUAN.

Fourth Week (ending June 30).—1. “The Heart Busy with Things About Us,” in THE CHAUTAUQUAN.

2. Sunday Readings for June 28, in THE CHAUTAUQUAN.

PROGRAMS FOR LOCAL CIRCLE WORK.

FIRST WEEK IN JUNE.

1. A Review Lesson—The History of Alexander.

2. Selection—“The Prayer of Agassiz.” By Whittier.

3. Reading—Story of “Perseus.” From “The Heroes.” By Charles Kingsley. [See “Talk About Books,” in THE CHAUTAUQUAN for May, 1885; also poem in present number.]

Music.

4. *Conversazione*—The Cause of the Trouble between England and Russia.

5. Selection—“Davie; an Epistle to a Brother Poet.” By Burns.

6. Critic’s Report.

SECOND WEEK IN JUNE.

1. Selection—Alcibiades. From “Plutarch’s Lives,” or from “The Young Folks’ Plutarch.”

2. A Paper on Our Local Birds.

3. Recitation—“Pegasus in Pound.” By Longfellow.

Music.

4. Essay—Alchemy. [Beside giving definition and history, allusion might be made to its introduction into fiction. See Goethe’s “Faust,” Scott’s “Kenilworth” and “Antiquary,” Bulwer’s “Last Days of Pompeii,” Hawthorne’s “Birth Mark,” Hoffman’s “Sand Man” in “Weird Tales,” and many other works which these will suggest.]

5. A General Talk on the Rebellion in the Canadian Provinces.

6. A Paper on Richard Grant White.

THIRD WEEK IN JUNE.—MONTHLY PARLOR MEETING.

Music.

1. A Paper on the Practical Education of American Girls.

2. A Character Sketch—Louis Agassiz.

Music.

3. Selection—"The Tragedy of the Night Moth." By Carlyle.

4. A Sketch of "Edie Ochiltre," the Beggar in Scott's "Antiquary."

Music.

5. Essay—The Parthenon; its History, Description, and Scattered Remains.

6. A Talk on Alaska.

Music.

FOURTH WEEK IN JUNE.—CLOSING EVENING.

1. A Report Presenting a Summary of the Year's Work.

2. Selection—"Song of the Greeks." By Thomas Campbell.

Music.

3. Essay—Science and Art in Housekeeping.

4. Recitation—The Dinner Hour. From "Lucille," Part I., Canto II., 23d and 24th stanzas.

Music.

5. A Paper on Schliemann's Researches in Troy.

6. A Half-hour Good-by Social.

The following suggestions are also offered for the closing exercises:

A C. L. S. C. banquet followed by toasts.

A luncheon party and charades.

Readings connected with any part of the year's work, illustrated by tableaux.

An evening of games, such as "Characters" or "Twenty Questions," in which one of the company selects a character or an object, and is then to be questioned by all the rest until they find out what he has in mind. The questions must be asked in such a way that the reply can be "yes," "no," or "I don't know." The company who can not guess rightly before the twentieth question is to be considered dull.

"Throwing Light" is also interesting. One of the party begins to tell a story, concealing all names. When any one thinks he knows what it is about he raises his hand, takes up the story, and goes on with it until a third is enlightened and proceeds with the narrative, and so on until it is evident to all in the room.

LOCAL CIRCLES.

C. L. S. C. MOTTOES.

“We Study the Word and the Works of God.”—“Let us keep our Heavenly Father in the Midst.”—“Never be Discouraged.”

C. L. S. C. MEMORIAL DAYS.

1. OPENING DAY—October 1.
2. BRYANT DAY—November 3.
3. SPECIAL SUNDAY—November, second Sunday.
4. MILTON DAY—December 9.
5. COLLEGE DAY—January, last Thursday.
6. SPECIAL SUNDAY—February, second Sunday.
7. FOUNDER’S DAY—February 23.
8. LONGFELLOW DAY—February 27.
9. SHAKSPERE DAY—April 23.
10. ADDISON DAY—May 1.
11. SPECIAL SUNDAY—May, second Sunday.
12. SPECIAL SUNDAY—July, second Sunday.
13. INAUGURATION DAY—August, first Saturday after first Tuesday; anniversary of C. L. S. C. at Chautauqua.

14. ST. PAUL'S DAY—August, second Saturday after first Tuesday; anniversary of the dedication of St. Paul's Grove at Chautauqua.

15. COMMENCEMENT DAY—August, third Tuesday.

16. GARFIELD DAY—September 19.

The great value in doing solid reading is that it enhances the value of all other experience. A member of the class of '87, writing from ROME, ITALY, says: "I think the course for this year particularly interesting, especially for our party, which is to spend the spring months among the ruins of that noble capital, Athens." The reading our friend has been doing has recommended itself to her foreign friends, who have made many inquiries about the course of reading, even wishing to be enrolled as members of our great Circle. The teacher over seas, like the tourist, finds benefit in the work. A member of the class of '83, writing from BULGARIA, says: "The time from six to seven in the morning is all that I can spare from my other work, but it makes the day brighter to begin it in this way, and so I read and study and think, and get charming glimpses of home circles. *Next year* I mean to have a circle here, and hope to know enough of the language to put some of the best things into Bulgarian."

On the home side of the Atlantic, circle life seems to be vigorous and growing. The organizations of past years hold to the work, and many new fields are opened monthly. At BROCKVILLE, ONTARIO, the "Island City" branch of thirteen members was organized last October. A simple but very effective method of work has been followed, that of appointing examiners on each subject, different members being appointed from time to time, so that all may be kept interested. They have had for president a college graduate to help them over tough places in their classics, and a practical chemist is proving the marvels of chemistry to them.

LIVERPOOL, NOVA SCOTIA, sends an encouraging word of the progress of their circle, formed in 1883, and now boasting twelve active members. The plan for the evening's work in this circle is very good. Introductory exercises of prayer, minutes of last meeting, roll call, responded to by quotations, and a select reading, precede the evening drill, which is an informal conversation on one or more of the C. L. S. C. studies, conducted

by a leader.—We quite agree with our friend at NIAGARA, who thinks that Chautauquans ought to know that there are other places in ONTARIO beside Toronto where circles are doing good work, and that their town is one of them. If all the Ontario towns have as bright and brave circles as Niagara, we most certainly hope we shall hear from them. “In December, 1882,” so their history runs, “we formed a triangle, with angles of various degrees of acuteness or obtuseness, but did not commence work till January, and being three months behind time we found the work rather heavy, but in proof of our zeal can report that not one of us that year ever missed any of our fortnightly meetings. In October, 1883, we were joined by another member, not less acute, and ‘stood four-square to all the winds that blow.’ We expected to form a real circle, with all our angularities smoothed off, but find ourselves this year with the original three, but from sickness and other reasons have had our meetings sadly broken in upon, so that we need to use our motto, ‘Never be discouraged.’ So far we have all enjoyed the work. We all lead busy lives, without our Chautauqua studies, but we hope that they will give to our lives sweetness and strength, and breadth and power. Many of us have done more reading each year than the course, but then such reading is apt to be desultory, and the fact of studying with others gives greater interest. We have generally kept the memorial days, and we all hope to visit Chautauqua some time.”

A step has been taken in western MAINE to form a Chautauqua association. AUBURN, LEWISTOWN and adjoining towns have many members of the C. L. S. C. In March a meeting was held in Auburn, and after a banquet a motion was passed providing for an associated circle of all the C. L. S. C.s of Western Maine. True to the Chautauqua instinct, they are going to have a summer meeting. MARANOCOOK is the chosen place, and June is the time. Western Maine Chautauquans have the hearty good wishes of us all for a delightful summer session. At their March meeting, these friends passed a just and appreciative resolution of gratitude to Chancellor Vincent for the happy thought that conceived the Chautauqua Idea, and the untiring and well directed zeal that has made it so efficient.—At WOODFORDS, MAINE, the “Arlington” circle of fourteen members held twenty-five meetings during the winter. Such zealous work justifies their claim that they possess the “enchanted number” for a circle. The Arlington proposes a parlor entertainment for a near day.

GREENLAND N. H. has two very strong Chautauqua organizations, the "Baketel" circle and the "Spare Minute Class." Founder's Day was celebrated with great *éclat* by these warm admirers of Chancellor Vincent. A public meeting was held in the town hall, with exercises of song, recitation, reading and tableaux. The last tableau was so characteristic it deserves a description. It was "The Chautauquans at Home," and represented the entire local circle and spare minute class at work. One was rocking the cradle and reading, another was at the ironing table with an open book, and several were sewing and studying at the same time. A happy close to the evening was a presentation to the honored president of the circle, the Rev. O. S. Baketel, of an elegant easy chair.

Among the senior circles it is pleasant to be able to count in that of NORTH GROTON, N. H. The secretary says: "The 'Angle' has kept silent since 1882, thinking that only two was a small number to report as a local circle, yet all this time we have met at every opportunity, for reading and questioning. We had long wished for guidance in home study, and the 'course' was eagerly taken as soon as heard of. This year two earnest '88s have joined us. With us 'Chautauqua has come to stay.' 'For,' as Dr. Vincent said at Framingham, 'goals yet grander wait our winning on the mountains by and by.'"

And now we have a nautical circle. The first mate, so we imagine, has sent us notes from the log-book, running over their whole course. Perhaps their sailing may guide another crew: "We have read in THE CHAUTAUQUAN, from month to month, many interesting reports from local circles, which have been like fresh breezes to our own sails. One would judge from reading that *success* was stamped upon *all* methods of circle work. We have thought sometimes that a part of the unwritten history of such work might be helpful. We suspect that—way back in the annals—some things were undertaken that did not turn out just right, and a few chapters from our that experience might save many a small boat from going to pieces in dangerous waters. In general, the 'Vincent' circle, of WEST BRATTLEBORO, VERMONT, has had fair sailing from port to port, set down on the C. L. S. C. line of travel. Four years ago a few of us began floating, not knowing enough ourselves of whither we were bound, to give a general invitation for 'all aboard.' At last, reinforced, we tried to go in two separate boats, which kept just near enough together, and just far enough apart, to render such a division of the

crew unnecessary, to say the least. One year we failed to set sail soon enough, and came in late at every port. This year we make one crew, under one efficient captain, a few only compelled by circumstances, or preferring to go in their own little skiffs. October 1, 1884, we set sail, and made directly for the shores of Greece. Landing under the direction of two well chosen generals, we scattered to spy out the land, bringing in such reports as we were able, at the time appointed. On the evening of November 3d we gathered to do reverence to the memory of one who had wished us 'Godspeed.' About that time experiments in chemistry furnished us with illumination. Since the observance of that memorial day, we have known that we were still landed upon the shores of Greece, but reports concerning the country and its people, past and present, have been few, for finding ourselves in danger of forgetting, rather never having known, our 'mother tongue,' we have spent some time in practice of the 'art of speech,' and, loyal to our native land, we have observed some of its festivals, and repeated to each other words from those we all delight to honor."

For faithful work few circles can exceed that in FRANKLIN, MASS. The circle numbers sixty-six members, nineteen of them being of the class of '88. Meetings are held fortnightly. Not one regular meeting has been omitted since October 1, 1883. The president, although pastor of a large church, has been absent but five times since the circle was formed—November, 1882. The work done at these meetings is solid review of past readings. "Founder's Day" was observed by accepting an invitation from the "Star" circle, FOXBORO, MASS., to visit them and engage in a "Question Match" on Greek History, after which a bountiful collation was served, and a social time enjoyed by all. Each heart felt grateful to Chancellor Vincent for the C. L. S. C.—At ROCKPORT, MASS., the circle has fallen from fifteen to five, but the five faithful seem to enjoy their work too well to need any commiserating. They are so interested in their studies they actually don't realize they have grown smaller—a method of taking things which removes the stings from all falls in fortune. The "Granite" circle turns each alternate meeting into a Round-Table, and finds that the plan works capitally. The circle is anticipating the pleasure of going in a body to Framingham Assembly this summer.

MASSACHUSETTS is the banner State again. At COCHESETT twenty-four members form a circle of "Plymouth Rocks." The Longfellow memorial

was their first experience with special days; a successful experiment, too, we judge, for they have decided to continue the plan for the remainder of the year. Cochesett is an Assembly offspring—a child of Framingham, which gave the first interest to its zealous founder.—The “Philomaths,” of NEW BEDFORD, send greetings to all our fraternity. Since their reorganization in October the circle has resolved itself into groups of six or seven, which hold weekly round-tables for thorough study.—It is wonderful what enormous interest some circles get on their investments. Here is the “Bryant,” of WORCESTER, MASS., actually making 100 per cent. in less than two years. In the fall of 1883 they began with twenty members, to-day they number forty. Is it far-fetched to attribute something of their success to the “question basket,” which forms an important part of their program?—The “Alphas,” thirty-two in number, of ATTLEBOROUGH, send us some capital hints for our programs. At a Greek evening recently, ten of their number were selected to give brief descriptions of Greek heroes—the rest of the company guessed the hero described. A half hour was also spent at the Mardigras, and a friend, fresh from the carnival scenes, described his experiences among the merry masqueraders. At an hour of electrical experiments a very happy device was exhibited recently before the circle. The electricity played over tin-foil, with grooves an eighth of an inch apart, and through a stencil-like arrangement showed to the surprise of the circle the illuminated letters, “C. L. S. C.”—*The Saturday Union*, of LYNN, MASS., is doing most effective work for the Chautauquans of its vicinity, and a strong body of workers it has to serve. By a late issue we notice that there are in the city six circles, and the list of Chautauquans, which appears with their residences in the same paper, includes 114 names. This “goodly companie” has given a course of lectures this winter; the seventh in the course was on “Electricity”—and by a lady who, we are happy to say, illustrated her talk by apparatus of her own making. Our Chautauqua women! How proud we are growing of their ability, their pluck, their womanliness!—NORTH BROOKFIELD’S circle has recently been favored by a poem on “The Chautauqua Idea,” also by a woman.—It would be unjust to allow a mistake, which found its way into last month’s “Local Circles,” to go uncorrected. At the Longfellow celebration held by the Chautauquans of Boston and vicinity, there were five hundred instead of fifty persons present.

The treasurer of the "Hall on the Hill" to be erected at Framingham this year, paid a well deserved compliment to the "Clio" circle, of PROVIDENCE, R. I., when he said that he wished there were more circles like the "Clio." These energetic friends took Chancellor Vincent at his word last summer, when he promised to lecture for any circle which would pledge \$200 to the "Hall on the Hill." They raised the money, had their lecture, and are satisfied. Their hard work seems to have only whetted their intellectual appetites, for they have had a long list of brilliant talks by distinguished men in addition to regular circle work. By the way, the New England branch of the class of '87 was to hold a meeting in Providence in April, and the "Clio" was to act as hostess. What was done? A Providence neighbor of the "Clio" is the "Channing" circle of twenty-five members. We notice that these Providence friends use a very pretty and appropriate heading on their letter paper. At the top of the sheet, in the corners, appear the names of president, vice president and secretary, and below "Headquarters of the Channing Circle of the C. L. S. C."

A party of DANBURY, CONN., Chautauquans went abroad one night not long ago to see "Athens in the Golden Age." A delightful time these tourists had. They made the passage of the Mediterranean Sea, and at Athens visited the Acropolis, went to the Areopagus to listen to Pericles, called on Xantippe, and did a hundred more interesting things, at last coming home via "Plymouth Rock." The "Nestors," however, do much beside travel. They have an excellent method of working the question box, which is an inevitable part of their program. The questions are gathered just before adjournment, and shuffled, each member drawing one. The first exercise after roll call at the next meeting is the answering of these queries. The imaginary trips which our Danbury friends like so well, the "Alpha" circle, of NORWICH, makes a part of each evening's work. When they journeyed from Boston to New York they went with Howells in a "Sleeping Car." They have lived over, on paper, all the preparations for the trip abroad, the life on the steamer, and have done the sight-seeing of the British Isles. These tourists have enlivened their travels by many a happy device. Once it was a *conversazione*, and again, in preparation, perhaps, for their visit to Athens, a pronouncing match on Greek proper names. What wonder they have had a phenomenal growth! Last year there was not a circle in Norwich, now there are six. When the "Alpha" organized last fall, it was

with eleven members, to-day they have fifty. Nothing to be surprised at, perhaps. It seems to have been "good growing weather" for the C. L. S. C., throughout New England.—MERIDEN, CONN., has had the common experience, the circle having increased to sixty-six members. They have found the key to the mastery of the Greek names and chemical terms. The pronouncing matches have unlocked the doors and the fortunate Meridenites are able to talk glibly on their Grecian rambles. A second circle, "Hanging Hills Class," was organized last fall in Meriden. It has grown to twenty members, who are doing superior work. This class observed Longfellow's Day appropriately. The '89 outlook must be very promising in Meriden.—"A small twig of the great New England branch," the "Endeavor," of STRATFORD, calls itself, and a healthy fruit-bearing twig, if small, we should call any circle that can double its membership, as it has done, in less than a year.

A splendid move has been made by the Chautauquans of NEW YORK CITY. The circles had never united there for public work until, April 9th, after a deal of planning and much labor the various local circles in the city, with one from JERSEY CITY, held a public meeting at the Broadway Tabernacle. Chancellor Vincent was secured to deliver the address. A large and enthusiastic audience, numbering about 1,200 persons, was present. "School after School, or the Every-day College" was the theme of the address. The friends who had prepared the meeting were more than jubilant over the way the interesting lecture "took" among their guests, and declared that though some of them had long been members of the C. L. S. C. all of them received new vistas of the work. After the lecture the Chancellor held a reception, at which the members of the following New York circles were received by him: The "Garfield," "Irving," "Unique," "Spare Moment," "Central," "Park Avenue," and the "Round-Table," of Jersey City.—BROOKLYN has a "Pierian" spring, at which twenty-five devotees of the muses "drink deep" and joyously. Essays, debates, recitations, quizzes, poems, and chemical experiments are the draughts these friends draw from their well. So happy are they in their festivities that another year they hope to see a sister welling up by their side.—At RANDALL'S ISLAND, NEW YORK CITY, there is one of those steady, hard-working circles, which by their fruits so favorably impress the people who watch them. The "Excelsior" has been in existence for two years and has a roll of seventeen persons. A

program full of good points is carried out at their regular sessions, which are interrupted only by Memorial services. The secretary finds, he writes, that the influence and example in regard to Chautauqua work is shown by an increased attendance and membership.—When circles increase in geometrical ratio—and a little over—year after year, it is not strange that a time should come when the leader inquires “What *shall* we do?” At GLENS FALLS four graduates, eleven ’86s, twenty-three Pansies, and fifty-nine Plymouth Rocks—ninety-seven in all—form the circle. It is an unwieldy number to study together, but, writes the secretary, “We are fearful that division into smaller circles will greatly lessen the membership. We are considering for next year this plan: We shall have our general meetings as at present and encourage the formation of sections for special meetings, making the leaders of the sections, together with the officers of the general circle, an executive committee.” The plan is good, and if the monthly joint meetings are made “state occasions,” there will be but little danger of the sections losing ground.—At TROY the monthly meetings are conducted in an admirable way—not one of the least of the secrets of their success is their habit of sending out cards with the program, and some such stirring word as this:

“We hope every member will be prepared with some facts on No. 6 [“No. 6” was a *conversazione* on “Our Territories”]. Let us make it a success. Our whole course is highly remunerative to one who reads. Make your hours count for profit. We have all the time there is. Not everything that comes to us and asks a little of our time should be granted audience or gratification.”

A similar plan is followed by the “Mettowee” circle, of GRANVILLE, N. Y. A “Round-Table” on a recent program was, “What we have seen (mentally or otherwise) during the past week.” If the Glens Falls people will make their monthly meetings big enough to arouse the pride of the members of the section, the result will not be doubtful.—SHUSHAN, N. Y., has two circles. The younger of the two is reported as “fast becoming one of the fixed and instructive institutions of the village,” a state that the three-year-old circle of DEANSVILLE is already in, we surmise, from the report of the work of its membership. The Deansville circle holds an annual public meeting, at which an entertainment is provided and refreshments served. Preparations are being made for this year’s meeting.—A “Crescent” has

appeared in KNOXBORO since the year 1885 began. Though so late in beginning that they are obliged to devote almost all their time to the readings, and are too busy to prepare elaborate programs, the interest has not flagged since the start. The “Crescent” has ten members.—SYRACUSE has always been one of the most interesting of the local centers of the C. L. S. C. Their graduates, of whom one is the genial Secretary of Chautauqua, Mr. W. A. Duncan, have organized a chapter of the “S. H. G.” They propose to hold monthly meetings, and to pursue one of the special courses of study. The membership of this new class is eleven now, and there will be yearly “more to follow.”—The “Philomathean,” of LANCASTER, N. Y., opening with 10 members, has grown to fifteen beside honorary and local members. Constant variety in programs, no “set way,” and hard work have been the maxims of their success.

The “Broadway” circle, of CAMDEN, N. J., puts in after a program of remarkably good timber, and the outline of their thorough organization, this healthy testimony: “‘Broadway’ circle is busily engaged in promoting this home study, and the older we grow the more we are able to discern the many blessings derived from it; the more we read and study the books the more does it stimulate our interest and thirst for knowledge.”—At PHILLIPSBURG, N. J., the circle has met this year with a sad loss in the death of Mrs. F. B. Holbert. Mrs. Holbert was a member of the class of '84, the president of the circle in Phillipsburg at the time of her death, and a most zealous friend of the C. L. S. C. To her the circle at Phillipsburg owes its existence.—At BRIDGEPORT a circle of eight was organized in October, 1884. A faithful leader has helped to keep them interested and alive, and already they write that a taste for solid reading has been acquired by the members.—A local circle has been organized at EAST ORANGE, with eighteen members. The favorite name, “Alpha,” has been given the class. One of their late meetings of unusual interest was the celebration of Longfellow’s birthday, at which, among other exercises, we notice what must have been a particularly pleasing number, “The Better Land,” illustrated by tableaux.

The genial, kindly associations of circles are one of their greatest charms. How the kindly attention, the pleasant surprises kindle the hearts and knit the affections! CARLISLE, PA., circle has recently experienced all the delight of doing one of these pleasant deeds. The birthday of their president,

Dr. Whitney, was celebrated by a genuine surprise. A game of chess at a neighbor's was the bait which enticed him from home, where, on his return, the Chautauquans of his circle, thirty-one strong, greeted him with good cheer and good wishes. It pays to slip into our Chautauqua life many of these pleasant little affairs.—Several notices of Longfellow celebrations held by PENNSYLVANIA circles reached us too late for the May issue of THE CHAUTAUQUAN. At BETHLEHEM, where there is a “thoroughly congenial” circle of ten, the day was appropriately observed. This circle has found a scheme of study which it reports works very well for them. The time from 7:30 to 10:00 p. m. on the evening of meeting is divided into half-hour periods. These periods are all but one devoted to quizzes on the subjects laid down in THE CHAUTAUQUAN outline, the extra period being given to a discussion of the works of some well known author.—At PITTSBURGH the “Hiawatha” observed the day with an excellent program, carried out before many friends. This circle—fifteen in number—is one of last fall's harvest. Their motto, “Bound to Win,” tells the stuff they're made of.—The wide-awake circle at NEW WILMINGTON, PA., prepared a program covering Founder's day, and Longfellow's, and reported a “royal good time” for their trouble. There are twenty-four members in the circle; their unanimous verdict is: “The C. L. S. C. has been a source of intellectual growth to us. And we have been led by it to take a wider view of the possibilities of life.”—A very good program of a Longfellow service comes from PLYMOUTH. We notice an analytical study among the numbers, and would commend such services to the circles as particularly profitable.—The LOCK HAVEN circle, at its Longfellow evening, paid a high compliment to the circle at RENOV, by reading the program carried out by the latter on a previous evening, and sending their greetings and congratulations to the Renovo Chautauquans, that they have grown so strong and enthusiastic in but one year's readings.—We are always sorry to miss in geography, but we will “own up;” we did in the April issue. The “Golden Flower” is not a Tennessee, but a Pennsylvania blossom, and HATBORO is a Keystone town. The “Golden Flower” has sent a series of really model programs recently.—From a friend at GILMORE, PA., we learn of the “Foster Brook” circle. This class was organized in October, 1882, with twenty-one members, but in a few months its course was rudely broken by the death of one of their young but zealous members, Mr. Henry Howe, of the class of '86. The work was again taken up, but February, '84, Mr. H. F. Howe, father of the former,

and a member of the same class, was laid away by the side of his son. Though so tried by sorrow our friends have bravely followed their work, saddened, yet rejoicing.—It is an experience that many circles have, we imagine, this of the TUNKHANNOCK circle, of finding that their second year's reading goes much more easily than the first, and that they have time for many things in their circle which once they did not have. One good thing that the "Tunkhannock" occasionally slips in is important items of news from the secular press.—The "Mountaineers," of CLEARFIELD, is one of the many, many circles brought into the field by Chancellor Vincent's kindling fervor. It is an '88 circle, and numbers fifteen members. Reviews, readings, and conversation supplement the programs of THE CHAUTAUQUAN. The reviews, particularly, they have found valuable. Each book is taken up after being read, questions made out on it, and answers given by the circle. Outlines of the books are also sometimes prepared. This latter plan we do not remember to have noticed in the reports before, but we know it to be a very effective method for reviewing the facts or arguments in a book.—We rarely open our monthly budget of letters without finding a WASHINGTON, D. C., representative. This month we have a program from the "Wesley Chapel" circle, a good one, of course, like all the Chautauqua work done in Washington. An interesting item on it is "The Public Buildings of Athens," illustrated by photographs.

A MIDDLETOWN, MD., letter suggests a new way of managing a local circle. "We are the 'Mayflower' circle, numbering five," our friend writes, "As we are active members of a literary society, which meets weekly, we have no circle meetings, but talk over our readings, and Chautauqua in general, when opportunities offer. We are earnest and interested, and expect to enter upon next year's work with increased knowledge, zeal, and numbers."

A bed of KENTUCKY Pansies is filling all the air of HARDINSBURG with sweetest fragrance. "Of all the red letter days in our circle calendar," writes one of them, "Longfellow's day is the brightest. We send you a program of our last celebration, and a delightful evening we had carrying it out. A year ago we thought we could not have another Longfellow's evening as pleasant as was that. We feel assured that the evening just passed was more nearly what such an evening should be. In looking back from this standpoint upon the past year, we are inspired for renewed energies and

work for the years to come.”——Another Pansy of Kentucky, from BEWLEYVILLE, writes: “The C. L. S. C. has been of immense benefit to me. My irregular habits of study and desultory reading, instead of strengthening, had enervated my mind.... My CHAUTAUQUAN and books are a great source of pleasure to some of my acquaintances who are not pursuing the course.”

CHILLICOTHE, OHIO, has, in the Walnut Street M. E. Church, an enthusiastic band of twenty-two workers, who report themselves as “progressing finely.” At a recent meeting a lecture on the “Art of Speech,” by the able Episcopalian rector of the city, was greatly enjoyed by the class.

It must be that a millennium has reached the circle of ERIE, MICHIGAN. If there is another circle in the country that can say as much, we should like to know where they are and how they do it. The Eries are forty-five in number, and though some of the members live six miles apart, and the meetings are held at the houses, yet through all the past long and dreary winter there was an average attendance of more than three fourths of the membership. “And,” thus writes our correspondent, “each member is expected to do whatever the committee may assign him. We never have a failure. The program is always carried out to the letter.”——At CASNOVIA, a circle of nine was organized in October last. Like nearly all our students, they are extremely busy people, but yet thorough in their work. They hold informal meetings for discussions and conversation on the readings of the month.——The first of January a circle was organized at MILFORD, Michigan, composed of twelve ladies, not all of them ’88s, there being a sprinkling of graduates and seniors to give direction to “Plymouth Rock” enthusiasm.——“One more Michigan circle” is heard of at JONESVILLE. The circle started out with twelve members, and is keeping up a working membership of nine. The meetings are very enjoyable, every one taking part in the most interesting and informal way.——Last October a wave of Chautauqua enthusiasm passed DUNDEE, Michigan, which resulted in organizing a C. L. S. C. of six members. They call the new organization “Longfellow’s” circle, and hope with the next year the numbers will increase. The circle has observed all memorial days, which have proved of great benefit and interest.——Evidently the “wave” has not left Michigan yet, for we have just received a card telling of the organization and election of officers of a new circle at GREENVILLE. Success to their efforts.——A sample of the work which the C. L. S. C. helps the student to do, the GRAND RAPIDS, Michigan,

circle furnishes in a recent program. The exercises included, in brief, papers which took "broad, quick views." Articles upon Thucydides, Socrates, Plato, Herodotus, Sappho, Greek Mythology, British Association for the Advancement of Science, American Soil, Life of Longfellow, and a review of Longfellow's novel, "Hyperion." This circle has made a big stride since last year, its membership having increased from twenty-three to thirty-seven.—The new pronoun which troubled some of our students has been employed by a correspondent from the circle at JACKSON, MICH. Perhaps his zeal to fill up this "long felt want" in language will make more clear to our readers the use of "thon." These circle notes are very suggestive: "We meet weekly at the home of one of the members, a system we have found to be an improvement on that of meeting in a public hall. Chautauqua's special sphere of influence is in the home; it is here 'thon' expects to do work which will be lasting, and eventually permeate every department of life. It will then be easily understood why 'thon' should find home and its associations a more congenial place of meeting than a public hall. Our meetings are conducted much in the usual manner, except, perhaps, our mode of managing a 'discussion.' To this feature we make special claim as being the originators. The leader of these talks, whose duty it is to 'keep the ball rolling,' is, with the subject, chosen a week beforehand. Each of the members is expected to inform 'thonselves' on the topic thus allotted, and to be prepared to give an opinion or ask questions. Richard Grant White's language articles, 'The Art of Speech,' and 'The Temperance Teachings of Science,' have already been dealt with in this profitable and interesting way."

We are glad to be able to introduce INDIANA this month with a clipping from a private letter to the lady here so honored:

"WABASH, IND.

"I want to tell you how our local circle of the C. L. S. C. honored itself last night in deciding to be called the 'Frances Willard' circle, and, furthermore, to observe your birthday as a memorial day.... We have in our circle, as you may suppose, a number of ladies active in the work of the W. C. T. U. Our circle numbers thirty-two. It was organized in '79, but never

christened until last night. The vote on the name was unanimous.”

From ALBION, INDIANA a friend writes: “We have organized a circle here consisting of twenty-four members. Prof. E. C. White, superintendent of schools, is our leader; and is greatly interested in our circle and its work. We meet each week, at the residences of the members, taking them alphabetically. The majority of the teachers in our public schools belong to the circle, and all the members are much interested in their work.”——And still another from the same state, from FOWLER, says: “We organized in October, 1884, with nine members; the circle now numbers fourteen members, of four different classes. One item of each program is a question drawer, questions to be taken from suggestions of the weekly readings. If any question is not answered by the circle, it is assigned to some member to be answered at the next meeting. I find this stimulates all to read carefully, as no one can know just what part of the week’s reading is to be investigated. This is a most interesting feature, and we can get those to engage in this who will do nothing else. We have two members living at a distance who are only present in person occasionally, but always present in manuscript when on program, so we call them our ‘paper members.’”——A Chautauqua circle was organized at LIGONIER, Indiana, in October, 1884, with a membership of seventeen, and named the “Ligonier Chautauqua Literary and Scientific Circle.” In it are represented three denominations. The programs are so varied as to avoid monotony in the exercises. One night the committee gave topics for brief essays, on another, questions to be answered at next meeting. The members of the circle pronounce the prescribed course of study excellent and interesting in its details, and feel grateful for a plan so far-reaching and so beneficent in its results. Bryant’s, Milton’s, and Longfellow’s days were observed by devoting to them and their works the entire evening. A knot of cardinal and old gold ribbon constitutes the circle’s badge. The circle is highly pleased with the motto “Let us be seen by our deeds,” and many of the members express themselves satisfied with their name, “Plymouth Rocks.”——We have received the list of names of the members, fifteen in all, which form the circle of BROOKVILLE, Ind. We are hoping to receive some of the Brookville circle’s experiences soon.

The "Peripatetics" of CHICAGO, ILL., is an organization formed last fall. There are twelve members. The circle, we fear, is not using the local circles' reports right, if, when they read them, they feel almost discouraged at the little they have accomplished. The whole spirit of Chautauqua is, "what has been done can be done," and our work is mainly to show by actual example what has been done. The "Peripatetics" have too many opportunities for doing superior work, living as they do, in such a city as Chicago, to grow discouraged.—The death of Richard Grant White has been a great loss to Chautauquans. His admirable "Studies in English" had met the warmest reception, and when they were cut short by his death all our readers felt they had met a personal loss. It was this feeling that led a Chicago circle to add to their resolutions of respect passed upon Mr. White's death, a clause stating that as a feeble expression of their regard and, as a token of respect to his memory, their local circle be hereafter known as the "Richard Grant White."—For novel diversions, commend us to the "Alpha," of QUINCY. This circle has sandwiched a great amount of fun in with its solid work this winter, nor has it been to the injury of the work. On College day they had a sleigh ride, not long after a mock trial; they have introduced a paper, the *Symposium*, which gives an opportunity for numberless hits, and on Valentine's day the *Symposium* furnished the entertainment, each member having sent to the editor an anonymous piece of poetry.—A circle was organized in STERLING last October, and has now nine members. They have no officers, and each meeting is conducted by the lady at whose house it is held. Their informal way of doing things has some advantages. An invalid member is mentioned, who, like many others, finds relief in affliction by having something to do, and doing it religiously.

On one of the last evenings of September, a few Morrisonians met and organized the "Alpha Society," of MORRISON, ILLINOIS. They number fifteen, and have accomplished much by the winter's work. The plan has been to take the outline of required reading, as laid down in THE CHAUTAUQUAN, and prepare the lesson as thoroughly as possible by the use of books and encyclopædias. At the meetings the class recites the lessons learned the week before. Occasionally they have an essay, or select reading.—A circle of twenty members was organized in JACKSONVILLE, Ill., last October, with the Rev. H. E. Butler, pastor of Congregational Church, as president. They have given special attention thus far to the Greek course, enlarging the

reading, and bringing in other authors as far as possible. Now that the Greek is nearly finished the circle is bringing chemistry into prominence.—The “Oakland” circle, of CHICAGO, Illinois, was organized in February, 1884, and belongs to the “Pansy” class. It has a good *constitution*, and hopes to survive till the graduation of its members. This circle is composed entirely of ladies, among whom are representatives of six churches in the South Division of the city. The meetings, which are held semi-monthly at the houses of the members, are well attended. Absent members are informed by postal cards of the place of the next meeting, the parts assigned them on the program, and the author from whom quotations are to be given. Those who are absent three times in succession are expected to present a suitable excuse to the society. At the close of the year a reception was given, to which each member invited a gentleman friend. A fine literary and musical program was rendered, and refreshments were served. The guests, among whom were ministers, lawyers, army officers, and members of the Chicago Board of Trade, were called upon to express their opinions of the C. L. S. C. A most enjoyable evening was spent.

A flourishing circle has existed at PORTAGE, WISCONSIN, for three years. It is only one of three, however, for last year the pastor of the M. E. Church organized a circle which is still active, and this year the Presbyterian pastor started a third. The *mother* circle is undenominational, as best becomes her position. One feature of this circle’s year is its annual reception. Let the secretary tell how it is managed: “We have heretofore held a reception at the close of each year, to which were invited the people of the place presumed to be most interested in education. The exercises have always been highly commended, and our work well appreciated by those in attendance—indeed, we consider our best missionary work has been done by this means—and many members added through the instrumentality of a sugar-coated dose of hard work (or its results, rather). Many people would attend a party, to whom a literary meeting of any kind is a ‘delusion and a snare.’”

THE C. L. S. C. CLASSES.

CLASS OF 1885.—“THE INVINCIBLES.”

“Press on, reaching after those things which are before.”

OFFICERS.

President—J. B. Underwood, Meriden, Conn.

Vice President—C. M. Nichols, Springfield, Ohio.

Treasurer—Miss Carrie Hart, Aurora, Ind.

Secretary—Miss M. M. Canfield, Washington, D. C.

Executive Committee—Officers of the class.

After Commencement, what? More of the same sort, undoubtedly—of a better sort, if possible. The members of the class of '85 are now near enough to the “finish” to arrive at some more or less mature conclusions as to the future. They find that the word “commencement” is to be no misnomer, and is not even to be criticised. For after four years of reading and study the students find that they have not come to the end, but only to the beginning. The matter selected for them has been so good, so substantial, and really so inspiring, as a whole, that an immortal, if not divine, sort of hunger for the best in literature, science, art, philosophy, has been created in their minds, and they could hardly stop if they desired. The aggregate amount of literary and scientific information which has been acquired is really large, and much of it has been retained as a permanent fund of knowledge, but this has proved to be but a small part of the benefit that has been derived. Thousands of people have just begun to find out how little they really know, and how much they ought yet to learn, to satisfy their own ideas and notions as to what is required to make one, not learned,

indeed, but reasonably well informed! And what they think they need to know, they feel sure that they have now discovered how to learn, and have acquired habits of reading and thinking which will make the processes easy and enjoyable, instead of laborious and tiresome. They have already looked beyond bounds into the green fields and pastures new of that which is best, most beautiful, and grandest in the domain of thought, and suggestion, and philosophical research and discovery, as brought out by poets, philosophers, statesmen and philanthropists, and they are likely to pass through the Golden Gate and out from the Temple on the hill into a still broader fraternity of thought and action, whose limits will correspond with those of the world itself.

A breezy letter from the plains of Dakota contains the following: "I have been studying entirely alone, and have found the course a blessing that can not be valued. Here, quite removed from society, while winter winds howl round my cabin home, I find help, companionship and pleasure in the studies of the dear 'Home College.' The most attractive corner of my little room is the one where my beloved books lie on the home-made shelves. My little ones love them too, and there is scarcely a time when I need help but what I can find it in my books."

One young lady from Massachusetts, who hopes to hear the "Chautauqua Chimes," writes: "I have a class of girls in Sunday-school, and want so much the help I think I shall be able to find at Chautauqua. The course has been just what I needed, and I know I have grown, mentally and morally, since joining the C. L. S. C."

This from a gentleman in Kansas: "If health will permit I hope to be one to enter in under the Arches among the 'Invincibles.' I must be one of the oldest of the '85s. I was fifteen days old at the battle of Waterloo, and, if I am spared till my next birthday, I shall have arrived at the bounds allotted to man in the Bible."

One '85 writes: "I hope to receive my diploma at Chautauqua, but, I am a busy mother with six children, and can not always plan so long ahead. I

have had the greatest pleasure and satisfaction in the course, for, with a family of wide-awake boys and girls about me, I have found it is very necessary to refresh myself and keep well informed on all subjects.”

One enthusiastic lady from Texas writes: “I have gathered through four years of delightful reading an intellectual bouquet, whose fragrance I hope to wear about me when I pass, not only through the Arches at Chautauqua, but when I pass through the ‘Beautiful Gates’ to the Celestial City.”

Another testifies: “I am one of the busy mothers and housekeepers who pursue the C. L. S. C. course under numerous and varied difficulties, but find my enthusiasm increasing as the four years draw to a close.”

From Massachusetts: “I intend to still ‘press on’ after I graduate—in fact, I hope always to be a Chautauquan.”

Wisconsin contributes: “I like our motto and our name, and I love the C. L. S. C. Though reading alone, it has always been an inspiring thought that many thousands are reading the same course.”

Another: “I regret that the course is so nearly finished, but the spirit it has awakened within me has enabled me to ‘Press on, reaching after those things which are before.’”

CLASS OF 1886.—“THE PROGRESSIVES.”

“We study for light, to bless with light.”

CLASS ORGANIZATION.

President—The Rev. B. P. Snow, Biddeford, Maine.

Vice Presidents—The Rev. J. T. Whitley, Salisbury, Maryland; Mr. L. F. Houghton, Peoria, Illinois; Mr. Walter Y.

Morgan, Cleveland, Ohio; Mrs. Delia Browne, Louisville, Kentucky; Miss Florence Finch, Palestine, Texas.

Secretary—The Rev. W. L. Austin, New Albany, Ind.

The “Progressives” of New England are true to their name, and most encouraging reports are received from circles and those who are studying alone. A young man teaching school in a remote village in Connecticut writes: “I feel far below the standard of our class, but am determined to do the best I can, God helping me. Leisure moments are delightfully spent in reading or meditation. Hope to complete the course in 1886, and then go on with extra readings.”

The class of 1886 is deeply bereaved by the removal to higher duties and joys of a most worthy member, Mrs. Emma Webster Darling, wife of the Rev. J. K. Darling, of Chelsea, Vermont. She died on the morning of Easter Sunday.

One of our busy workers, A. M. T., of Ontario, Canada, has made an attractive little devotional book, “My Work, or Conditional Promises,” for every day in the month.

A young lady from Boston writes: “I have devoted to C. L. S. C. work at least forty minutes every day since I have been a member, and would gladly do more if time would allow.”

From the snow hills of Maine comes this cheerful testimony: “I sometimes envy people their riches, but am thankful for the C. L. S. C. every day of my life, for I am a farmer’s daughter, and so situated that I am debarred from the enjoyments of most young people, and would often be very lonely were it not for the books of the C. L. S. C.”

The Hopkinton tent, at Framingham, has been secured for headquarters, and will be made comfortable. If the ladies of ’86 who contemplate visiting Framingham next summer will remember that they are a “committee of the

whole” on decorations, the tent can doubtless be made homelike and attractive at little expense. Bring something to brighten it, if only a penny Japanese fan.

CLASS OF 1887.—“THE PANSIES.”

“Neglect not the gift that is in thee.”

OFFICERS.

President—The Rev. Frank Russell, Mansfield, Ohio.

Western Secretary—K. A. Burnell, Esq., 150 Madison Street, Chicago, Ill.

Eastern Secretary—J. A. Steven, M.D., 164 High Street, Hartford, Conn.

Treasurer—Either Secretary, from either of whom badges may be obtained.

Executive Committee—The officers of the class.

The New England Pansies seem to be more active of late than their fellow blossoms farther west. The following report of their reunion represents something of their enthusiasm: The New England branch of the class of 1887 C. L. S. C. held its spring meeting in the chapel of Union Congregational Church, Providence, April 3d, 1885. About one hundred members were present. After a short time spent in social intercourse, the meeting was called to order by the president, the Rev. F. M. Gardner, for Rhode Island. The C. L. S. C. study song was sung, which was followed by the secretary's report of the meeting in Boston; the minutes were duly approved. It was voted that the committee on headquarters be increased by the addition of Mr. Jeffers, of Pawtucket, and Mrs. Morrill, of Allston. Mr. Gardner, in his own bright manner, gave some account of the efforts of the committee in preparing for class headquarters at Lakeview, and stated reasons why the plan suggested at the Boston meeting should be postponed,

though not abandoned. Inasmuch as several members of our class have been afflicted by the loss of loved ones from their homes, it was voted that a committee be appointed to present resolutions at this meeting expressing the sympathy of the class with them in their bereavement. The musical and literary exercises were opened with a fine piano solo, which was followed by a pleasing quartette. Then an address on "The C. L. S. C. vs. Social Pastimes," by the Rev. N. T. Dyer, of Middleboro, was delivered. Mr. Dyer being unable to be present because of illness, Mrs. Dyer most creditably took his place. The address was a convincing statement of the advantages of the C. L. S. C., and could it be circulated among those not interested in the course, would undoubtedly influence many to enroll in the Circle. Mrs. Emily C. Fletcher, of Pawtucket, read a poem written for the occasion, from which we extract the following, referring to the influence of the C. L. S. C.:

"It has cleared the brow of discontent,
Made happy the lowly one,
Cheering the home and its social hall,
Enliv'ning the tasks begun.

"It takes from age the mournful thoughts,
That often the heart will shroud,
It lifts the life to a higher sphere,
Silvering ev'ry dark cloud."

After music, and an address on Lakeview, resolutions of sympathy to those of the circle who had met with bereavements were adopted. The association then adjourned, after which a delightful reunion was enjoyed by the members.

Two members of the class of '87 have recently left us for the "better life:" Miss Grace F. Cook, who died March 22, at Vilas, Wisconsin, after a protracted illness, and Mrs. Rev. E. S. Osborne, of Kingston, New York, who died at her home, March 16th.

CLASS OF 1888.—"THE PLYMOUTH ROCKS."

“Let us be seen by our deeds.”

CLASS ORGANIZATION.

President—The Rev. A. E. Dunning, D.D., Boston, Mass.

Vice Presidents—Prof. W. N. Ellis, 108 Gates Avenue, Brooklyn, N. Y.; the Rev. Wm. G. Roberts, Bellevue, Ohio.

Secretary—Miss M. E. Taylor, Cleveland, Ohio.

Treasurer—Miss M. E. Taylor, Cleveland, Ohio.

All items for this column should be sent, in condensed form, to the Rev. C. C. McLean, St. Augustine, Florida.

Class badges may be procured of either President or Treasurer.

Our circle in Ouray, Colorado, in the midst of the Rocky Mountains, numbers ten. The picturesque spot where live these '88s is about a day's journey from the railroad. They name themselves after the Indians, "Uncompahgee," who once wigwamed there. They have their "round-table," and keep up their weekly meetings.—Another circle, of three, has been organized among the "Rockies," at Gunnison, Colorado, and meets weekly. Our '88s have reached the Indian Territory. At Chouteau we have one hard worker, who, having commenced in February, has nearly caught up with the class.—At the confluence of the Missouri and Vermillion Rivers we have an enthusiastic class of ten. Lawyers, teachers, journalists, milliners, and busy wives, with a "Pansy" for the president, compose the class. They conduct their class weekly, on the "conversational plan," which they claim affords grand opportunity for interchange of opinions and sentiments.—A circle of five ladies and one lone gentleman compose the "Clio," of Clark, Dakota. So delighted are they that they never adjourn for any other engagements.—The "Kankakee," of Illinois, thirty-eight regular and four honorary members, meets fortnightly. Their Shakspeare program was so full that a portion was postponed until the next meeting. An honorary member has delighted them with an address upon "Water."—The twelve members of "Calumet," Carthage, Ill., were favored with a visit and

instructive talk from Chancellor Vincent.—We were greatly surprised to receive a letter from our old friend, the Rev. W. H. Hyatt, president of our circle in Whiteland, Ind.—Ten young persons of Dubuque, Iowa, compose the “Circle of Athens.” An excellent motto have they selected: *Sapientiam petimus*. That their search for wisdom is eager is evinced by the fact that they have in a most interesting manner pursued the studies in spite of all allurements. The memorial days have been appropriately observed, and Longfellow’s day celebrated by a banquet.—Seven constitute “Alpha,” of Barnesville, Ohio. They began January 1, and have nearly completed their studies. They expect to begin on time next year.—The “Athena,” of Wanskuck, Providence, R. I., is composed of fifteen busy people, who are delighted with the studies.—From the programs of the “Hamilton,” of Lowell, Mass., we are satisfied that the forty members are truly among our liveliest coming Chautauquans. They have largely experimented in chemistry and electricity. This circle mourns the loss of one of its best members in Mrs. Benjamin Robinson, who endeavored to brave a New England storm, in order to attend one of the meetings, and lost her life.—Seventeen regular and three honorary members represent a circle in Joplin, Mo. Nothing but illness has caused an absence at “roll call.” Once a week they follow Chautauqua program. Success has marked their public as well as private meetings.—One from Maine has taken fresh courage since reading Chancellor Vincent’s article in the April CHAUTAUQUAN, “How to Work Alone.”—The “Riversides,” of Milford, N. H. (eight members), have finished the year’s studies.—“Zeta Phi,” of Buffalo, N. Y., are seven “zealous learners.” They observe all special days, having essays upon given subjects.—A zealous lady of ’87 class organized seven earnest and hopefuls into the “I. X. L.,” of Newport, Ky.—Clamida (state not named) boasts of two enthusiastic circles. The secretary of one strongly objects to our name, repudiating the idea that we have anything in common with the “Pilgrim Fathers.” She is even tired of a reference to their trials, and believes, with another, that the “Pilgrim Mothers” are more worthy of “toasting,” closing with “Seriously, why were we thus afflicted?”

THE CHAUTAUQUA UNIVERSITY.

A TEACHING METHOD.

BY PROF. RICHARD S. HOLMES, A.M.

I desire in this paper to make some very plain answers in a very plain way to a question which has come to me in varying forms, from various sources. It is a practical question, and concerns the possibilities of that department of Chautauqua work which aims to bring the advantages of the higher education within the reach of those large classes of our population which have been hitherto debarred from them. Naturally, the question originates with the very people whom the enterprise seeks to aid, and strangely enough is shared by those whose culture and education should have been a barrier to such a doubt.

Men who would gladly avail themselves of any real advantages for education brought within their reach, and within their means, yet unwilling to make the pecuniary outlay which the effort might involve, until convinced that the correspondence system offers real advantages, hesitate, and say: "We are favorably impressed with the idea as given in your announcements, but are not sure that it can be put into practical operation; before attempting the work it may demand, we are compelled to ask, how is the work to be done? how can teaching by correspondence be made practically successful? Show us the method, that we may understand." Still others, men of advanced education, of approved excellence of judgment, men engaged in professional life, have said, "We concede that education by the means you propose is possible under certain favorable conditions, but we doubt the practicability of attempting by such a means to cover the wide field of general education." Straightway they fall to instancing particular subjects as illustrations of the truth of their statement. Now the proof of the pudding is in the eating; and if this paper succeed in furnishing tastes of this particular pudding which shall be palatable and shall create a favorable opinion as to the worth of the whole, the service rendered to the cause will

be valuable. I propose, therefore, without invading the province of any of the gifted teachers who act as Directors of the different Departments and Schools in the Chautauqua School of Liberal Arts, as we shall hereafter call what has been known as the University, to show, if possible, how a person of good natural endowment, at the maximum of his mental strength, and with earnest devotion to his work can acquire a knowledge of a language, literature, or science by correspondence alone.

I will make three preliminary remarks. First, the student must bring to this work the same earnestness that he gives to that pursuit of his daily life upon which he has been or may be dependent for his livelihood. Second, in the study of language by correspondence, the path marked out by the experience of the ages is the path in which the correspondence student must go. The gateway of that path is the grammar of the language; and no student can pass through it till he possess the key which shall unlock its bars. To own a grammar is therefore a necessity. I am ready to believe that in oral teaching of a language, actual study of grammar, as grammar, may be put over into the final years of the course, giving the early years to the undisputed control of synthetic methods; but for the correspondence student, a grammar is an essential. Third, the student must be willing to follow the most minute directions of his teacher, without question, no matter how simple or how difficult a matter their performances may seem to be. To obey is the first essential to success.

Let us now look at a *method* for learning a language or science. It is not given as *the method* in use in the schools, but only as a means of showing that the thing proposed is possible. There should be for the beginner four papers for every lesson; or four kinds of work to be done.

First—There should be a paper *stating principles to be learned*, and adding complete references to the text-book upon which they are based, that the student may add to his teacher's dictum, the confirmation of his own research. Let it be distinctly noted that this paper is *to contain statements of principles to be learned*, and is not to be a mere budget of directions to paragraphs and sections of a text-book. The advantages to be gained by such a method of study are too obvious to need elaboration here.

Second—There should be a paper giving abundant and apt illustrations of those principles, derived from the best sources, adapted to the pupil's knowledge, and different from any which have been otherwise brought to his notice. These illustrations of principles should be memorized by the student and should form the basis of the paper containing the test of the student's work.

Third—There should be a paper giving examples for practice in these same principles; examples for transliteration, phonic representation, or translation in case of a foreign language, examples for experimentation, classification, or analysis in case of a science or literature.

Fourth—There should be a paper of examination or question, for the purpose of testing the student and revealing the character of the work he has done. These questions should be framed with the utmost care and skill of which the teacher is master, and should act as a quickening impulse to the student. This paper should be in a sealed envelope, and should not be opened till all the work of the other three papers has been done, and the student feels that his lesson is learned. In addition to what has been suggested, there should be required in the study of language, as soon as the student can correctly pronounce, a regular exercise in memorizing from some standard author, and daily repetitions aloud of what is thus given to the memory. In the case of English, Latin, and Greek, this seems to me indispensable. This last suggestion, it will be noticed, contains a hint that the pronunciation of a language can be taught by correspondence. It is a hint which I am prepared to assert as a proposition, and to defend as far as the Latin and Greek languages are concerned. The amount of matter given in the lesson should be enough to require one week for its preparation by a student able to devote from one to two hours daily to study.

When the papers of a lesson have been fully mastered, and the student feels that all he can do upon it is done, the whole work should be at once sent back to the teacher. Now, to guard against loss of time, such as would occur were the student compelled to wait without work after he has forwarded his lesson recitation to the teacher until the necessary exchange by post has been made, two lessons should be sent out by the teacher at the first assignment. This plan would wholly avoid what might be costly delay

where student and teacher were separated by the width of the continent or the ocean.

As soon as the first recitation paper reaches the teacher's hand, his immediate duty is to forward the next lesson of the series, and so regularly through the whole course of instruction. He will now at his leisure examine the paper which has come into his possession, while the student is engaged upon the second of his lessons. What shall be the teacher's work with this returned paper? Certainly not one of correction. Now begins his real work of teaching. First, there must be a careful and painstaking inspection of each line of the student's work. Second, every error must be plainly marked, so that the eye of the student will not fail to observe it. Third, plain reference should be made to those sections and paragraphs of the grammar or text-book which have been violated. Fourth, a word of encouragement, advice, suggestion, or warning should be added to each paper, drawn from the teacher's wide and varied experience, and which will be practically helpful to the pupil. It must be carefully noted that in this treatment of the recitation paper, the teacher has made no correction, has told nothing, but has simply indicated errors, and thrown the student back upon his own resources to correct his own work. This is one of the elements of true teaching.

The return of this critically marked paper to the student brings us to consider another important process in this work, and that is the review by the student of his first lesson work, or his second period of study upon it. There has enough time elapsed since it was last in his hands to have it come now with all the force of a new lesson, and to enable him to look at it judicially. The critical investigation which follows has a three-fold value. First, it is a review. Second, it is a means for accurate self-test. Third, it is a monitor, under whose warning all future lesson work is subjected to the careful scrutiny which the former criticism suggests. Two things still remain to be done with the returned lesson paper. One to make a separate classified list of the errors it contained; the other to date it, file it, and lay it carefully away for reference. The classified list of errors will serve as a check against the commission of like errors, or an aid in detecting any that may have been carelessly made. At first the list will be large, but after a little it will grow less and less rapidly, till finally its utter lack of growth will be the surest mark of the pupil's excellence of attainment. Such is an outline for a possible method of conducting educational work by correspondence. It

presents a method which I believe is practical, which is drawn from an experience of years in the class room, and which is in harmony with established principles of educational philosophy.

A touching bit of experience has been sent us by a member of the class of '88. The writer had persuaded his son to join a circle, but, as he writes, "He attended one meeting of the circle and came home very much discouraged, declaring that he would not attend another meeting, urging as his reasons that he compared unfavorably with others, and that he would never be able to pronounce those horrid Greek names, etc. I tried to encourage him and advanced several arguments trying to show him what a great advantage this course of reading would be to him, but finally gave it up, fearing if I urged him so strongly he would become disgusted. I determined then to take the four years' course of study myself, thinking that by having the books in our home, and sometimes relating anecdotes, incidents and historical facts gathered from these readings, that my boys might become interested for themselves. It is impossible for me to give my children the advantages of a liberal education, as my heart longs to do, and by getting them interested in the C. L. S. C., I hope to make up to them in some degree their loss of a college education."

EDITOR'S OUTLOOK.



AN AMERICAN DIPLOMAT.

There is very general regret, at home and abroad, that the new administration has removed Professor James Russell Lowell from the office of American minister at the court of St. James. There is no disposition to complain; but there is some natural wonderment. Mr. Lowell was an ideal American diplomat; he represented worthily the people as well as the government of the United States. It is no disparagement to his successor to say that no other American can quite fill the place Mr. Lowell has made for himself. It should be remembered that we, fortunately, have very little proper diplomatic business anywhere in the world; and whenever any serious negotiation is to be undertaken, it can be done at Washington. Our important treaties are made in the national capital; and our gravest foreign affairs are always directly administered by the Secretary of State. Since Franklin it has seldom happened that a minister has been entrusted with any grave duties or burdened with any serious responsibilities. Even during the civil war Mr. Seward managed at Washington the more serious business of the foreign department.

In this generation, we have had some successful foreign ministers; but their success has in every case been in non-official or extra-official lines. Mr. E. B. Washburne, our minister in Paris during the Franco-German war, won a high reputation, not as a diplomat of his country, but as an American minister entrusted, by an act of international courtesy, with the rights and welfare of Prussians in Paris. As the agent of the Berlin government during the war and siege, Mr. Washburne endeared himself to the large German population of Paris by his kindness, common sense, and energy in caring for a body of subjects of a hostile country. No one but a typical American could have done this work at all well. A man trained to diplomacy would have failed. It needed a man who could put his character and American office into a breach made by war, and devise means of providing for an

extemporaneous necessity. Most men would have failed; Washburne succeeded because he was a typical American of the largest pattern—able, frank, tireless, resourceful.

In England, Mr. Lowell has, under different circumstances, developed a new line of diplomatic representation. He has represented the character and culture of the American people. The average politician supposes himself to be the typical American; the fact that we are ashamed of him is the sufficient proof that he is thoroughly mistaken. In what do Americans broadly differ from most other, if not all other, peoples? Is it not in this, that we are the great reading nation of the world? Our culture goes down to the bottom and reaches out to the extremities of our life. We have no class distinctions, no titled magnates, no rights of birth-privilege; and in the school rooms of the land the rich and the poor meet together, and our great newspapers go into the hands of both rich and poor. We are not so distinctly a nation of commercial people as we are a nation of people under an unexampled influence of general culture. A Professor who was much more than a professor; a man of the world who was also a man of books; an alert and quick, practical man who was also a poet; a gentleman who was, first of all, manly; a diplomat who was at home in a school room, could lecture on any literary theme, could instruct the educated Englishman in the history of his own men of letters; an American who knew Shakspeare, Milton, Bunyan, the English Bible, and every shred of English art and song and event by heart—such a man has Mr. Lowell been in England. He was at home there on those levels of life and heights of achievement which are common to us and Englishmen; and his leadership there in literature and learning conferred honor on us in the precisely most honorable things in our character as a people. We simply are not philistines—hard dogmatists of dollars and precedents—we are a thoughtful, informed, studious and brainy people who despise wealth which is not held for the service of truth, well-being and progress. No man could represent us so well as a poet, teacher, essayist and scholar like Mr. Lowell. He could represent us at the level of those ideals and achievements of which we are not ashamed. We despise nothing so heartily as illiterate wealth; no other people on earth equal us in contempt for the commerce which has only sordid aims and results. The American merchant builds schools, museums, churches, asylums, hospitals—if he does not we despise him.

The problem in selecting the best foreign representatives is to secure men who will represent our national character. Our diplomacy is unimportant. Such as we have to do can be carried on at Washington. We want ministers abroad who will be typical Americans, whose conspicuous position will display in them the best and truest results of our unique social and political system. If they are honored abroad, as Mr. Lowell has been in unusual ways and measure, the honor is given to us and glorifies us. There is always a fool to say that our minister is more English than American—simply because Englishmen respect him. It is a poor kind of criticism, and fortunately there is less and less of it in the press. We all stand better abroad, command more honest respect, are better understood in our best characteristics, because James Russell Lowell has represented us in London. May his successor succeed to the full measure of this representative office.

The most distinguished honor ever conferred upon an American has come to Mr. Lowell. The English press, representing the best public opinion of that nation, invites him to remain., in England “as the unofficial representative of American literature, learning, manners, and knowledge of the world.” Nothing like that can be found in the long history of diplomacy. “He has been,” says one great journal, “a sort of guest-friend of England,” and it then describes him as “the most eminent American of this generation,” and adds, “Englishmen of all ranks and stations recognize in Mr. Lowell a faithful and jealous guardian of the interests of his country, and a type of all that is best in its intellectual and moral character.” There is not a word in all these encomiums which is not a eulogy of the American people.

THE CHOLERA.

The disease called Asiatic Cholera is at home in India. It travels, at long intervals, into Europe. There have been eight: or nine of these visitations during the century. The disease travels with man and his belongings; and since intercourse between Asia and Europe has become more swift and abundant, there is a tendency to more frequent visits of the dreaded scourge. In fact, however, this tendency has been overcome by sanitary science.

Until last year, cholera had not been in Europe for sixteen years. It arrived the last previous time in 1866 and tarried into the next year. During all the intervening years, there was cholera on the Ganges, and an increasing flow of humanity between the two continents; yet sixteen years elapsed between the visits. Last year cholera landed from ships at Toulon and Marseilles in southern France, and produced general alarm in Europe. It swept around all the European coasts of the Mediterranean, and in the especially filthy towns caused a large mortality. It has doubtless wintered in Europe, and its second year is usually the worst. It can begin early and use the long summer for its desolating work. This is the second summer. It is probable that cholera will be a large feature of the health and mortuary reports of the year. European travel will be restricted by the caution which the prevalence of this disease inspires. American resorts for Americans will be unusually popular.

Will cholera visit us? In previous European visits it has always looked in upon us, sometimes the first, sometimes the second year. In 1866, it did not come, but did come in 1867. But the visit of that year was less baneful than any other we have had. In 1884, we escaped; can we escape in 1885? No doubt exists that we *can*. It is only a question of effective quarantine. The faithful discharge of their duties by all health officers at ports would exclude the unwelcome guest. And, at our principal port, where the danger is greatest, there is good reason to believe that the fidelity, wisdom and vigilance are equal to the emergency. Cholera is more likely to penetrate to us by some little-used door; the front gate will be securely guarded. At the smaller doors, there ought to be no danger. But there *is* danger, and it is probably too much to hope that there will be adequate watchfulness. There is, even at New York, a long line of accidents to be reckoned with. It would not be surprising, however, if cholera got a foothold among us from the West Indies, landing in some southern port.

If it comes, what then? It should not be very successful in its work of death. It could be stamped out in any city with a well organized health force. In New York, a few cases would not be cause for any alarm—not half so much as if the cases were in some small and careless community. Still it is most probable that if it effects a landing it will travel over the country. We have had warning enough. It is a filth disease which can be cornered and killed by cleanliness, if the cleanliness foreruns it. If we wait to clean up filthy quarters in our towns until the disease arrives, it will then be too late

to clean. We shall have a good measure of the sanitary condition of our cities, if the cholera visits them. It will do its terrible work in the unclean quarters of unclean towns. It will not stop there. Once established in a filthy quarter, cholera easily thrusts its arms into adjacent clean places. If the long and loud warning has been heeded, there will be little to fear from cholera. The cleanliness of our towns will discourage and expel the intruder.

What can individuals do? Keep cool and in good health. All epidemic diseases fix upon the infirm, the debilitated, and the fearful. The mortality from cholera is only about one death for four cases. In some epidemics of it, the rate is one in three cases; but among people who are well fed and in fair conditions of comfort, and have proper care, there are five chances of recovery to one of death. The high rate of mortality in some towns results from want of care and medicine. Popular rumor exaggerates the danger of death from cholera. It is a case of *ogne ignotum pro magnifico*—we know so little about it that we magnify the danger beyond all warrant of the actual facts. Some of our danger—perhaps most of it—will come from the enthusiasm of the reporters. Last January these enthusiasts discovered genuine Asiatic cholera in St. Louis. They were sure of it. The evidence was perfect, they said. Of course there never was, and never will be, a case of Asiatic cholera in St. Louis in January. But after summer begins, the “Lo here” will begin to alarm the timid. There are sure to be many false reports; the true one may be in the bundle. Let us hope it will not be—and keep cool.

We hope that inland quarantines will not be resorted to if the disease appears among us. They are useless as well as inhuman. They shut the intruder out of the gate, and he crawls under the fence. If cholera gets upon this continent, safety will be secured only by cleanliness of streets and houses. Clean people may die in clean streets, but it will be because there are unclean people in neighboring streets to receive and breed the disease. Nor should the well fly from the sick. The nurses and priests are safest in Italian cities. Those who fly, do so when it is too late, and carry the disease with them. And, after all, cholera may not visit us in 1885.

A REVIEW OF THE CHAUTAUQUA YEAR.

A very marked indication of the success of the Chautauqua Idea is the increase of imitations of our work; and this runs parallel with an increase of efforts to promote systematic culture among grown people. Dr. Samuel Johnson used to say that he did all his hard study when he was a boy, and he very properly lamented it. It is one of the strangest things in modern civilization that, except among a small body of professors and specialists, the world does its studying entirely in early life. If the chance to study, or the compulsion to study, be wanting in the first twenty years, the door is supposed to be shut forever. The Chautauqua movement tried to expose the folly of this method, and to show that persons whose education was neglected in youth may secure culture in middle life, or even in old age. History gave us examples enough; but Chautauqua has made thousands of new examples to illustrate the perfect practicalness of adult study. The success of Chautauqua has drawn general attention to the subject. A variety of plans for promoting the education of adults are coming before the public. We do not regard any of them as rivals; they will enlarge our public as well as make smaller centers of such culture.

The members of the C. L. S. C. who have gone with us over the subjects of this year will do well to look back over the road and see precisely what they have gone over. They will probably notice that their study and reading has not interfered with their regular pursuits and duties. They have been able to add these studies to their customary tasks and interests. It is a kind of gratuity, therefore, which they have received. It is so much in addition to other results of annual effort. They will further see that the amount of this study and reading is considerable. Good method has made odd minutes yield a bulk which would require weeks of consecutive and unbroken effort. The effect on the mind is better and more permanent because the study has been continued through a year. A college man of our acquaintance says that Professor Time does better work than any of his colleagues. Our C. L. S. C. members have had the instruction of Professor Time. Those who are finishing the course will do well to remember that they have learned how to learn without a living teacher. This is the best thing the C. L. S. C. has done for you—this helping you to study alone. A power of this sort ought to be cultivated and kept. The contents of any course of education should be small in comparison with the attainments which it renders possible. This large, broad, life-long self-education lies in the power to study alone

without the fear of a master or the ordeal of an examination before one. This power colleges fail to give, and it is one of their weaknesses that they can not give it. When the taskmaster stopped, Samuel Johnson said that he stopped learning. We may doubt it, but in a sense Johnson was right. If any reader has acquired the power to *go on alone* by pursuing the course, he has a rare piece of wealth, a capital which may produce the widest and best culture.

“LIBERTY ENLIGHTENING THE WORLD.”

The New York *World* has done wondrously well to raise \$30,000 by a popular subscription for the setting up of the Bartholdi statue in New York harbor. Our readers know that this colossal statue of “Liberty Enlightening the World” is a present from the French people, but that there has been no provision made for the cost of setting up the great work of French art. Various attempts to raise the necessary \$100,000 have yielded but small results, and when the *World* took hold of the matter the failure had become a very disagreeable joke for the press. Other newspapers have engaged in public enterprises. The New York *Herald* sent a ship into the Arctic Ocean in search of the Pole. The New York *Tribune* maintains by the subscriptions of its readers a “Fresh-Air Fund” which gives thousands of children an outing in the country during the hot months. The enterprise of the *World* is, if possible, more praiseworthy, and its success in rescuing a lost cause is as honorable as it is unexpected. The subscriptions have been made by people of all conditions and fortunes, by boys and girls, by merchants, artists and laborers. A popular interest in the subject has been created, a great deal of information imparted, many mistakes corrected, and a popular support secured for a work of art. The work is in the line of large and healthy philanthropy, and the *World* deserves unstinted credit. Let us all hope that it will carry the cause through to entire success.

We are, as a nation, still poor in public statuary. We have some good pieces and not a few failures. Perhaps many persons have feared that the Bartholdi statue is another monstrosity. Let them take courage; the *World* has presented excellent evidence that the work is equal to its mission. The French people are to have a small edition of it set up in Paris, and eminent

French critics are unsparing in their praise of it. Of course we can not be too confident of the effect of a statue higher than Trinity steeple or the pillars of the East River bridge, set up so as to face the sea and meet the eyes of the world entering the harbor of New York. On that point there is excellent artistic judgment in favor of the success of the audacious conception of the artist. We may now regard the plan as certain to be carried out at no distant day, and if it succeeds it will be one of the wonders of the world. A statue rising to the height of three hundred and seventeen feet above the waves—the familiar figure of Liberty confronting the world in our chief American city—will be a unique and impressive piece of art, if it be not an utter failure. The *if* is not a large doubt; only the small doubt which attends the most perfect human work which is as yet untried. The *World's* popular subscription had passed \$35,000 when these lines were sent to press, on the 28th of April.

EDITOR'S NOTE-BOOK.

A highly successful "Chautauqua Day" has been carried out at the New Orleans Exposition, under the leadership of Prof. E. A. Spring, Prof. W. F. Sherwin and the Rev. A. H. Gillet. An audience of over 5,000 people met in Music Hall to hear Bishop Mallalieu on "The Relation of Chautauqua to the Home and Society," Prof. Sherwin on "The Story of Chautauqua," talks from Wallace Bruce and the Rev. Mr. Gillet, and music from the Mexican Band. The great crowd was given Chautauqua badges and C. L. S. C. circulars, and taught the Chautauqua salute—the latter in compliment to Señor Payen, the leader of the Mexican Band, and the pet of the Exposition. One of the audience remarked on leaving the hall: "I have been at every day entertainment here since the beginning, and this has been the most successful, the best managed, the most interesting occasion there has been."

There has been an almost breathless attention given throughout the world during the past month to the attitude of England and Russia. Words have been weighed and steps measured with the nicest exactness. Affairs which in other times would take up columns are given corners. The prevalent opinion has been that war must come, though peace negotiations are being vigorously pushed. Could war be held off twenty years it might be that we should be so much wiser that arbitration might be made to prevail.

The most active business which the United States Navy has had for years was caused by the Isthmus troubles. Our treaty with New Granada guarantees free and uninterrupted passage across the Isthmus. To secure this a force of 500 men and four ships was placed at Aspinwall after the insurgents had burned the city. They thence proceeded to Panama, which they succeeded in restoring to order. Not an easy thing to do, with the natives sympathizing generally with the rebels, and very suspicious of the "Gringos," as they call us, and with the French determined to have the credit of whatever restoration could be made.

M. de Lesseps is firmly convinced of the advantages to America of the Panama Canal, prophesying in a recent letter that it will reanimate our merchant marine, will greatly enlarge our internal business, and make us, in short, commercial kings. It can hardly be doubted that were any one of the great schemes for interoceanic communication in operation, our whole system of exchange would be modified and our business multiplied. The chief question seems to be now, is the Panama Canal, Mr. Eads's Tehuantepec Ship Railway, or the Nicaragua Canal likely to be the best for the United States?

The recent conviction of two "saints" charged with polygamy is having a wholesome effect. The law which makes such a decision possible in Utah is the Edmunds bill, by which a man believing it right to have more than one living or undivorced wife at a time maybe challenged as a juryman. Under this law it is extremely difficult for a Mormon brought to trial to escape sentence, and the sentence is such that the fear of it will more than probably take a great deal of charm out of the doctrine of "spiritual wives."

The biographical department of the scrap-book compiler has had great opportunities to grow rich in anecdotes of General Grant this past month. His prolonged illness and the wonderfully sustained interest and sympathy of the public have led to many letters and stories getting into print which otherwise might never have been known. All of this matter but shows more and more clearly how just is our love and reverence for "our hero."

For the benefit of the international copyright there were held in New York recently, on two afternoons, Authors' Readings. A notable galaxy of literary characters gathered on the stage. Among them were George William Curtis, Professor Charles Carroll, Julian Hawthorne, Will Carleton, W. D. Howells, the Rev. Robert Collyer, Prof. H. H. Boyesen, Bishop Potter, Mark Twain, Edward Eggleston, Henry Ward Beecher, and others. The audience which greeted the readers was large and appreciative.

One of the most horrible accidents of the month of April was the collapse of eight nearly completed tenement houses in New York City.

Twelve persons were injured in the fall. The investigation discloses a deliberate intention on the part of the contractor to use the poorest material obtainable, and a criminal—whether intentional or not—neglect on the part of building authorities to prevent his plan. Indeed, contractors who are honest in their work, state the commission cares more for the fee which puts a contract through than for the material put into the building. A vigorous public sentiment which would put a few such criminals into State prison might prove beneficial.

“Arbor Day” is extending its conquests. Its latest subject is none other than the Province of Ontario, whose Minister of Education sends out a communication to all teachers asking that in the interest of sanitation and æsthetics the 8th of May be set apart as a holiday in every rural and village school, for grading school grounds, laying out walks, and setting out trees.

Hon. B. G. Northrop, in a letter dated April 4th, writes: “Sixteen states have adopted ‘Arbor Day,’ and to that result the article in THE CHAUTAUQUAN, and extracts quoted from it, have greatly contributed.”

The Bureau of Education sends out a pamphlet on “Arbor Day.” It will help many a teacher to an interest in the scheme, and will tell him what trees to select, how to plant them, and how to arrange a program suitable for the holiday.

The trouble which we have with foreign tongues sometimes enables us to heartily appreciate the blunders of the foreigner who tries our English tongue. We can not possibly make worse blunders with French than a late number of the aristocratic *Révue des deux Mondes*, when in its attempts to express in English the idea of a candidate favoring a railway it said, “*My politic is railway.*”

One might almost call Lord Tennyson’s verses on “The Fleet” the literary sensation of the month. It is rather difficult to see just why the verses deserve the unstinted ridicule American papers have given them. Certainly they are tame, but Lord Tennyson is an old man—a man whom

this generation ought to honor for the noble pleasure he has left to it and to its posterity. Were his lines much worse, courtesy demands that we remember that with age comes decay.

The new Minister of Education in Greece, where the four Gospels are used as a reading book by the advanced classes in the primary department of the public schools, proposes to introduce them into the higher schools. The purity of the diction of the four Gospels makes them, regardless of other considerations, most desirable reading for children who are just forming their literary style.

It is a sort of fashion which common sense people will be glad to adopt—the run on plain food which London society is said to be having. The Prince of Wales began it by cutting down every *ménu* over which he presided, and he has had a larger following than usual—both the people who always follow and those whose good sense decides what they do, taking up the custom. It would be a wise lesson for both dinner givers and hotel keepers to learn that the quality of a dinner can not possibly depend upon the number of dishes.

Bringing out a successful novelty in flowers or vegetables, is to a nurseryman what a bonanza mine is to a westerner, or a flowing well to an oil producer. The present season has several: There is a new lettuce with leaves like the oak, a red celery, a celery which blanches naturally and bears leaves like an ostrich plume; a pansy with blossoms two and one fourth inches in diameter; a zinnia round as a ball, and a mignonette with spikes twelve to fifteen inches in length, and of pure white. Perhaps the greatest novelty about the business this spring is the fact that in the cities the safe deposit vaults are receiving thousands of pounds of invaluable seeds for keeping—treasures quite as priceless to the seedsmen as the jewels which repose beside them are to their owners.

Another compliment to America! When M. de Lesseps, the new member of the French Academy, was installed, in April, he delivered, it is said, the shortest speech ever delivered by an incoming Academician. Thereupon M.

de Rénan complimented him on adopting the pithy, pointed style of *l'Amerique*.

We are very glad to introduce the Chautauqua Town and Country Club, a new branch of the C. L. S. C., for the study of agriculture. Everybody is expected to do something in this novel society of practical out-of-door work. Everybody must raise a plant, or cultivate a bed, or care for an animal. A well known Orange Co., N. Y., farm is to be the working headquarters of the new organization, and its course of reading and experiment is under the control of Mr. Charles Barnard. Chancellor Vincent honors the organization with his supervision, a guarantee of success, and the headquarters of the C. L. S. C. have been extended to take in the business of the C. T. C. C. In July we hope to give our readers a broader look at this charming club.

We called attention recently to Dr. Warren's entertaining theory of the whereabouts of the garden of Eden. And now we have another explorer for this land-one Moritz Engel, of Dresden, who locates it about seventy miles southwest of Damascus. Herr Engel makes his theory almost as entertaining and plausible as does Dr. Warren his.

The article on the "Natural History and People of Borneo," which appears in this issue of THE CHAUTAUQUAN, is in reality an outline of a book, "Ten Years in a Jungle," written by Mr. Hornaday, and soon to be published by Messrs. Scribner's Sons. This work is to be fully illustrated and furnished with maps, and will give much desirable information on many of the points but lightly touched in the article.

One of the most magnificent pieces of architecture in the world is about to be erected in St. Petersburg as a votive chapel commemorating the murder of the Emperor Alexander II., of Russia. It stands over the spot where he fell, is to be erected by donations from the entire nation, will cost \$10,000,000, and will be completed, it is expected, in 1891.

Mr. James Anthony Froude says in a recent article that the best histories are those which are written by men who hate “moral evil” and love “moral good,” and who are “afraid to tell lies” to defend their theories; as examples, he mentions, Herodotus, Thucydides, Tacitus, and Carlyle.

The readers of THE CHAUTAUQUAN are familiar with the *Chautauqua Assembly Daily Herald*, the organ of the Chautauqua meetings. The forthcoming volume will contain all the best things said on the Chautauqua platform and in all the meetings during the coming session, with comments on the great men who frequent Chautauqua during the months of July and August. In short, it will depict in all its interesting details the unique life of the great Assembly. This paper, with its nineteen issues, is the necessary supplement to the THE CHAUTAUQUAN, just as the Assembly is the supplement to the work of the year. Every reader of the C. L. S. C., every reader of THE CHAUTAUQUAN, every lover of Chautauqua should have the *Assembly Daily Herald*. For rates and address see advertisement in this impression.

Evening High Schools are becoming a permanent feature in the school system of a number of our large cities. They deserve the heartiest support of educators and municipal authorities. In Cincinnati there is an evening high school similar in requirements to the upper grade grammar school; in St. Louis one which prepares students for the Polytechnic school of Washington University; one in New York which, in 1883, had an average attendance of 951 pupils; another in Boston which, in October 1884, enrolled 1,592 pupils. The character and the patronage are proofs sufficient that such institutions are in demand. A late circular from the Bureau of Education declares that “it is an institution which has come to stay, and that it has a more important future than can now be understood.”

A hopeful sign to temperance workers is the favor with which the scientific temperance education bill has been meeting in the legislatures of various states. During the winter of 1884-5 this bill became a law in Nevada, Alabama, Wisconsin, Missouri, Maine, New Jersey, Oregon, Nebraska, Pennsylvania, and Kansas, making in all fourteen states which

require instruction in public schools concerning the physiological effects of stimulants and narcotics.

C. L. S. C. NOTES ON REQUIRED READINGS FOR JUNE.

Inoculation for cholera is the latest preventive. A number of people have tried it in Valencia, Spain, and with perfect success, the authorities contend. Cholera virus is used. The patient becomes prostrated with a tumor in about twenty-four hours, but usually recovers in a couple of days. Certainly, if as sure as speedy, humanity has found a boon. It will be wiser, however, to wait until the cholera attacks Valencia before trusting too implicitly to inoculation for cholera.

Unbounded curiosity has been displayed at the New Orleans Exposition, over the "Woman's Department." In spite of fate Mrs. Julia Ward Howe and her co-workers have succeeded in making it a success, and in getting a fair representation of what women can do. They have a literary department filled with scores of volumes written by women, a woman's exchange, samples of ten thousand kinds of fancy work, of course, and, better than that, of engraving, wood-carving, drawing, silk culture and weaving; a novel niche is made by a display of patents taken out by women, and by blacksmith work and forging done by women. Could they transport a cottage with its inmates and let the work of home-making go on day after day, they would add the crown to their department.

Mrs. Elizabeth Cady Stanton in the *North American* makes the remarkable statement that women have never been benefited by Christianity, but that their elevation is altogether due to the spread of Teutonic respect for noble women. One must naturally conclude that Mrs. Stanton has not studied her history thoroughly. Her statements sound very much like an attempt to build up a theory without respect to the facts in the case. Certainly no fact is better established than that with the spread of the doctrines of Christ came the elevation of women.

CHEMISTRY.

1. “De Saussure,” sō-sür, Horace Benedict. (1740-1799.) A Swiss naturalist who traveled extensively and made valuable contributions to science. Much of his time and attention were given to mountain researches. He constructed the best hygroscope, the instrument which indicates a change in the degree of moisture in the atmosphere. The cyanometer, an instrument for measuring the degrees of blueness in the sky, was also invented by him.

2. “Osmose.” The tendency of fluids, when placed in contact, to mix.

3. “Cotyledons.” The seed lobes which surround and nourish young plants. Endogenous plants, or inside growers, such as Indian corn, have only one cotyledon, while exogens, or plants that grow by making external additions, have two, as seen in the young pea.

4. “Hales,” Stephen. (1677-1761.) An English clergyman who gave much time to scientific study. He was the author of several works, among which was a book on the physiology of vegetables, entitled “Statical Essays,” which has been translated into several languages.

5. “Dean Trench,” Richard Chevenix. (1807- .) A British clergyman. In 1856 he was made Dean of Westminster, and in 1864 Archbishop of Dublin. His publications are numerous, including poetical, theological, philological, and historical works.

6. “Venus fly-trap,” or dionæa. This curious plant is a native of the sandy bogs of North Carolina. It is a perennial plant, whose leaves grow from its roots. In the midst of them there grows a leafless stem, about six inches high, which terminates in a cluster of white flowers. The long, slender leaves bear at their extremities a second leaf, which has been compared to two eyelids joined at their bases; this leaf is fringed with stiff, bristly hairs, and on the upper surface there are three very delicate bristles on each side, so placed that a fly can not walk over the leaf without touching one of them. As soon as touched the two parts of the leaf instantly close over the victim and hold it in their embrace until death follows, and the insect is partly absorbed. The sensitiveness of the plant resides entirely in these bristles on the surface of the leaves. The juice from the glands attracts the insects, and

the plant receives nourishment from their bodies, which are partly dissolved in a liquid which exudes from the leaves.

7. “Nepenthes,” or pitcher plants. A genus of plants whose leaves form receptacles for water. The plants are natives of the swamps in the East Indies and Australia. The linear-lanceolate leaves have at their extremities a long, spiral stem, at the end of which is attached an urn, or pitcher, of narrow cylindrical form, from six to twelve inches long, in the different species, which are of the same color as the leaves, in some varieties, in others spotted with red, and still others are of a blood red color. The pitcher is terminated by a lid which is sometimes closed. The pitcher is always partly filled with watery liquid. Honey is secreted around the mouth of the pitcher, and insects are found within it, which have been lured by the honey. These are supposed to contribute to the nourishment of the plant. For further description, see the article on “Borneo,” in the present number of THE CHAUTAUQUAN. There are some varieties of the pitcher plant found in the United States.

8. “Diatoms,” A species of microscopic plants belonging to the Algæ or sea weeds, growing in both salt and fresh water. For a long time they were thought to belong to the animal kingdom, and were classed among the animalcules, as they had the strange power of secreting from the water silicious shells. They grew in some places in such abundance as to form large beds of their shelly remains, the material of which is used for the polishing powders known under the name of tripoli. The species found now are exactly like the fossils. Ehrenberg says: “Species are to be found, in a living state, in situations where they have been propagated from times far anterior to the existence of man.” They may be obtained by allowing the water in which they grow to stand for a time, and then pouring off all but the slimy part at the bottom. The diatoms may often be seen to move a little in the water, from which it was supposed they belonged to the animal kingdom. Professor Bailey describes one very interesting species, as follows: “At one moment the needle-shaped frustules lie side by side, forming a rectangular plate; suddenly one of the frustules slides forward a little way, the next slides a little also, and so on through the whole number. These motions are constantly going on, and with such rapidity that the eye can scarcely follow them. The cause of the motion is wholly unknown, but it is probably mechanical and not vital.” One species in its growth takes on

a fanlike appearance and is exceedingly beautiful. Fossil diatoms have been found in all the strata of rock formations. These with some of the minute creatures of the animal kingdom have left greater records of themselves than most of the higher forms of life. Extensive deposits of their silicious shells are found in various parts of the world. The city of Richmond, Va., stands on a layer of them which is eighteen feet in thickness.

ERRATA.—The first line of the second experiment, page 441 of THE CHAUTAUQUAN for May, should read: “A body is buoyed up in water by a force equal to the weight of the water it displaces.”

SUNDAY READINGS.

1. “Summerfield,” John. (1798-1825.) An American divine. He was born in England, and moved to New York in 1821. He visited many of the large cities, where his eloquence drew crowds to listen to his sermons. He founded the American Tract Society.

2. “Krumacher,” Friedrich Adolph. (1768-1845.) An eminent German theologian, pastor of the Reformed Church at Crefeld, and later pastor at Bremen. He was the author of many works, both in prose and poetry.

ANIMAL BIOLOGY.

1. “Batrachia,” bā-trā'kĩ-ä.

2. “Ophidia,” ō-fid'ĩ-ä.

3. “Lacertilia,” lā-ser-til'ĩ-ä.

4. “Iguanodon,” i-guā'nō-don. This gigantic fossil reptile was discovered in the year 1882, in the Wealden beds of Kent and Sussex, also in the Isle of Wight, by Dr. Mantell. The enormous bones were very abundant, and from a close study of them, it has been estimated that the extreme length of the reptile must have been twenty-eight or thirty feet, of which the head was three, and the tail thirteen feet. It stood higher above the ground than any reptile now in existence. The teeth, which bear a remarkable resemblance to

the teeth of the American lizard, the iguana, from which it was named, show that the animal was herbivorous. It probably fed on the trees growing along the borders of swamps and streams, and was able to lift its body for this purpose on its hind legs.

5. “Meg, a-lo-sau'rus.” The word comes from the Greek, as do nearly all the words in this list, and means great lizard. The bones found in the rocks of the Oolite formation, the one next below the Wealden, show the animal to have been terrestrial and carnivorous. It fed upon smaller reptiles and the young of the larger orders. Its huge body was supported on four large, strong legs, the hind ones being estimated to have had a length of nearly six feet. The length of the animal must have been thirty or forty feet. In the first estimates made of all these saurians, their size was overestimated, the body of the iguanodon having been given as seventy feet in length, and the megalosaurus as sixty or seventy feet.

6. “Ple, si-o-sau'rus.” The word means like a lizard. It is a remarkable fossil sea-reptile, which shows a long, snake-like neck, in which there are thirty-three vertebræ, ten more than in the longest neck of any bird. The body and head are comparatively short. Its whole length, when living, was probably twenty-five or thirty feet. Buckland describes it as follows: “To the head of a lizard it united the teeth of a crocodile, a neck of enormous length, resembling the body of a serpent, a trunk and tail having the proportions of an ordinary quadruped; the ribs of a chameleon, and the paddles of a whale.” There were fifty teeth in each jaw. Conybeare says of it: “That it was marine is evident from its paddles; that it may have occasionally visited the shore, the resemblance of its extremities to those of the turtle may lead us to conjecture; its motion must have been, however, very awkward on land; its neck must have impeded its progress through the water. May it not therefore be concluded—since, in addition to these circumstances, its respiration must have required frequent access to the air—that it swam upon or near the surface, arching back its neck like the swan, and occasionally darting it down at the fish which happened to float within its reach? It may, perhaps, have lurked in shoal water along the coast, concealed among the sea weed, and, raising its nostrils to the surface from a considerable depth, may have found a sure retreat from the assaults of dangerous enemies, while the flexibility of its neck may have compensated for the want of strength in its jaws and its incapacity for swift

motion through the water, by the suddenness and agility of the attack which it enabled it to make on every animal fitted for its prey which came within its reach.”

7. “Ich'thy-o-sau'rus” (fish-lizard). This animal, which also is known only by its fossil remains, must have been about thirty feet long. It had enormous eyes, a long and large head, a mouth armed with powerful teeth, and a short neck. The body was fish-like in form, and the four paddles, resembling those of a whale, were comparatively small. The tail was long and gradually flattened toward the end; it was the principal organ of locomotion. They were very active in their movements, and consequently were dangerous enemies to all other animals living in the sea. Their food consisted chiefly of fishes.

8. “Chelonia,” kē-lo'nĭ-ä.

9. “Loricata,” lör,i-cā'tä.

10. “Aves,” ā'vēs.

11. “Anchylosed,” ang'kī-losed. United; made fast; stiffened.

12. “Esophagus,” e-sof'a-gus. The passage leading to the stomach, through which the food and drink pass.

13. “Raptores,” rap-tō'res.

14. “Insessores,” in-ses-sō'res.

15. “Cursores,” cur-sō'res.

16. “Grallatores,” gral,-lā-to'res.

17. “Natatores,” nat'-a-tō'res.

18. “Monotremata,” mon'o-trem'a-tä.

19. “Edentata,” ē-den'ta-tä.

20. “Armadillos,” ar-ma-dil'lōs.

21. “Sirenia,” sī-rē'ni-ä.

- [22.](#) “Cetacea,” sē-tā'sē-ä.
 - [23.](#) “Cheiroptera,” kī-rop'te-rä.
 - [24.](#) “Rodentia,” rō-den'shī-a.
 - [25.](#) “Ungulates,” un'gū-lates.
 - [26.](#) “Pinnigrades,” pin'ni-grades.
 - [27.](#) “Plantigrades,” plant'ī-grades.
 - [28.](#) “Digitigrades,” dig'i-ti-grades.
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Those who wish to carry on further the study of Animal Biology will find a wide range of reading on the subject awaiting them. We append a list, but are unable to give publishers and prices: “Comparative Zoölogy,” by James Orton; “Methods of Study in Natural History,” by Agassiz; “Principles of Zoölogy,” by Agassiz and Gould; “Elementary Biology,” by Huxley and Martin; Wood’s “Illustrated Natural History;” Jones’s “Animal Creation;” “Elements of Zoölogy,” by Holder; Packard’s “Zoölogy;” Nicholson’s “Manual of Zoölogy;” “Winners in Life’s Race: or, The Great Backboned Family,” by Arabella B. Buckley; “Life and her Children: or, Glimpses of Animal Life from the Amœba to the Insects,” by Miss Buckley.

For special studies: Lyell’s “Antiquity of Man;” Blumenbach’s “Natural History of Man;” Huxley’s “Elementary Lessons in Physiology;” “Sea-side Studies in Natural History,” by Agassiz; Taylor’s “Half-hours at the Seaside;” Dana’s “Corals and Coral Islands;” Duncan’s “Transformation of Insects;” Packard’s “Guide to the Study of Insects;” Coues’s “Key to North American Birds;” Jordan’s “Popular Key to the Birds, etc., of the Northern United States;” “Birds of North America,” by Baird, Brewer and Ridgeway; Baird’s “Mammals of North America;” Scammon’s “Marine Mammals of North Pacific;” Coues’s “Fur-bearing Animals of North America;” Huxley’s “Manual of Vertebrates;” “Tropical Nature,” by Wallace (a work on reptiles); “Check List of North American Reptiles and Batrachians,” by E. D. Cope; “Game Fishes of the United States,” by G. Brown Goode; “Blind Fishes of the Mammoth Cave,” in “American Naturalist,” vol. vi., p. 6; Holder’s “American Fauna;” Agassiz’s “Development of Osseous Fishes;”

Gunther's "Introduction to the Study of Fishes;" Packard's "Guide to the Study of Insects," also his "Half Hours with Insects;" Wood's "Strange Dwellings," "Natural History," and "Homes Without Hands;" Burmeister's "Entomology;" Lubbock's "Ants, Bees, and Wasps;" McCook's "Agricultural Ants of Texas;" "Crustacea of the United States Exploring Expedition," by J. D. Dana; "The Lobster and Lobster Fishing," by W. W. Wheildon; "Barnacles," by J. S. Kingsley; "The Cray Fish," by Huxley; "Terrestrial Air-breathing Mollusks of the United States," by W. G. Binney; "Our Sea Anemones," by A. E. Verrill; "The Atlantic and Depths of the Sea," by Thompson; Leidy's "Fresh Water Rhizopods;" Pritchard's "Infusoria."

Serial publications containing much matter in regard to these subjects are *Popular Science Monthly*, *American Journal of Science*, *American Naturalist*, *Smithsonian Contributions*, *Annals and Magazine of Natural History*, and *Nature*.

COURSE OF READING FOR 1885-6.

Roman History and Latin Literature.

Studies in Italian History, Biography, and Art.

Political Economy.

Studies in Human Nature.

Studies in Robert Browning.

Electricity.

Philosophy, International Law, etc., etc.

The books for the course will soon be announced.

J. H. VINCENT.

NEW HAVEN, CT., April 27, 1885.

PARAGRAPHS FROM NEW BOOKS.

“Look there, my dearies!” and she pointed to a little blue and white tea-pot on the high mantel-shelf above the hearth, on which they were sitting. “Last night, when they’d taken Ben away, and I couldn’t finish t’ psalm, and I couldn’t do much more praying than a little bairn thet’s flayed and troubled in t’ dark night, I lifted my eyes to thet tea-pot, and I knew t’ words thet was on it, and they wer’ like an order and a promise—a’ in one; and I said, ‘There! thet’s enough, Lord!’ and I went to my bed and slept, for I knew there ’ud be a deal to do to-day, and nothing weakens me like missing my sleep.”

“And did you sleep, Martha?”

“Ay, I slept. It wasn’t hard wi’ t’ promise I’d got.”

Then Phyllis took a chair and stood upon it and carefully lifted down the tea-pot. It was of coarse blue and white pottery, and had been made in Staffordshire when the art was emerging from its rudeness, and when the people were half barbarous and wholly irreligious—one of half a dozen that are now worth more than if made of the rarest china, the “Blue Wesley Tea-pot;” rude little objects, yet formed by loving, reverential hands to commemorate the apostolic labors of John Wesley in that almost savage district. His likeness was on one side, and on the other the words, so often in his mouth, “In God we trust.”—*From “The Hallam Succession,” by Amelia E. Barr.*

Discriminate in the use of AND and TO. Instead of saying “Go *and* see them before you leave;” “Try *and* help him obtain a place;” “Come *and* meet our friends at my house;” say “Go *to*,” “Try *to*,” and “Come *to*.”

Discriminate in the use of the word ARTIST. Keep *artist* to designate the higher order of workmen; as painters, sculptors, musicians, architects, and the like. Don’t use it to designate barbers, laundrymen, tailors, etc.

Discriminate in the use of *BAD*. Don't say "I have a *bad* cold;" say "a *severe* cold." As colds are never *good*, we should not say they are *bad*. We can have *slight* colds, or *severe* colds, but not *bad* colds.

Discriminate between *BAD* and *BADLY*. Don't make the mistake, so frequently made, of saying "I feel very *badly*." Use *bad*. *Badly* is an adverb, and should not be employed. One might as well say "I feel *happily*," instead of *happy*.

Discriminate between *BALANCE* and *REMAINDER* OR *REST*. Don't say "The *balance* of the library remained unsold;" "He spent the *balance* of the evening at home;" "The *balance* of the money he left in their keeping;" "We will now have the *balance* of the toasts." Use *rest* or *remainder*. Balance denotes the excess of one thing over another.

Discriminate in the use of *DEPOT*. The best critics contend that we should not call a *railway-station* a *depot*. A *depot* is a place where goods or stores of any kind are kept.

Discriminate between *HAD RATHER* and *WOULD RATHER*. Don't say I *had rather* not do it; say I *would rather* not do it.

Discriminate in the use of *POLITE* and *KIND*. Don't say "your *polite* invitation was received;" "You are very *polite* in being so obliging;" "They gave us a *polite* reception." Use *kind*.

Discriminate in the use of *SUCH* and *SO*. Don't say, "*Such* a handsome bonnet;" "*Such* a lovely girl;" "*Such* a rough road." Use *so* handsome, *so* lovely, etc.—From "*Discriminate*." [M]

ONE MEANS OF HELPING RUSSIA.—But there is no obstacle which can not be overcome by energy, spirit of sacrifice, and courage. The Russian despotism must and will be destroyed; for it is not permitted to the stupid obstinacy of one, nor to the infamous egotism of a few, to arrest the progress and light of a nation of a hundred million souls. We can only wish the mode of execution of the unavoidable may be the least disastrous, least sanguinary, and most humane. And there is a force which can strongly contribute to this—it is the public opinion of European countries. It is strange, but quite true; Russian governmental circles are much more

impressed by what is said about them in Europe than by the wailing of all Russia from the White Sea down to the Euxine.—*From “Russia Under the Tzars,” by Stepniak.*

THE GOLDEN PRIME OF '49.—A knowledge of the characteristic features of the mining days of 1849 is essential to a full appreciation of the good sense and political wisdom shown by the miners as a class. Merchants, mechanics, farmers existed but to supply the miners; and the gold of the mines was the chief resource of California. Four fifths of the able-bodied male population were living in the mineral belt, or were on their way thither, when the working season of 1849 opened. Only four years before there had been but five hundred Americans in California; in February, 1848, but two thousand; by December, 1848, this number had grown to six thousand; by July, 1849, to fifteen thousand; and by December, 1849, to fifty-three thousand. Chiefly owing to the gold rush of 1848-53 the center of population of the United States moved eighty-one miles farther west. Within four years after the spring of 1849 the population of the new state was 300,000, and more than \$260,000,000 had been dug from the gold fields.—*From Charles Howard Shinn's “Mining Camps.”*

THE WISE MAN'S VIEW OF LIFE.

“Look on the Spirit as the rider! take
The Body for the chariot, and the Will
As charioteer! regard the Mind as reins,
The Senses the steeds, and the things of sense
The ways they trample on. So is the Soul
The Lord that owneth spirit, body, will,
Mind, senses, all; itself unowned. Thus think
The wise!

“He who is unwise drives with reins
Slack on the neck o’ the senses; then they romp,
Like restive horses of a charioteer.

“He that is wise, with watchful mind and firm,
Calms those wild Fire, so they go fair and straight,
Like well trained horses of a charioteer.”

—Edwin Arnold’s *“The Secret of Death.”*

AMY ALLSTON’S REPORT OF STRAWBERRY CULTURE.—On the 27th of May Janet and I picked our first berries for market, and from that time on we were truly busybodies. We worked early and late, being careful to have dresses short enough not to injure the vines. It was hard work for our bent, weary backs, as it grew warm, but we knew it must be done. Mother, Janet, Will and Pete did good service, and little Cecy cheered us from her wagon under a neighboring tree.

In about five weeks the harvest was over, and after taking breath, we counted our gains.

	DR.	
Cost of plants (3,630)		\$11 80
Cost of picking		10 25
Total		<u>\$22 05</u>

CR.

By 1,178 quarts of berries, at an average of 15c. per qt.	\$176 70
Net profit on one quarter acre	<u>\$154 65</u>

If we had to pay expenses of plowing, weeding, manure, etc., it would have greatly lessened the profits. On the other hand the season was backward, and the yield was not large, they tell me, in consideration of the fine condition of the plants and the cultivation they received, which was better than we could give a larger plot. At all events we set out half an acre more the August after.—*From Helen Campbell's "What-to-do Club."*

[M] Discriminate. A Companion to "Don't." A Manual for Guidance in the Use of Correct Words and Phrases in Ordinary Speech. By Critic. New York: D. Appleton & Co. 1885. Price, 30 cents.

TALK ABOUT BOOKS.

A happy thought was carried into successful execution in the publication of “The Hundred Greatest Men.”[\[N\]](#) The editor conceived the idea that a “natural history of civilization ought to be, not so much a narration of events, as a description of men and things,” and hence for many years busied himself in making a collection of portraits and antiquities as a preparation for his work. The great men of all ages have been grouped into eight classes, and treated under the following divisions of the book: Poetry, art, religion, philosophy, history, science, politics, and industry. Introductions have been written for these different departments by Matthew Arnold, Taine, Max Müller, Rénan, Noah Porter, Dean Stanley, Prof. Helmholtz, J. A. Froude, and John Fiske. The general introduction is by Emerson. Only those characters have been selected who have stood the test of time without diminution of fame. A portrait accompanies each sketch, and many of these have been obtained at great expense of time and trouble. The sketches themselves are word-paintings of the character, the life, and the times of the men, and the wonder is that in such brief space so much information can be given. The volume is a veritable art gallery.

A story of much delicacy of touch, and with some strong delineation of character, is “The Hallam Succession”[\[O\]](#) of Amelia C. Barr. The placing of the story could hardly be more picturesque. Mrs. Barr opens with English life of the first of the present century, carries us to a plantation near New Orleans, through “Sam” Houston’s Texan struggles, and finally to the long, low, verandahed house of a Texas plantation. The varied setting gives the variety to a story of faithful love and devotion. The many pleasing touches given in the descriptions of the characters are the finest features of the book. A fair example is the paragraph quoted on another page of this impression, which in its true and delicate pathos is rarely equaled.

One or two selections taken from “The Harnt that Walks the Chilhowee,” one of the eight stories that make up the volume, “In The Tennessee Mountains,”[\[P\]](#) will show the style and aim of these unique writings, which

are just now attracting so much attention. This “harnt,” or ghost, was supposed to be that of a poor, deformed little creature, who some time before had been shot, and, as every one thought, killed. He himself had been falsely accused of taking a man’s life. His death, however, had been a mistake, and it was the real man who walked the mountain in search of food. Simon Burney, an “uncouth and densely ignorant” mountaineer, sought out the helpless, forlorn little specter, for whose capture \$200 reward had been offered, and said: “I ain’t a-goin’ ter help no man break the law an’ hender jestic; but ef ye will go an’ stand yer trial I’ll take keer of ye. An’ arter the jury hev done let ye off, ye are welcome ter live along o’ me at my house till ye die.” And after the trial and the clearance, this rough friend was as good as his word, and “ungrudgingly gave of his best ... and worked early and late that there might be enough to divide.” “A prince could not have dispensed hospitality with a more royal hand.” The author closes by saying: “The grace of culture is, in its way, a fine thing, but the best that art can do—the polish of a gentleman—is hardly equal to the best that nature can do in her highest moods.” All of the stories are full of interest, and show great love for humanity. The author, in a bright, cheery way, true to the life which she represents, portrays the strong and noble natures found in the lowest walks of life.

“A Book of Sermons,”[\[Q\]](#) by Bishop Simpson, will meet with a warm welcome everywhere. To those who have heard him preach, who have listened to his eager, impassioned words, and have been borne onward by the tide of his eloquence, it may seem that to merely read the words—to have them without the powerful influence, the magnetic personality, behind them—is to possess the letter without the spirit. But such will be happily surprised. The book bears witness that Bishop Simpson was a great sermonizer, as well as a great speaker. By the aid of a little imagination as one reads, he can see again the spare form, and the face lighted up with enthusiasm and love; can hear the voice rapidly and earnestly uttering the words which point out plainly and simply the way of life. While, without this help, those who never heard him will find great pleasure and profit in reading the book.

No one who knows Dr. Buckley but expects good things from his pen, and he has given to young men a particularly useful volume in “Oats or Wild-Oats?”[\[R\]](#) Advice is as difficult a thing to prescribe judiciously as it is

to take pleasantly. It is very liable to be insipid, preachy, to lack verve qualities, which young men particularly abominate; it is very little of it readable. But these talks of Dr. Buckley's abound in readable qualities. They are directly to the point, and full of clear-cut common sense. They advise young men in the practical matter of choosing a profession, and in making a success of it. The talks come from a man who, even if he "never worked a half day on a farm in his life," has observed and conversed with hundreds of farmers; who knows from what he has heard and learned from their confessions to him just where they make their business, social, intellectual, and religious mistakes; and who is not afraid to express what he believes to be their dangers, and their strong points. As with farming, so with all the trades and professions. He knows his ground, and treats it in a tone a trifle blunt, perhaps, but nevertheless full of force and conviction.

Mr. Marvin, an Englishman who for several years has lived in Russia, and is thoroughly acquainted with the character, the policy, and the wishes of that government, has, in writing "The Russians at the Gates of Herat,"^[S] tried to impress more deeply upon the English people the necessity of immediate effort in order to secure themselves in the possession of the entrance to India. The steady advance of Russia for years past toward this empire has been carefully traced, and all the treacherous devices by which she has gained ground are exposed. The popular opinion that colossal mountain ranges lie between the Russian possessions and India has been shown to be an error; the new route over which they are passing is such that one could "drive all the way four-in-hand." "No surrender," must be the motto of every Englishman as regards Penjdeh, and "Hands off," in respect to all places leading directly to Herat, says the author. A careful study of this book will give one a clear idea of this complicated movement.

The "Old Farm"^[T] was one that had been for years without a tenant, for its owner feared to lease it lest its old fashioned house and belongings might not suit the new comer, and he would want it changed. At last one was found willing, glad to take it just as it was. One bright October day a business man from the great city, broken down in health, took possession of it with his family. "A year of retirement and rest will restore his vigor and save him for the future," said the doctor. Very soon the early tastes of the naturalist, for such he had been, awoke, and the days were spent in seeking, gathering and studying specimens of minute animal life, or "the other

tenants of the farm,” as he called them. The conversations held regarding them make up the main part of the work. Spiders, moths, bees, wasps, and ants are among those described at length; their history and habits are very accurately given. The blending of the story with the natural history makes an exceedingly interesting book.

[N] The Hundred Greatest Men. With Portraits, reproduced from Fine and Rare Steel Engravings. Edited by Wallace Wood, M.D. New York: D. Appleton & Co. 1885. Price, \$6.00.

[O] The Hallam Succession. A Tale of Methodist Life in Two Countries. By Amelia E. Barr. New York: Phillips & Hunt. 1885.

[P] In the Tennessee Mountains. By Charles Egbert Craddock. Boston: Houghton, Mifflin & Co. Price, \$1.25.

[Q] Sermons by Bishop Matthew Simpson. New York: Harper & Brothers, Franklin Square. 1885.

[R] Oats or Wild-Oats? Common Sense for Young Men. By J. M. Buckley, LL.D. New York: Phillips & Hunt. 1885. Price, \$1.50.

[S] The Russians at the Gates of Herat. By Charles Marvin. With Maps and Portraits. New York: Charles Scribner’s Sons. 1885. Price, 50 cents.

[T] Tenants of an Old Farm. By Henry C. McCook, D.D. Illustrated from Nature. New York: Fords, Howard & Hulbert. 1885. Price, \$2.50.

BOOKS RECEIVED.

The Women of the Reformation. By Mrs. Annie Wittenmyer. New York: Phillips & Hunt. Cincinnati: Cranston & Stowe. 1885. Price, \$2.00.

Preparation for Reading Xenophon. By James M. Whiton, Ph.D., and Mary B. Whiton, A.B. New York: D. Appleton and Company. 1885.

Assyriology; Its Use and Abuse. By Francis Brown. New York: Charles Scribner's Sons. 1885.

Elias Power, of Ease-in-Zion. By John M. Bamford. New York: Phillips & Hunt. Cincinnati: Cranston & Stowe. 1885. Price, 85c.

Wondrous Love. A Collection of Songs and Services for Sunday-schools. By Geo. F. Root and C. C. Case. Cincinnati: The John Church Publishing Co., 74 West Fourth Street.

The *Century Magazine*, Vol. XXIX. New Series, Vol. VIII. New York: The Century Co.

Composition and Rhetoric. By G. P. Quackenbos, LL.D. New York: D. Appleton and Company. 1885.

The Spinning Wheel of Tamworth. By the Rev. W. A. Smith. New York: National Temperance Society and Publishing House, 58 Reade Street. 1884.

The Secret of Death and Other Poems. By Edwin Arnold. Boston: Roberts Brothers. 1885. Price, \$1.00.

Delivered from Afar. By Ralph Roberts. New York: Phillips & Hunt. Cincinnati: Cranston & Stowe. 1885. Price, \$1.50.

A Child's Garden of Verses. By Robert Louis Stevenson. New York: Charles Scribner's Sons. 1885.

Harriet Martineau. By Mrs. F. Fenwick Miller. Boston: Roberts Brothers. 1885.

Grammar of Old English. By Edward Sievers, Ph.D. Boston: Ginn, Heath & Co. 1885.

Hygiene for Young People. By A. B. Palmer, M.D., LL.D. New York: A. S. Barnes & Co. 1885.

Krusi's Drawing Tablets. New York: D. Appleton & Co. 1885.

Select Spelling and Pronouncing Lessons. New York; D. Appleton & Co.

Hygiene and Physiology. By J. D. Steele, Ph.D. New York: A. S. Barnes & Co. 1885.

One of the Duanes. By Alice King Hamilton. Philadelphia: J. B. Lippincott & Co. 1885. Price, \$1.25.

Harriet Martineau's Autobiography. Edited by Maria Weston Chapman. Boston: Houghton, Mifflin and Company. 1885.

Russia Under the Tzars. By Stepniak. New York: Charles Scribner's Sons. 1885.

Twenty-eight Breakfasts. By Emily Raymond. Toledo: Blade Printing and Paper Co. 1885. Price, 10 cents.

Graded Review Questions. By W. M. Giffin and David Maclure. New York: A. Lovell & Co., 16 Astor Place. 1885.

Leonard and Gertrude. By Pestalozzi. Translated and abridged by Eva Channing. Boston: Ginn, Heath & Co. 1885.

Herbert Spencer's Philosophy. Examined by James McCosh, D.D., LL.D. New York: Charles Scribner's Sons.

Redemption in Prophecy. By John G. Wilson. Philadelphia: J. B. Lippincott & Co. 1885. Price, \$1.25.

The Handy Companion. J. R. Holcombe & Co., Publishers. Cleveland, O.

Personal Traits of British Authors. By E. T. Mason. New York: Charles Scribner's Sons. 1885. Price, \$1.50.

A Hand-Book of the United Brethren in Christ. By E. L. Shuey, A.M. Dayton, O.: United Brethren Publishing House. 1885.

Anthè. By Mrs. G. W. Chandler. New York: Phillips & Hunt. Cincinnati: Cranston & Stowe. 1885. Price, \$1.00.

At the Sign of the Blue Boar. A Story of the Reign of Charles II. By Emma Leslie. New York: Phillips & Hunt. 1885.

Life of Edward Thompson, D.D., LL.D., late a Bishop of the Methodist Episcopal Church. By his son, Edward Thompson, M.A. Cincinnati: Cranston & Stowe. 1885.

SPECIAL NOTES.

C. L. S. C. STATIONERY.—An entirely new design in stationery will be sold at Chautauqua and other Assemblies this season. It is the only authorized C. L. S. C. stationery. A new and uniform Chautauqua badge has also been devised, and will be on sale at the various Assemblies. This badge bears different emblematic designs, suitable for the different orders in the C. L. S. C. It will take the place of all other badges, except the garnet badge of the graduates, worn on Commencement day.—R. S. HOLMES.

In the April issue of THE CHAUTAUQUAN we gave the credit of the management of the School of Cookery at Lake de Funiak to Mrs. Emma P. Ewing. We were wrong. Mrs. Ewing was not at the Florida Chautauqua, but Mrs. Sophie W. Knight, of Grand Rapids, Michigan, was, and a very delightful series of lessons in the art of cookery she gave.

Chautauqua never fails to provide something new for its devotees. This summer the novelty is a mineral spring. The waters have been in the laboratory of an eminent chemist, and are pronounced rich in healthful qualities.

ERRATA.—The name of L. W. Sabin appeared among the graduates in THE CHAUTAUQUAN for February under Missouri; it should have appeared under Iowa.

TO THE CLASS OF '88.—So much having been written *pro* and *con*, respecting our class name, it is proposed to have the entire class vote for or against the name. The circles will send their vote, giving the number in favor and against present name. Those who are not in circles can send their votes as individuals. In order to insure insertion the vote must be sent to the Rev. C. C. McLean, St. Augustine, Fla., before the 1st of June.

CHAUTAUQUA DAY AT OCEAN GROVE.—At Ocean Grove, N. J., arrangements have been made to graduate all Chautauquans who have completed the course of study and prefer to receive their diploma there; and the 29th of July next has been set apart as a “Chautauqua day,” for the graduation ceremonies and other exercises appropriate to the occasion. Chancellor Vincent, Superintendent of Instruction, and other officers of the C. L. S. C. will be present, and a large number of the pupils of the young and vigorous “People’s College,” representing all parts of the country, from the ocean to the Rocky Mountains, are expected to witness and take part in the interesting exercises while enjoying the amenities of that charming seaside town. There is, at Ocean Grove, a local circle comprising a membership of nearly sixty. Many of the members are as full of enthusiasm as our honored superintendent, with all his deep devotion to the institution he has conducted to such a high degree of prosperity and usefulness, can desire, and the necessary preparations for the “Chautauqua day” will be made under its direction. The president of the Ocean Grove Association, the Rev. Dr. E. H. Stokes, is also president of the local circle. He is a zealous Chautauquan, and to his influence and example must be attributed much of the zeal, industry and enthusiasm that have made the local circle so interesting and profitable to members and visitors. All members of the class of ’85 who can not go to Chautauqua to receive their diplomas within the sacred shades of our beloved *alma mater*, but may be able to make a visit for the purpose to Ocean Grove, will be cordially welcomed, and it is suggested that they will do well to give as early notice as practicable of their intentions in that regard. Communications on the subject should be addressed to the Secretary of the Ocean Grove Local Circle, Mrs. Lulu Pile Little.

IMPORTANT TO MEMBERS OF THE C. L. S. C.

The *Chautauqua Assembly Daily Herald* will be printed in the grove at Chautauqua, and issued as a morning paper in August, 1885. It will begin Saturday, August 1, and continue every day for nineteen days. Nobody can keep well informed on the Chautauqua movement without the *Assembly Herald*. It will tell you nearly everything that is done. It will cost \$1.00 for the season, and it will contain stenographic reports of all the scientific, philosophical, and other lectures delivered at the Chautauqua meetings, together with elaborate reports of all meetings held there the coming season. The price is very low. The *Assembly Daily Herald* and THE CHAUTAUQUAN at one time, \$2.25. This offer is good only till August 1st. Members of the C. L. S. C. will find a great many valuable hints and suggestions in the reports of meetings published in the *Assembly Daily Herald*, also more than seventy splendid lectures. Some "local circles" use the *Herald* to assist them in conducting their work during the winter months. Send in your names early, before the press of business is upon us at Chautauqua.

CHAUTAUQUA SCHOOL OF LANGUAGES, 1885.

LEWIS MILLER, *Pres't.* J. H. VINCENT, D.D., *Sup't of Instruction.*

CHAUTAUQUA SCHOOL OF HEBREW.

INSTRUCTORS AND LECTURERS.

WILLIAM M. HARPER, Ph.D., Principal. Professor in Baptist Union Theological Seminary, Morgan Park, Illinois.

CHARLES R. BROWN, Professor in Newton Theological Institution, Newton Center, Massachusetts.

DAVID A. M'CLENAHAN, M.A., Pastor of United Presbyterian Church, New York City.

J. J. ANDERSON, M.A., Professor in Theological Institute, Tuscaloosa, Alabama.

REGULAR COURSES OF INSTRUCTION.

1. *The Elementary Class* will be made up of those who have no knowledge of the language. It will make about sixty recitations, three each day (one on Saturday).

2. *The Intermediate Class* will be made up of those who have once studied Hebrew, but desire to renew this study from the beginning. The class will begin with the first principles, and the work will differ from that of the Elementary Class rather in amount than in character.

3. *The Progressive Class* is intended for those who have had training in the elementary principles of Hebrew, but who wish to become familiar with

the details of the language.

4. *The Advanced Class* will do work of an advanced character. It is intended for those who are thoroughly familiar with the grammar of the language, who have at command a large vocabulary, and who have read large portions of the Hebrew Bible.

The members of the Intermediate, Progressive and Advanced Classes will be divided into four Extempore sections, each of which will read, during the four weeks, forty hours under the care of an instructor.

I. Special Courses.

In the Chautauqua School of Hebrew, instruction will be given, if desired, in the following special courses:

1. *Aramaic*.—One hour a day with Professor William R. Harper, Ph.D., Morgan Park, Ill.

2. *Advanced Hebrew Grammar*.—One hour a day with Professor William R. Harper, Ph.D.

3. *Hebrew New Testament*.—One hour a day with Professor J. J. Anderson, of Tuscaloosa, Ala.

II. Lectures.

In the various courses of lectures given at Chautauqua, during the session of the Summer School, there will be a number bearing upon topics relating to Old Testament study.

The principal of the school will, in addition, give a series of ten studies on “Modern Criticism of the Hexateuch.”

III. General Information.

Attention is invited to the following items:

Time.—The Chautauqua School of Hebrew will open August 4th, at 2 p. m., and continue until August 31st, at 12 m.

Tuition Fee.—The tuition fee will be \$10.00, payable to the Secretary of the Chautauqua Assembly. This sum includes admission to the grounds. Boarding can be obtained at from \$4 to \$6 per week.

Books.—The books needed for the several classes will be furnished at the opening of the school, or may be obtained in advance from the American Publication Society of Hebrew, Morgan Park, Ill. Books for the Special Courses must be ordered in advance.

Correspondence.—Those who desire further information, are requested to correspond with the *Principal*,

WILLIAM R. HARPER, Morgan Park, Ill.

COLLEGE OF ENGLISH.

W. D. M'CLINTOCK, A.M., Director.

Summer Term, 1885.

I. ENGLISH LANGUAGE.

1. Anglo-Saxon.

(a) Beginners.—Sweet's "Anglo-Saxon Primer."

(b) Advanced.—Sweet's "Anglo-Saxon Reader,"
and Lounsbury's "English Language."

2. Essentials of English. Introduction of the Scientific Study
of English Grammar.

Ten lectures during Teachers' Retreat.

II. ENGLISH LITERATURE.

1. Shakspeare, Hamlet. Four weeks.

2. English Lyric Poetry. Studies in the Analysis and Criticism
of Poetry. Two weeks.

THE CHAUTAUQUA COLLEGE OF MODERN LANGUAGES.

PROF. J. H. WORMAN, PH.D., Director.

PROF. A. LALANDE, Associate, School of French.

The College of Modern Languages, under the direction of the distinguished teacher and author, Dr. J. H. Worman, will open July 11th, and continue in session for six weeks.

For full information concerning the COLLEGE OF MODERN LANGUAGES for the coming season, address as follows: German, Italian, and Spanish, Dr. J. H. Worman, Vanderbilt University, Nashville, Tenn., for French, Prof. A. Lalande, Bridgeport, Conn.

ACADEMIA OF LATIN AND GREEK.

EDGAR L. SHUMWAY—Principal.

For information concerning the “Academia of Latin and Greek,” see the announcements in THE CHAUTAUQUAN for April and May, or address

EDGAR L. SHUMWAY,
Rutgers College, N. J.

Transcriber's Notes:

Obvious punctuation errors repaired.

Page 513 (illustration caption), “paletta” changed to “patella” (S, femur; T, patella; U, tibia;)

Page 519, “nterspaces” changed to “interspaces” (cavernous interspaces where trickling springs)

Page 521, “be” added (from which it will be seen)

Page 524, “Picadilly” changed to “Piccadilly”
(Egyptian Hall, Piccadilly, London)

Page 528, “Calvanism” changed to “Calvinism”
(Calvinism was revolutionary)

Page 534, “durion” changed to “durian” (we can only
refer to the durian)

Page 541, repeated word “of” removed (a method of
taking things)

Page 545, “grapest” changed to “grandest” (best, most
beautiful, and grandest)

Page 548, “unfaverably” changed to “unfavorably” (he
compared unfavorably with others)

Page 558, “Chautauquo” changed to “Chautauqua”
(printed in the grove at Chautauqua)

*** END OF THE PROJECT GUTENBERG EBOOK THE
CHAUTAUQUAN, VOL. 05, JUNE 1885, NO. 9 ***

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